Подключение библиотек

Основные уравнения, алгоритмы, зависимости и ф-и

Параметры расчета

Исходные данные

Коэф. запаса: $safety = 1.3$

Горючее: $Fuel = \text{"Керосин"}$

$turbine = \text{"ТНД"}$

Ресурс (с): $\tau = 5000 \cdot 3600$

Высота движения (м): $H_{\upsilon} = 0$

Массовый расход перед Т (кг/с):

Массовый расход утечек Т (кг/с):

Массовый расход на охл Т (кг/с):

$$\begin{pmatrix} G_{г} \\ G_{leak} \\ G_{cooling} \end{pmatrix} = \begin{cases} \begin{pmatrix} 32.30 \\ 106.96 \cdot 10^{-3} \\ 3240.8 \cdot 10^{-3} \end{pmatrix} & \text{if } turbine = \text{"ТВД"} \\ \begin{pmatrix} 35.43 \\ 35.65 \cdot 10^{-3} \\ 810.2 \cdot 10^{-3} \end{pmatrix} & \text{if } turbine = \text{"ТНД"} \end{cases} =$$

| | 1 |
|---|---|
| 1 | 35.43 |
| 2 | 0.04 |
| 3 | 0.81 |

Мощность Т (Вт): $N_T = 10^6 \cdot \begin{cases} 14.893 & \text{if } turbine = \text{"ТВД"} \\ 15.181 & \text{if } turbine = \text{"ТНД"} \end{cases} = 15.181 \cdot 10^6$

Полное давление перед Т (Па): $P^*_{г} = 10^3 \cdot \begin{cases} 2731.8 & \text{if } turbine = \text{"ТВД"} \\ 927.5 & \text{if } turbine = \text{"ТНД"} \end{cases} = 927.5 \cdot 10^3$

Полная температура перед Т (К): $T^*_{г} = \begin{cases} 1773 & \text{if } turbine = \text{"ТВД"} \\ 1368.9 & \text{if } turbine = \text{"ТНД"} \end{cases} = 1368.9$

Коэф. избытка воздуха в Т: $\alpha_{ox} = \begin{cases} 2.267 & \text{if } turbine = \text{"ТВД"} \\ 2.493 & \text{if } turbine = \text{"ТНД"} \end{cases} = 2.493$

Полное давление отбора охлаждающего воздуха (К): $P^*_{cooling} = 10^3 \cdot \begin{cases} 2845.6 & \text{if } turbine = \text{"ТВД"} \\ 319.4 & \text{if } turbine = \text{"ТНД"} \end{cases} = 319.4 \cdot 10^3$

Полная температура отбора охлаждающего воздуха (К): $T^*_{cooling} = \begin{cases} 806.9 & \text{if } turbine = \text{"ТВД"} \\ 418.2 & \text{if } turbine = \text{"ТНД"} \end{cases} = 418.2$

Коэф. сохранения полного давления охлаждения: $\sigma_{cooling} = 0.97$

Подогрев охл. от КС [К]: $\Delta T_{охл.подогрев} = 40$

Газовая постоянная (Дж/кг/К): $R_{газ}(\alpha_{ox}, Fuel) = 288.5$

Допустимая температура Л (К): $T_{Л.доп} = 1373$

Абс. скорость перед Т (м/с):

Абс. скорость после Т (м/с):

$$\begin{pmatrix} c_Г \\ c_Т \end{pmatrix} = \begin{cases} \begin{pmatrix} 100 \\ 180 \end{pmatrix} & \text{if turbine} = \text{"ТВД"} \\ \begin{pmatrix} 180 \\ 260 \end{pmatrix} & \text{if turbine} = \text{"ТНД"} \end{cases} =$$

| | 1 |
|---|---|
| 1 | 180.0 |
| 2 | 260.0 |

[1, с.15] $80 \le c_Т \le 400 = 1$

Лопаточный КПД Т: $\eta_Л = 88\%$

$88\% \le \eta_Л \le 95\% = 1$

Угол входа в Т: $\alpha_Г = 90 \cdot °$

Окр. скорость Л последней ступени на ср. диаметре Т (м/с):

$$u_Т = \begin{cases} 520 & \text{if turbine} = \text{"ТВД"} \\ 260 & \text{if turbine} = \text{"ТНД"} \end{cases} = 260.0$$

Исходные данные

Основне размеры Т

$z = ORIGIN .. N_r$

Полное давление отбора охлаждающего воздуха (К): $P^*_{cooling} = P^*_{cooling} \cdot \sigma_{cooling} = 309.8 \cdot 10^3$

Полная температура отбора охлаждающего воздуха (К): $T^*_{cooling} = T^*_{cooling} + \Delta T_{охл.подогрев} = 458.2$

Массовый расход перед Т (кг/с): $G_T = G_Г - G_{leak} = 35.4$

Массовый расход после Т (кг/с): $G_T = G_Г + G_{cooling} = 36.2$

Удельная работа Т (Дж/кг): $L^*_T = \dfrac{N_T}{mean(G_Г, G_T)} = 424.1 \cdot 10^3$

$L^*_T \le 550 \cdot 10^3 = 1$

Располагаемый теплоперепад в Т (Дж/кг): $H_T = \dfrac{L^*_T + 0.5 c_T^2}{\eta_Л} = 520.3 \cdot 10^3$

$$\begin{pmatrix} \text{iteration} \\ k_г \\ P_г \\ T_г \end{pmatrix} = \left| \begin{array}{l} \text{iteration} = 0 \\ k_г = k_{ад}\left(Cp_{газ}\left(P^*_г, T^*_г, \alpha_{ox}, \text{Fuel}\right), R_{газ}\left(\alpha_{ox}, \text{Fuel}\right)\right) \\ \text{while } 1 > 0 \\ \quad \left| \begin{array}{l} \text{iteration} = \text{iteration} + 1 \\ Cp_г = \dfrac{k_г}{k_г - 1} \cdot R_{газ}\left(\alpha_{ox}, \text{Fuel}\right) \\ T_г = T^*_г - \dfrac{c_г^{\,2}}{2 \cdot Cp_г} \\ P_г = P^*_г \cdot \left(\dfrac{T_г}{T^*_г}\right)^{\frac{k_г}{k_г - 1}} \\ k'_г = k_{ад}\left(Cp_{газ}\left(P_г, T_г, \alpha_{ox}, \text{Fuel}\right), R_{газ}\left(\alpha_{ox}, \text{Fuel}\right)\right) \\ \text{if } \left|\text{eps}\left(\text{"rel"}, k_г, k'_г\right)\right| \le \text{epsilon} \\ \quad \left| \begin{array}{l} k_г = k'_г \\ \text{break} \end{array} \right. \\ k_г = k'_г \end{array} \right. \\ \left(\text{iteration} \;\; k_г \;\; P_г \;\; T_г\right)^T \end{array} \right. =$$

| | 1 |
|---|---|
| 1 | 1.0 |
| 2 | 1.3 |
| 3 | 890047.3 |
| 4 | 1356.0 |

Количество итераций: $\text{iteration} = 1$

Показатель адиабаты перед Т: $k_г = 1.298$

Статическое давление перед Т (Па): $P_г = 890 \cdot 10^3$

Статическая температура перед Т (К): $T_г = 1356$

Теплоемкость перед Т (Дж/кг/К): $Cp_г = Cp_{газ}\left(P_г, T_г, \alpha_{ox}, \text{Fuel}\right) = 1256$

$$\begin{pmatrix} iteration \\ k_T \\ P_T \\ T_T \end{pmatrix} = \left| \begin{array}{l} iteration = 0 \\ k_T = k_Г \\ \text{while } 1 > 0 \\ \quad \left| \begin{array}{l} iteration = iteration + 1 \\ k_{cp} = mean(k_Г, k_T) \\ Cp = \dfrac{k_{cp}}{k_{cp} - 1} \cdot R_{газ}(\alpha_{ox}, Fuel) \\ P_T = P^*_Г \cdot \left(1 - \dfrac{H_T}{Cp \cdot T^*_Г}\right)^{\frac{k_{cp}}{k_{cp}-1}} \\ T_T = T^*_Г - \dfrac{H_T \cdot \eta_л}{Cp} \\ k'_T = k_{ад}\left(Cp_{газ}(P_T, T_T, \alpha_{ox}, Fuel), R_{газ}(\alpha_{ox}, Fuel)\right) \\ \text{if } \left|eps(\text{"rel"}, k_T, k'_T)\right| \le epsilon \\ \quad \left| \begin{array}{l} k_T = k'_T \\ \text{break} \end{array} \right. \\ k_T = k'_T \end{array} \right. \\ \left(iteration \;\; k_T \;\; P_T \;\; T_T\right)^T \end{array} \right. =$$

| | 1 |
|---|---|
| 1 | 2 |
| 2 | 1.32 |
| 3 | 191463.061 |
| 4 | 994.672 |

Количество итераций: $iteration = 2$

Показатель адиабаты после Т: $k_T = 1.320$

Статическое давление после Т (Па): $P_T = 191.5 \cdot 10^3$ $\quad P_T \ge P_{атм}(H_\upsilon) = 1$

Статическая температура после Т (К): $T_T = 994.7$

Теплоемкость после Т (Дж/кг/К): $Cp_T = Cp_{газ}(P_T, T_T, \alpha_{ox}, Fuel) = 1190.6$

Ср. показатель адиабаты Т: $k = \mathrm{mean}(k_Г, k_Т) = 1.309$

Ср. теплоемкость Т (Дж/кг/К): $Cp = \frac{k}{k-1} \cdot R_{газ}(\alpha_{ox}, Fuel) = 1222.3$

Степень понижения давления: $\pi_T = \frac{P^*_Г}{P_Т} = 4.84$

Удельный объём перед Т (м³/кг):

Удельный объём после Т (м³/кг):

$$\begin{pmatrix} v_Г \\ v_Т \end{pmatrix} = R_{газ}(\alpha_{ox}, Fuel) \cdot \begin{pmatrix} \frac{T_Г}{P_Г} \\ \frac{T_Т}{P_Т} \end{pmatrix} =$$

| | 1 |
|---|---|
| 1 | 0.440 |
| 2 | 1.499 |

Площадь кольцевого сечения перед Т (м²):

Площадь кольцевого сечения после Т (м²):

$$\begin{pmatrix} F_Г \\ F_Т \end{pmatrix} = \begin{pmatrix} \frac{G_Г \cdot v_Г}{c_Г} \\ \frac{G_Т \cdot v_Т}{c_Т} \end{pmatrix} =$$

| | 1 |
|---|---|
| 1 | 86421 |
| 2 | 208684 |

$\cdot 10^{-6}$

Основне размеры Т

Коэф. формы: $k_n = 6.8$

Выбранный материал Л: $Material = \begin{cases} \text{"ВКНА-1В"} & \text{if } turbine = \text{"ТВД"} \\ \text{"ЭИ464"} & \text{otherwise} \end{cases}$ = "ЭИ464"

ВКНА-1В
ВКНА-4У
ВКНА-25

Плотность материала Л (кг/м^3): $\rho_Л = \begin{cases} 7938 & \text{if } Material = \text{"ВКНА-1В"} \\ NaN & \text{otherwise} \end{cases}$ =

Коэф. Пуассона материала Л (): $\mu_Л = 0.3$

Модуль Юнга I рода материала Л (Па): $E_Л = 210 \cdot 10^9$

Предел длительной прочности Л РК (Па): $\sigma_{дл.Л} = 10^6 \cdot \begin{cases} 205 & \text{if } Material = \text{"ВКНА-1В"} \\ 120 & \text{otherwise} \end{cases} = 120 \cdot 10^6$

## Частота вращения Т

Мах частота вращения ротора на входе (об/мин): $\sqrt{\dfrac{\sigma_{дл.Л}}{safety \cdot k_n \cdot F_г}} = 12533$

Мах частота вращения ротора на выходе (об/мин): $n_{max} = \sqrt{\dfrac{\sigma_{дл.Л}}{safety \cdot k_n \cdot F_т}} = 8065$

Рекомендукмая ном. частота вращения (об/мин): $n = n_{max} \cdot 0.95 = 7662$

$n = \begin{cases} 15000 & \text{if } turbine = \text{"ТВД"} \\ 5300 & \text{if } turbine = \text{"ТНД"} \end{cases} = 5300$

Ном. частота вращения (рад/с): $\omega = \dfrac{2 \cdot \pi \cdot n}{60} = 555.0$

Ср. диаметр перед Т (м):
Ср. диаметр после Т (м):

$$\begin{pmatrix} D_{г.ср} \\ D_{т.ср} \end{pmatrix} = \frac{2}{\omega} \cdot \begin{pmatrix} u_г \\ u_т \end{pmatrix} = \begin{array}{c|c} & 1 \\ \hline 1 & 936.9 \\ 2 & 936.9 \end{array} \cdot 10^{-3}$$

Длина Л первой ступени Т (м):
Длина Л последней ступени Т (м):

$$\begin{pmatrix} l_г \\ l_т \end{pmatrix} = \frac{1}{\pi} \cdot \begin{pmatrix} \dfrac{F_г}{D_{г.ср}} \\ \dfrac{F_т}{D_{т.ср}} \end{pmatrix} = \begin{array}{c|c} & 1 \\ \hline 1 & 29.36 \\ 2 & 70.90 \end{array} \cdot 10^{-3}$$

$$\frac{l_г}{D_{г.ср}} = \frac{1}{31} \qquad \frac{l_т}{D_{т.ср}} = \frac{1}{13}$$

Диаметр периферии после Т (м):
Диаметр корня после Т (м):

$$\begin{pmatrix} D_{т.пер} \\ D_{т.кор} \end{pmatrix} = \begin{pmatrix} D_{т.ср} + l_т \\ D_{т.ср} - l_т \end{pmatrix} = \begin{array}{c|c} & 1 \\ \hline 1 & 1007.8 \\ 2 & 866.0 \end{array} \cdot 10^{-3}$$

Частота вращения Т

$y_0 = 0.55$

Коэф. использования скорости: $\mu_c = \mathrm{mean}(0.7, 1) = 0.9$ $\quad 0.7 \le \mu_c \le 1 = 1$

Определение количества ступеней Т

$$\begin{pmatrix} Z_{recomend} \\ \alpha_{возв} \end{pmatrix} = \left| \begin{array}{l} c_{cp} = \mathrm{mean}(c_г, c_т) \\ \alpha_{возв} = 0.025 \\ \text{while } 1 > 0 \\ \quad \left| \begin{array}{l} Z = \mathrm{round}\left[ \dfrac{2 \cdot H_T \cdot \dfrac{(1+\alpha_{возв})}{(\mu_c \cdot c_{cp})^2} - 1}{\dfrac{u_T^{\,2}}{(\mu_c \cdot c_{cp})^2 \cdot y_0^{\,2}} - 1} \right] \\ \text{break if } \left| \mathrm{eps}\left[ \text{"rel"}, \alpha_{возв}, \dfrac{Z-1}{2 \cdot Z} \cdot \left( \pi_T^{\frac{k-1}{k}} - 1 \right) \cdot (1 - \eta_л) \right] \right| < \mathrm{epsilon} \\ \alpha_{возв} = \dfrac{Z-1}{2 \cdot Z} \cdot \left( \pi_T^{\frac{k-1}{k}} - 1 \right) \cdot (1 - \eta_л) \\ \text{if } \alpha_{возв} = 0 \\ \quad \left| \begin{array}{l} \begin{pmatrix} Z \\ \alpha_{возв} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \text{break} \end{array} \right. \end{array} \right. \\ \begin{pmatrix} Z \\ \alpha_{возв} \end{pmatrix} \end{array} \right. =$$

| | 1 |
|---|---|
| 1 | 5.000 |
| 2 | 0.022 |

Рекомендуемое количество ступеней: $Z_{recomend} = 5$

Количество ступеней: $Z = \left| \begin{array}{l} 1 \text{ if turbine} = \text{"ТВД"} \\ 4 \text{ if turbine} = \text{"ТНД"} \end{array} \right. = 4$

Дискретизация ступеней: $i = 1..Z$

Дискретизация сечений: $ii = 1..2 \cdot Z + 1$

Определение количества ступеней Т

Равномерное распределение мощности Т по ступеням (Вт): $N_{ст_i} = \frac{N_T}{Z}$

$N_{ст}{}^T =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.80 | 3.80 | 3.80 | 3.80 |

$\cdot 10^6$

Вид проточной части:
("const", "кор", "ср", "пер","доля от предыдущего диаметра периферии")

$$ЗППЧ = \begin{pmatrix} \begin{cases} \text{"const"} & \text{if } Z = 1 \\ \text{"кор"} & \text{otherwise} \end{cases} & \text{"1.07"} & \text{"1.065"} & \text{"1.03"} & \text{"пер"} & \text{"пер"} \\ \begin{cases} \text{"пер"} & \text{if } Z = 1 \\ \text{"1.055"} & \text{otherwise} \end{cases} & \text{"1.07"} & \text{"1.05"} & \text{"кор"} & \text{"пер"} & \text{"пер"} \end{pmatrix}^T$$

Определение проточной части ОТ

Линейное распределение кольцевых площадей по сечениям:

$$F = \left| \begin{array}{l} \text{for } i \in 1..2Z+1 \\ \quad F_i = \frac{F_T - F_Г}{st(Z,3)-1} \cdot i + \left( F_Г - \frac{F_T - F_Г}{st(Z,3)-1} \right) \\ \text{for } i \in 1..Z \\ \quad \text{for } a \in 2..3 \\ \quad\quad F_{st(i,a)} = F_{st(i,a-1)} \text{ if } ЗППЧ_{i,a-1} = \text{"const"} \\ F \end{array} \right.$$

$F^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 86421 | 101704 | 116987 | 132270 | 147553 | 162836 | 178118 | 193401 | 208684 |

$\cdot 10^{-6}$

$$
D = \left|
\begin{array}{l}
\text{for } i \in 2Z+1 \\
\quad \text{for } r \in 1..N_r \\
\quad\quad D_{i,r} = \left|
\begin{array}{ll}
D_{\text{Т.кор}} & \text{if } r = 1 \\
D_{\text{Т.ср}} & \text{if } r = av(N_r) \\
D_{\text{Т.пер}} & \text{if } r = N_r
\end{array}
\right. \\
\text{for } i \in Z..1 \\
\quad \text{for } a \in 2..1 \\
\quad\quad \text{for } r \in 1..N_r \\
\quad\quad\quad D_{st(i,a),r} = \left|
\begin{array}{l}
\text{if } \text{ЗППЧ}_{i,a} = \text{"const"} \\
\quad \left|
\begin{array}{ll}
D_{st(i,a+1),av(N_r)} - \dfrac{F_{st(i,a)}}{\pi \cdot D_{st(i,a+1),av(N_r)}} & \text{if } r = 1 \\
D_{st(i,a+1),av(N_r)} & \text{if } r = av(N_r) \\
D_{st(i,a+1),av(N_r)} + \dfrac{F_{st(i,a)}}{\pi \cdot D_{st(i,a+1),av(N_r)}} & \text{if } r = N_r
\end{array}
\right. \\
\text{if } \text{ЗППЧ}_{i,a} = \text{"кор"} \\
\quad \left|
\begin{array}{ll}
D_{st(i,a+1),1} & \text{if } r = 1 \\
\dfrac{1}{2} \cdot \left[ D_{st(i,a+1),1} + \sqrt{\left(D_{st(i,a+1),1}\right)^2 + \dfrac{4 \cdot F_{st(i,a)}}{\pi}} \right] & \text{if } r = av(N_r) \\
\sqrt{\left(D_{st(i,a+1),1}\right)^2 + \dfrac{4 \cdot F_{st(i,a)}}{\pi}} & \text{if } r = N_r
\end{array}
\right. \\
\text{if } \text{ЗППЧ}_{i,a} = \text{"ср"} \\
\quad \left|
\begin{array}{ll}
D_{st(i,a+1),av(N_r)} - \dfrac{F_{st(i,a)}}{\pi \cdot D_{st(i,a+1),av(N_r)}} & \text{if } r = 1 \\
D_{st(i,a+1),av(N_r)} & \text{if } r = av(N_r) \\
D_{st(i,a+1),av(N_r)} + \dfrac{F_{st(i,a)}}{\pi \cdot D_{st(i,a+1),av(N_r)}} & \text{if } r = N_r
\end{array}
\right. \\
\text{if } \text{ЗППЧ}_{i,a} = \text{"пер"} \\
\quad \left|
\begin{array}{ll}
\sqrt{\left(D_{st(i,a+1),N_r}\right)^2 - \dfrac{4 \cdot F_{st(i,a)}}{\pi}} & \text{if } r = 1 \\
\dfrac{1}{2} \cdot \left[ \sqrt{\left(D_{st(i,a+1),N_r}\right)^2 - \dfrac{4 \cdot F_{st(i,a)}}{\pi}} + D_{st(i,a+1),N_r} \right] & \text{if } r = av(N_r) \\
D \quad \text{if } r = N
\end{array}
\right.
\end{array}
\right.
\end{array}
\right.
$$

$D^T =$ $\cdot 10^{-}$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 620.7 | 620.7 | 651.1 | 698.2 | 750.4 | 802.7 | 843.9 | 866.0 | 866.0 |
| 2 | 662.2 | 669.0 | 704.0 | 754.0 | 808.5 | 862.8 | 906.5 | 932.1 | 936.9 |
| 3 | 703.7 | 717.4 | 756.9 | 809.9 | 866.6 | 922.9 | 969.0 | 998.1 | 1007.8 |

$$R = \frac{D}{2}$$

$R^T =$ $\cdot 10^{-3}$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 310.3 | 310.3 | 325.6 | 349.1 | 375.2 | 401.4 | 422.0 | 433.0 | 433.0 |
| 2 | 331.1 | 334.5 | 352.0 | 377.0 | 404.2 | 431.4 | 453.2 | 466.0 | 468.5 |
| 3 | 351.9 | 358.7 | 378.4 | 404.9 | 433.3 | 461.4 | 484.5 | 499.1 | 503.9 |

$$
\bar{d} = \left|
\begin{array}{l}
\text{for } i \in 1..Z \\
\quad \text{for } a \in 1..3 \\
\quad\quad \bar{d}_{st(i,a)} = \dfrac{D_{st(i,a),1}}{D_{st(i,a),N_r}} \\
\bar{d}
\end{array}
\right.
$$

$\bar{d}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.8819 | 0.8651 | 0.8602 | 0.8621 | 0.8659 | 0.8698 | 0.8709 | 0.8677 | 0.8593 |

$\bar{d}^T \le 0.9 =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$$
h = \left|
\begin{array}{l}
\text{for } i \in 1..2Z+1 \\
\quad h_i = \dfrac{F_i}{\pi \cdot D_{i,av(N_r)}} \\
h
\end{array}
\right.
$$

$h^T =$ $\cdot 10^{-3}$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 41.54 | 48.39 | 52.90 | 55.84 | 58.09 | 60.07 | 62.55 | 66.05 | 70.90 |

$$
\begin{cases}
D_{st(i,a+1),N_r} & \text{if } r = N_r \\
\text{if } \left(ЗППЧ_{i,a} \neq \text{"const"}\right) \wedge \left(ЗППЧ_{i,a} \neq \text{"кор"}\right) \wedge \left(ЗППЧ_{i,a} \neq \text{"ср"}\right) \wedge \left(ЗППЧ_{i,a} \neq \text{"пер"}\right) \\
\quad \begin{cases}
\sqrt{\left(\dfrac{D_{st(i,a+1),N_r}}{str2num\left(ЗППЧ_{i,a}\right)}\right)^2 - \dfrac{4 \cdot F_{st(i,a)}}{\pi}} & \text{if } r = 1 \\
\dfrac{1}{2} \cdot \left[\sqrt{\left(\dfrac{D_{st(i,a+1),N_r}}{str2num\left(ЗППЧ_{i,a}\right)}\right)^2 - \dfrac{4 \cdot F_{st(i,a)}}{\pi}} + \dfrac{D_{st(i,a+1),N_r}}{str2num\left(ЗППЧ_{i,a}\right)}\right] & \text{if } r = av\left(N_r\right) \\
\dfrac{D_{st(i,a+1),N_r}}{str2num\left(ЗППЧ_{i,a}\right)} & \text{if } r = N_r
\end{cases} \\
\text{NaN} \quad \text{otherwise}
\end{cases}
$$

$D$

$$
u = \left| \begin{array}{l} \text{for } i \in 1..2 \cdot Z + 1 \\ \quad \text{for } r \in 1..N_r \\ \quad\quad u_{i,r} = \dfrac{\pi \cdot D_{i,r} \cdot n}{60} \\ u \end{array} \right.
$$

$u^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 172.2 | 172.2 | 180.7 | 193.8 | 208.2 | 222.8 | 234.2 | 240.3 | 240.3 |
| 2 | 183.8 | 185.7 | 195.4 | 209.3 | 224.4 | 239.4 | 251.6 | 258.7 | 260.0 |
| 3 | 195.3 | 199.1 | 210.0 | 224.7 | 240.5 | 256.1 | 268.9 | 277.0 | 279.7 |

$F^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 86421 | 101704 | 116987 | 132270 | 147553 | 162836 | 178118 | 193401 | 208684 |

$\cdot 10^{-6}$

$\overline{d}_1 = 0.8819$ $\quad$ $\overline{d}_1 \le 0.9 = 1$

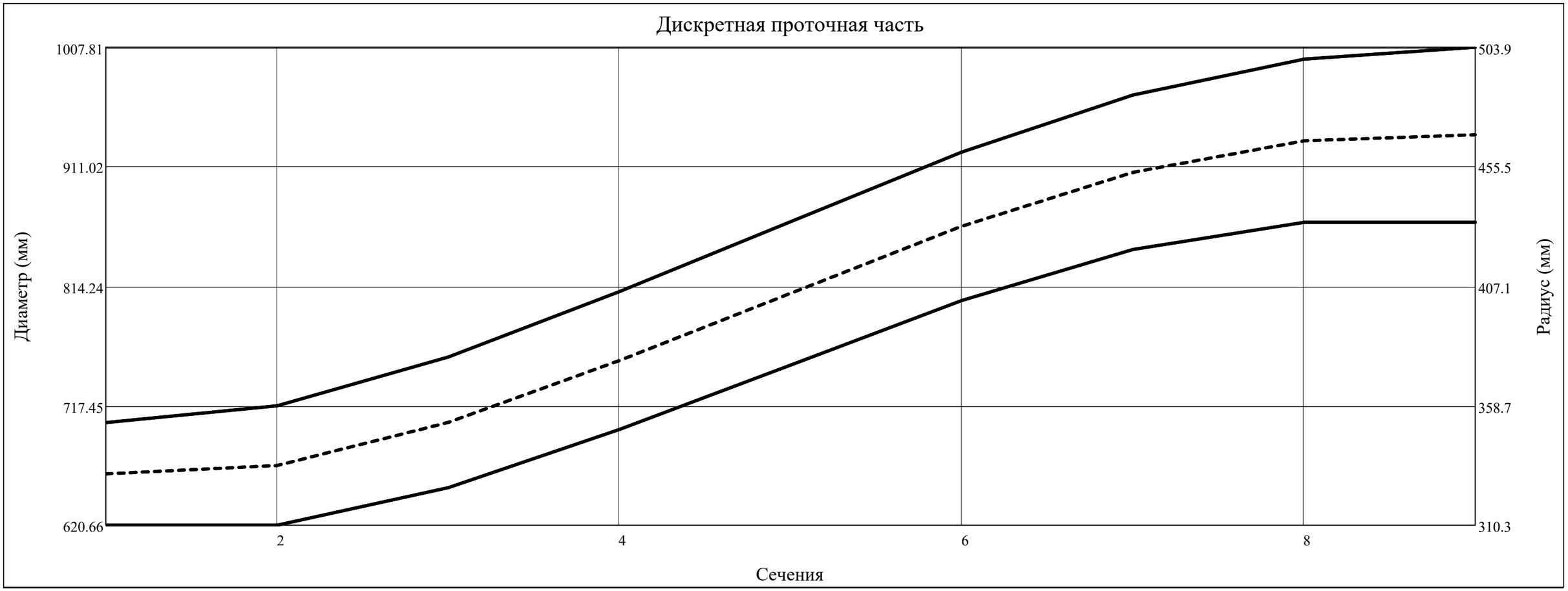
$\overline{d}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.8819 | 0.8651 | 0.8602 | 0.8621 | 0.8659 | 0.8698 | 0.8709 | 0.8677 | 0.8593 |

$D^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 620.7 | 620.7 | 651.1 | 698.2 | 750.4 | 802.7 | 843.9 | 866.0 |
| 2 | 662.2 | 669.0 | 704.0 | 754.0 | 808.5 | 862.8 | 906.5 | 932.1 |
| 3 | 703.7 | 717.4 | 756.9 | 809.9 | 866.6 | 922.9 | 969.0 | ... |

$\cdot 10^{-3}$




$h^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 41.54 | 48.39 | 52.90 | 55.84 | 58.09 | 60.07 | 62.55 | 66.05 | 70.90 |

$\cdot 10^{-3}$

Осевая ширина Л СА и РК [1, с.183]:

$$\begin{pmatrix} B_{CA} \\ B_{PK} \end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \begin{pmatrix} B_{CA_i} \\ B_{PK_i} \end{pmatrix} = \begin{pmatrix} \dfrac{D_{st(i,2),av}(N_r) - 0.25 \cdot h_{st(i,2)}}{16.4} \\ \dfrac{D_{st(i,3),av}(N_r)}{24} \end{pmatrix} \\ \begin{pmatrix} B_{CA} \\ B_{PK} \end{pmatrix} \end{array} \right.$$

$\mathrm{stack}\left(B_{CA}^{\,T}, B_{PK}^{\,T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 40.1 | 45.1 | 51.7 | 55.8 |
| 2 | 29.3 | 33.7 | 37.8 | 39.0 |

$\cdot 10^{-3}$

Радиальный зазор (м):
Осевой зазор (м):
(Лучше выбирать большее значение)

$$\begin{pmatrix} \Delta_r \\ \Delta_a \end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } a \in 1..3 \\ \quad\quad \left| \begin{array}{l} \Delta_{r_{st(i,a)}} = 0.001 \cdot D_{st(i,a),N_r} \\ \Delta_{a_i} = 0.25 \cdot B_{CA_i} \end{array} \right. \\ \begin{pmatrix} \Delta_r \\ \Delta_a \end{pmatrix} \end{array} \right.$$

$\Delta_r^{\,T} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.704 | 0.717 | 0.757 | 0.810 | 0.867 | 0.923 | 0.969 | 0.998 | 1.008 |

$\cdot 10^{-3}$

$\Delta_a^{\,T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 10.014 | 11.282 | 12.924 | 13.957 |

$\cdot 10^{-3}$

$0.2 \le \dfrac{\Delta_{a_i}}{B_{CA_i}} \le 0.25 =$

| |
|---|
| 1 |
| 1 |
| 1 |
| 1 |

$$\begin{pmatrix}\gamma_{ПЧпер}\\ \gamma_{ПЧ}\\ \gamma_{ПЧкор}\end{pmatrix} = \left|\begin{array}{l}
\text{for } i \in 1..Z \\
\quad \text{for } a \in 1..2 \\
\quad\quad \text{for } r \in N_r \\
\quad\quad\quad \left|\begin{array}{l}
\begin{pmatrix}k2\\ k1\end{pmatrix} = \dfrac{0.5}{B_{CA_i}}\cdot\begin{pmatrix}D_{st(i,2),r} - D_{st(i,1),r}\\ 0\end{pmatrix} \text{ if } a = 1 \\
\begin{pmatrix}k2\\ k1\end{pmatrix} = \dfrac{0.5}{B_{PK_i}}\cdot\begin{pmatrix}D_{st(i,3),r} - D_{st(i,2),r}\\ 0\end{pmatrix} \text{ if } a = 2 \\
\gamma_{ПЧпер_{st(i,a)}} = \operatorname{atan}\left(\dfrac{k2 - k1}{1 + k2\cdot k1}\right)
\end{array}\right. \\
\text{for } i \in 1..Z \\
\quad \text{for } a \in 1..2 \\
\quad\quad \left|\begin{array}{l}
\begin{pmatrix}k2\\ k1\end{pmatrix} = \dfrac{0.5}{B_{CA_i}}\cdot\begin{pmatrix}D_{st(i,2),N_r} - D_{st(i,1),N_r}\\ D_{st(i,2),1} - D_{st(i,1),1}\end{pmatrix} \text{ if } a = 1 \\
\begin{pmatrix}k2\\ k1\end{pmatrix} = \dfrac{0.5}{B_{PK_i}}\cdot\begin{pmatrix}D_{st(i,3),N_r} - D_{st(i,2),N_r}\\ D_{st(i,3),1} - D_{st(i,2),1}\end{pmatrix} \text{ if } a = 2 \\
\gamma_{ПЧ_{st(i,a)}} = \operatorname{atan}\left(\dfrac{k2 - k1}{1 + k2\cdot k1}\right)
\end{array}\right. \\
\text{for } i \in 1..Z \\
\quad \text{for } a \in 1..2 \\
\quad\quad \text{for } r \in 1 \\
\quad\quad\quad \left|\begin{array}{l}
\begin{pmatrix}k2\\ k1\end{pmatrix} = \dfrac{0.5}{B_{CA_i}}\cdot\begin{pmatrix}D_{st(i,2),r} - D_{st(i,1),r}\\ 0\end{pmatrix} \text{ if } a = 1 \\
\begin{pmatrix}k2\\ k1\end{pmatrix} = \dfrac{0.5}{B_{PK_i}}\cdot\begin{pmatrix}D_{st(i,3),r} - D_{st(i,2),r}\\ 0\end{pmatrix} \text{ if } a = 2 \\
\gamma_{ПЧкор_{st(i,a)}} = \operatorname{atan}\left(\dfrac{k2 - k1}{1 + k2\cdot k1}\right)
\end{array}\right. \\
\begin{pmatrix}\gamma_{ПЧпер}\\ \gamma_{ПЧ}\\ \gamma_{ПЧкор}\end{pmatrix}
\end{array}\right.$$

$\mathrm{stack}\left(\gamma_{ПЧпер}^{T}, \gamma_{ПЧ}^{T}, \gamma_{ПЧкор}^{T}\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 9.70 | 33.92 | 30.41 | 40.08 | 28.58 | 31.42 | 14.59 | 7.08 |
| 2 | 9.70 | 6.50 | 2.86 | 2.32 | 1.72 | 2.81 | 3.42 | 7.08 |
| 3 | 0.00 | 27.43 | 27.56 | 37.76 | 26.86 | 28.61 | 11.18 | 0.00 |

$.^{\circ}$

$\gamma_{ПЧ}^{T} \le 20\cdot^{\circ} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$\gamma_{ПЧ}^{T} \le 25\cdot^{\circ} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$\gamma_{ПЧкор}^{T} > -12\cdot^{\circ} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$\gamma_{ПЧкор}^{T} > -15\cdot^{\circ} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

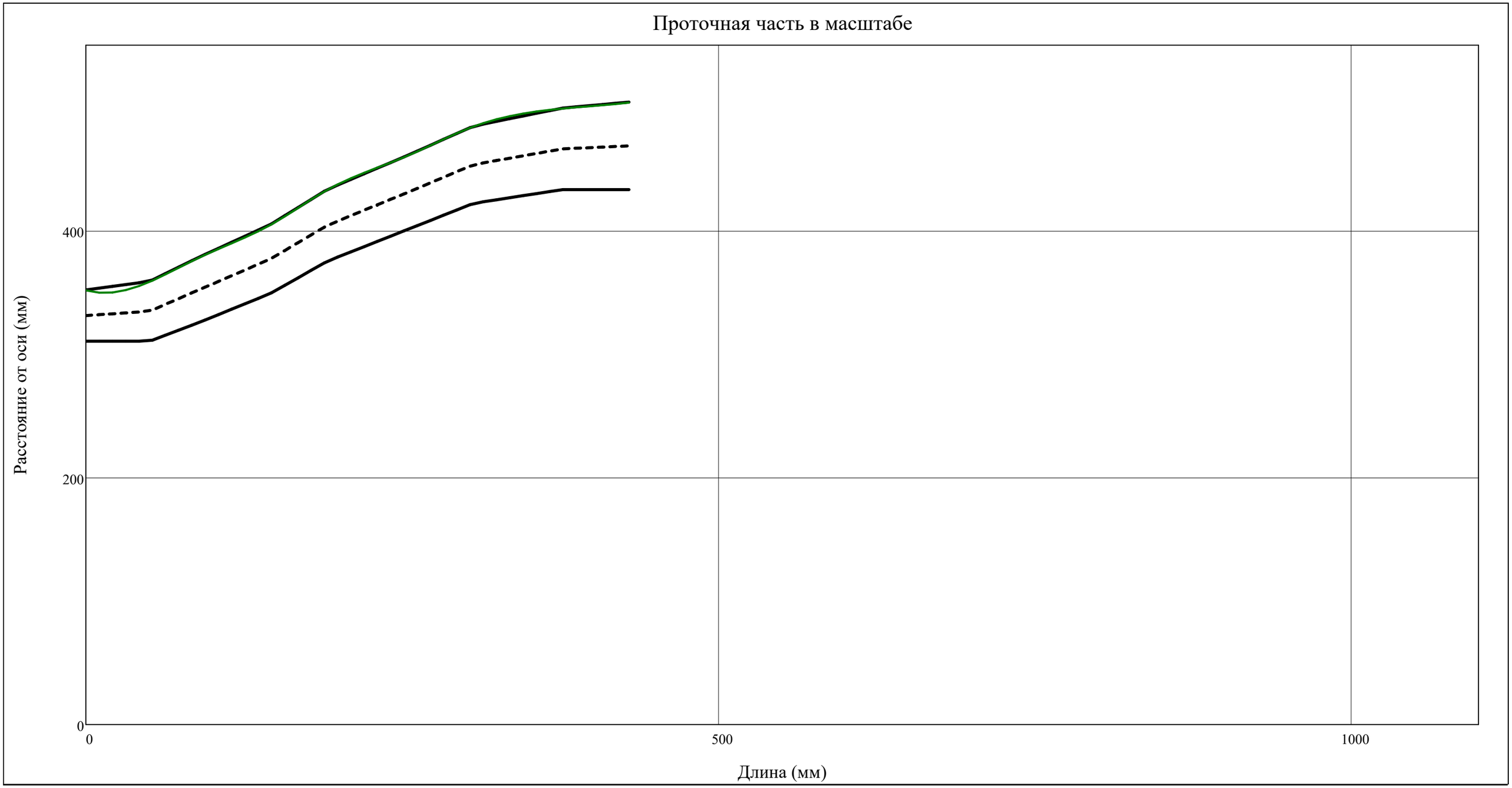
$$
\begin{pmatrix} x_{ПЧ} \\ y_{ПЧпер} \\ y_{ПЧср} \\ y_{ПЧкор} \\ y_{Лпер} \end{pmatrix} =
\left|
\begin{array}{l}
c = 1 \\
x_{ПЧ_c} = 0 \\
y_{ПЧпер_c} = D_{st(c,1),N_r} \\
y_{Лпер_c} = y_{ПЧпер_c} - \Delta_{r_c} \\
y_{ПЧср_c} = D_{st(c,1),av(N_r)} \\
y_{ПЧкор_c} = D_{st(c,1),1} \\
\text{for } i \in 1..Z \\
\quad \left|
\begin{array}{l}
c = c + 1 \\
x_{ПЧ_c} = x_{ПЧ_{c-1}} + 0.5\cdot\Delta_{a_i} + B_{CA_i} + 0.5\cdot\Delta_{a_i} \\
\begin{pmatrix} y_{ПЧпер_c} \\ y_{ПЧср_c} \\ y_{ПЧкор_c} \end{pmatrix} = \begin{pmatrix} D_{st(i,2),N_r} \\ D_{st(i,2),av(N_r)} \\ D_{st(i,2),1} \end{pmatrix} \\
y_{Лпер_c} = y_{ПЧпер_c} - \Delta_{r_i} \\
c = c + 1 \\
x_{ПЧ_c} = x_{ПЧ_{c-1}} + 0.5\cdot\Delta_{a_i} + B_{PK_i} + 0.5\cdot\Delta_{a_i} \\
\begin{pmatrix} y_{ПЧпер_c} \\ y_{ПЧср_c} \\ y_{ПЧкор_c} \end{pmatrix} = \begin{pmatrix} D_{st(i+1,1),N_r} \\ D_{st(i+1,1),av(N_r)} \\ D_{st(i+1,1),1} \end{pmatrix} \\
y_{Лпер_c} = y_{ПЧпер_c} - \Delta_{r_i}
\end{array}
\right. \\
\begin{pmatrix} x_{ПЧ} \\ y_{ПЧпер} \\ y_{ПЧср} \\ y_{ПЧкор} \\ y_{Лпер} \end{pmatrix}
\end{array}
\right.
$$

$$
Length = \sum_{i=1}^{Z} B_{CA_i} + \sum_{i=1}^{Z} \Delta_{a_i} + \sum_{i=1}^{Z} B_{PK_i} = 380.7\cdot 10^{-3}
$$

$$
x = \min(x_{ПЧ}), \min(x_{ПЧ}) + \frac{\max(x_{ПЧ}) - \min(x_{ПЧ})}{N_{dis}} .. \max(x_{ПЧ})
$$

$y_{ПЧпер}(l) = \mathrm{linterp}\left(x_{ПЧ}, 0.5 \cdot y_{ПЧпер}, l\right)$ $y_{ПЧср}(l) = \mathrm{linterp}\left(x_{ПЧ}, 0.5 \cdot y_{ПЧср}, l\right)$ $y_{ПЧкор}(l) = \mathrm{linterp}\left(x_{ПЧ}, 0.5 \cdot y_{ПЧкор}, l\right)$

$y_{Лпер}(l) = \mathrm{interp}\left(\mathrm{cspline}\left(x_{ПЧ}, 0.5 \cdot y_{Лпер}\right), x_{ПЧ}, 0.5 \cdot y_{Лпер}, l\right)$



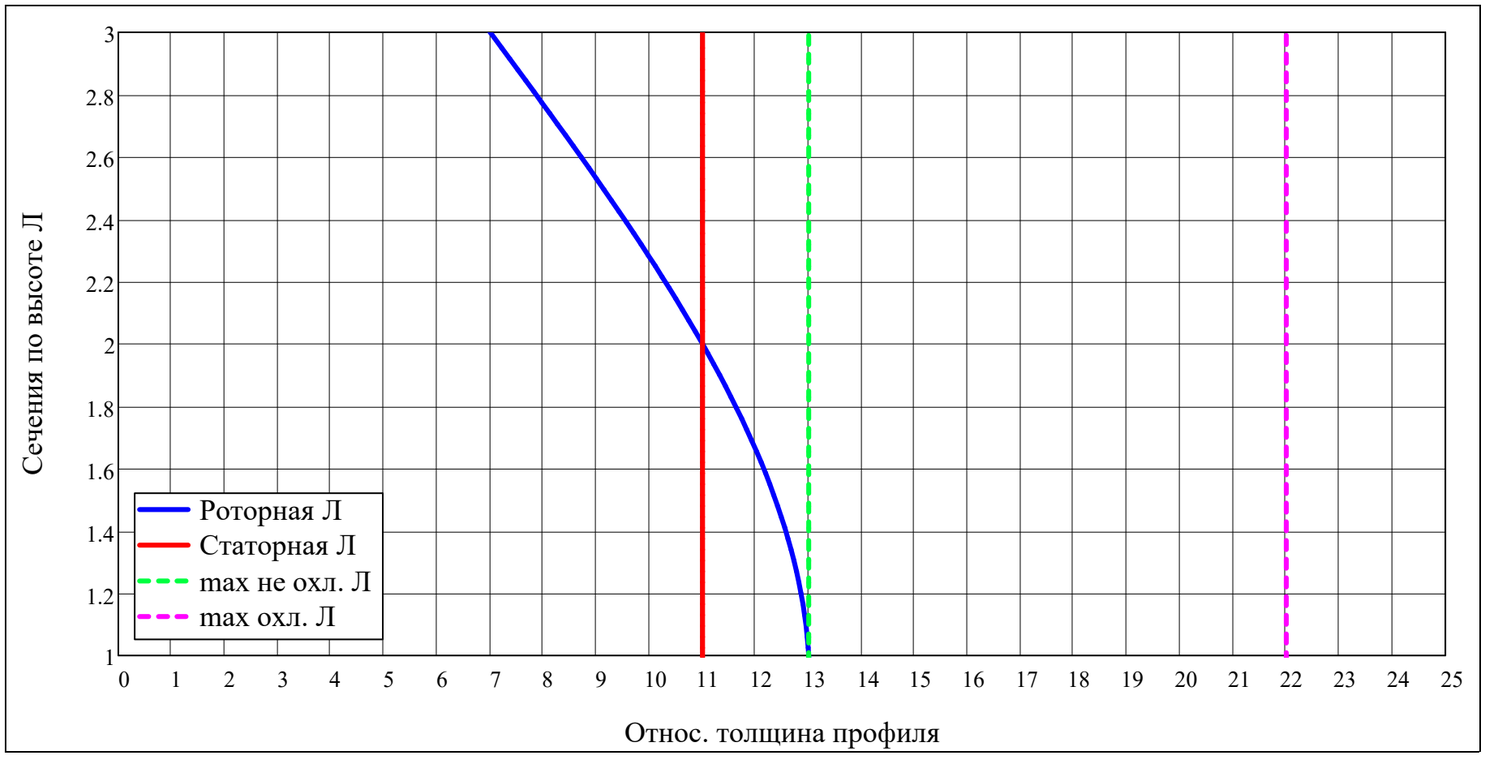

Определение проточной части ОТ

Относ. толщины и радиусы профилей

Относ. толщины Л РК и СА:

$$\overline{c}_{stator.}(r) = \begin{cases} \mathrm{interp}\left[\mathrm{cspline}\left[\begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}15\\15\\15\end{pmatrix}\%\right], \begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}15\\15\\15\end{pmatrix}\%, r\right] & \text{if } T_{Л.доп} < T^*_г \\ \mathrm{interp}\left[\mathrm{cspline}\left[\begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}11\\11\\11\end{pmatrix}\%\right], \begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}11\\11\\11\end{pmatrix}\%, r\right] & \text{otherwise} \end{cases}$$

$$\overline{c}_{rotor.}(r) = \begin{cases} \mathrm{interp}\left[\mathrm{cspline}\left[\begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}17\\15\\11\end{pmatrix}\%\right], \begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}17\\15\\11\end{pmatrix}\%, r\right] & \text{if } T_{Л.доп} < T^*_г \\ \mathrm{interp}\left[\mathrm{cspline}\left[\begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}13\\11\\7\end{pmatrix}\%\right], \begin{pmatrix}1\\ av(N_r)\\ N_r\end{pmatrix}, \begin{pmatrix}13\\11\\7\end{pmatrix}\%, r\right] & \text{otherwise} \end{cases}$$




$$\begin{pmatrix}\overline{c}_{stator}\\ \overline{c}_{rotor}\end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } r \in 1..N_r \\ \quad\quad \begin{pmatrix}\overline{c}_{stator_{i,r}}\\ \overline{c}_{rotor_{i,r}}\end{pmatrix} = \begin{pmatrix}\overline{c}_{stator.}(r)\\ \overline{c}_{rotor.}(r)\end{pmatrix} \\ \begin{pmatrix}\overline{c}_{stator}\\ \overline{c}_{rotor}\end{pmatrix} \end{array} \right.$$

$\overline{c}_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 11.00 | 11.00 | 11.00 | 11.00 |
| 2 | 11.00 | 11.00 | 11.00 | 11.00 |
| 3 | 11.00 | 11.00 | 11.00 | 11.00 |

$\cdot\%$

$\overline{c}_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 13.00 | 13.00 | 13.00 | 13.00 |
| 2 | 11.00 | 11.00 | 11.00 | 11.00 |
| 3 | 7.00 | 7.00 | 7.00 | 7.00 |

$\cdot\%$

$$\begin{pmatrix} \overline{r_inlet}_{rotor} & \overline{r_inlet}_{stator} \\ \overline{r_outlet}_{rotor} & \overline{r_outlet}_{stator} \end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } r \in 1..N_r \\ \quad \left| \begin{array}{l} \begin{pmatrix} \overline{r_inlet}_{stator_{i,r}} \\ \overline{r_outlet}_{stator_{i,r}} \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0.2 \end{pmatrix} \cdot \overline{c}_{stator.}(r) \\ \begin{pmatrix} \overline{r_inlet}_{rotor_{i,r}} \\ \overline{r_outlet}_{rotor_{i,r}} \end{pmatrix} = \begin{pmatrix} 0.3 \\ 0.1 \end{pmatrix} \cdot \overline{c}_{rotor.}(r) \end{array} \right. \\ \begin{pmatrix} \overline{r_inlet}_{rotor} & \overline{r_inlet}_{stator} \\ \overline{r_outlet}_{rotor} & \overline{r_outlet}_{stator} \end{pmatrix} \end{array} \right.$$

$\overline{r_inlet}_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 4.400 | 4.400 | 4.400 | 4.400 |
| 2 | 4.400 | 4.400 | 4.400 | 4.400 |
| 3 | 4.400 | 4.400 | 4.400 | 4.400 |

$\cdot\%$

$\overline{r_inlet}_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.900 | 3.900 | 3.900 | 3.900 |
| 2 | 3.300 | 3.300 | 3.300 | 3.300 |
| 3 | 2.100 | 2.100 | 2.100 | 2.100 |

$\cdot\%$

$\overline{r_outlet}_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2.200 | 2.200 | 2.200 | 2.200 |
| 2 | 2.200 | 2.200 | 2.200 | 2.200 |
| 3 | 2.200 | 2.200 | 2.200 | 2.200 |

$\cdot\%$

$\overline{r_outlet}_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.300 | 1.300 | 1.300 | 1.300 |
| 2 | 1.100 | 1.100 | 1.100 | 1.100 |
| 3 | 0.700 | 0.700 | 0.700 | 0.700 |

$\cdot\%$

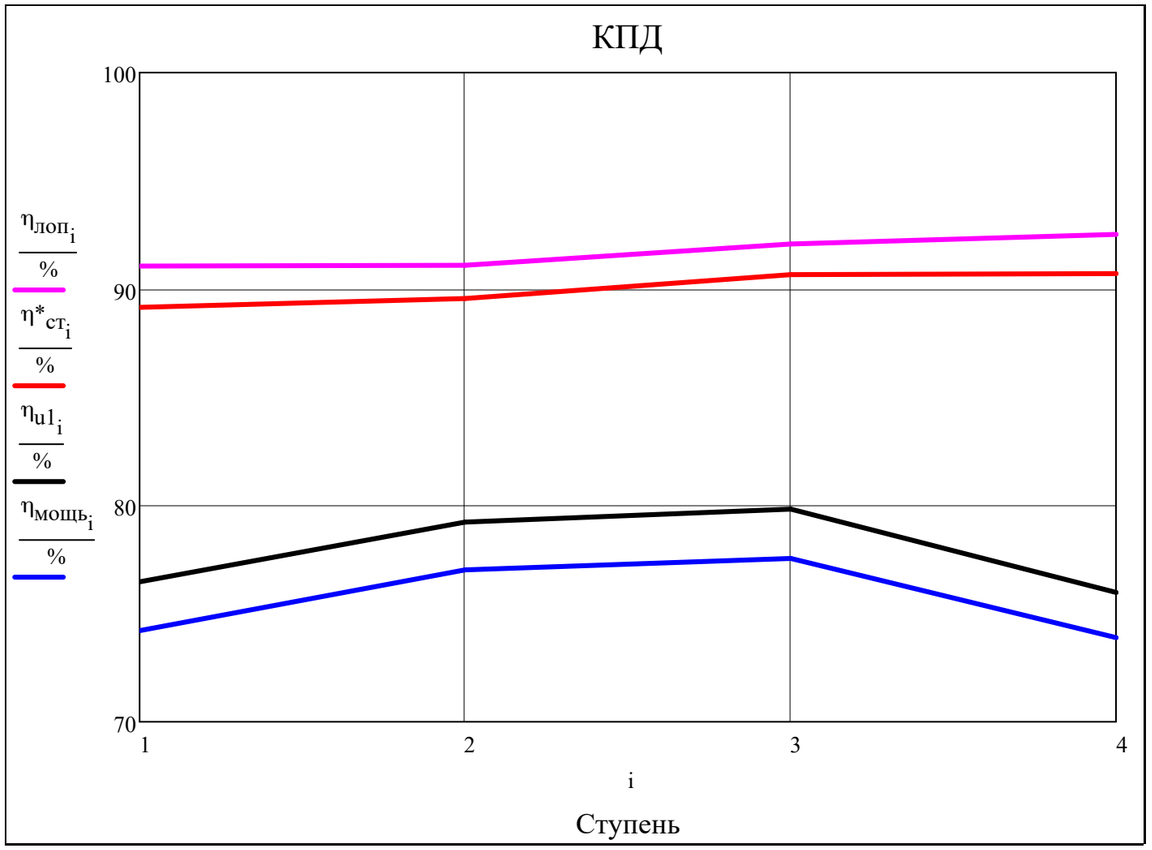
Относ. толщины и радиусы профилей

Вывод результатов поступенчатого расчета продольной геометрии ОТ в EXCEL:

$$R_{L.cp} = \begin{pmatrix} \begin{cases} 0.16 & \text{if turbine} = \text{"ТВД"} \\ 0.13 & \text{otherwise} \end{cases} & 0.15 & 0.18 & 0.185 & 0.5 & 0.5 \end{pmatrix}^T$$

Поступенчатый расчет ОТ

Параметры турбины




$\eta_{лоп}{}^T =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 91.06 | 91.09 | 92.08 | 92.53 |

$\cdot\%$

$\eta^*_{ст}{}^T =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 89.16 | 89.56 | 90.67 | 90.72 |

$\cdot\%$

$\mathrm{stack}\left(\eta_{u1}{}^T, \eta_{u2}{}^T\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 76.46 | 79.22 | 79.83 | 75.98 |
| 2 | 76.16 | 79.02 | 79.67 | 75.79 |

$\cdot\%$

$\eta_{мощь}{}^T =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 74.21 | 77.02 | 77.55 | 73.89 |

$\cdot\%$

$\eta_{мощь_i} \le \eta_{u1_i} \le \eta^*_{ст_i} \le \eta_{лоп_i} =$

| |
|---|
| 1 |
| 1 |
| 1 |
| 1 |
| 1 |
| 1 |

Степень понижения полного давления Т:

Степень понижения давления Т:

$$\begin{pmatrix} \pi^*_T \\ \pi_T \end{pmatrix} = P^*_{st(1,1),av(N_r)} \cdot \begin{bmatrix} \left(P^*_{st(Z,3),av(N_r)}\right)^{-1} \\ \left(P_{st(Z,3),av(N_r)}\right)^{-1} \end{bmatrix} =$$

| | 1 |
|---|---|
| 1 | 4.06 |
| 2 | 4.46 |

Теплоперепад по параметрам торможения (Дж/кг):

Располагаемый теплоперепад (Дж/кг):

$$\begin{pmatrix} H^*_T \\ H_T \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{Z} H^*_{ст_i} \\ \sum_{i=1}^{Z} H_{ст_i} \end{pmatrix} =$$

| | 1 |
|---|---|
| 1 | 476.4 |
| 2 | 567.0 |

$\cdot 10^3$

Мощность Т (Вт): $\sum_{i=1}^{Z} N_{ст_i} = 15.18 \cdot 10^6$ $\quad eps\left(\text{"rel"}, N_T, \sum_{i=1}^{Z} N_{ст_i}\right) = 0.000 \cdot \%$

Удельная поступенчатая работа Т [Дж/кг]: $L_T = \sum_{i=1}^{Z} \frac{N_{ст_i}}{mean\left(G_{st(i,2)}, G_{st(i,3)}\right)} = 428.9 \cdot 10^3$

Лопаточный КПД Т: $\eta_{Tлоп} = \frac{\sum_{i=1}^{Z} \frac{N_{ст_i}}{mean\left(G_{st(i,2)}, G_{st(i,3)}\right)} + \frac{\left(c_{st(Z,3),av(N_r)}\right)^2}{2}}{H_T} = 80.42 \cdot \%$

$k_{T.cp} = k_{ад}\left(Cp_{газ.cp}\left(P_{st(1,1),av(N_r)}, P_{st(Z,3),av(N_r)}, T_{st(1,1),av(N_r)}, T_{st(Z,3),av(N_r)}, \alpha_{ox_{st(1,1)}}, \alpha_{ox_{st(Z,3)}}, Fuel\right), R_{газ.cp}\left(\alpha_{ox_{st(1,1)}}, \alpha_{ox_{st(Z,3)}}, Fuel\right)\right) = 1.308$

Адиабатный КПД Т: $\eta^*_T = \frac{L_T}{H^*_T} = 90.04 \cdot \%$

Политропический КПД Т: $\eta^*_{T.п} = \eta^*_n\left(\text{"расширение"}, \eta^*_T, \pi^*_T, k_{T.cp}\right) = 88.42 \cdot \%$

Мощностной КПД Т: $\eta_{Tмощь} = \frac{\sum_{i=1}^{Z} \frac{N_{ст_i}}{mean\left(G_{st(i,2)}, G_{st(i,3)}\right)}}{H_T} = 75.65 \cdot \%$

$L_{ст}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 107.1 | 107.2 | 107.2 | 107.2 |

$\cdot 10^{3}$

$N_{ст}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.8 | 3.8 | 3.8 | 3.8 |

$\cdot 10^{6}$

$Lu_{ст}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 110.4 | 110.3 | 110.4 | 110.3 |

$\cdot 10^{3}$

$Lu_{нагрузка}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.0 | 2.3 | 1.8 | 1.6 |

$H_{ст}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 144.4 | 139.2 | 138.3 | 145.1 |

$\cdot 10^{3}$

$\mathrm{stack}\left(H_{stator}^{T}, H_{rotor}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 125.6 | 118.3 | 113.4 | 118.3 |
| 2 | 18.8 | 21.0 | 25.0 | 27.0 |

$\cdot 10^{3}$

$\mathrm{submatrix}\left(R_L^{T}, \mathrm{av}(N_r), \mathrm{av}(N_r), 1, Z\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.1 | 0.2 | 0.2 | 0.2 |

$G^{T} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 35.394 | 35.394 | 35.394 | 35.394 | 35.394 | 35.394 | 35.394 | ... |

$\alpha_{ox}^{T} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.493 | 2.493 | 2.493 | 2.493 | 2.493 | 2.493 | 2.493 | 2.493 | 2.493 |

$\mathrm{stack}\left(\theta_{CA}^{T}, \theta_{PK}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | -0.005 | -0.112 | -0.246 | -0.420 |
| 2 | -0.066 | -0.192 | -0.353 | -0.568 |

$\mathrm{stack}\left(g_{охлCA}^{T}, g_{охлPK}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2 | 0.00 | 0.00 | 0.00 | 0.00 |

$\cdot 10^{-3}$

$G_{охлCA_i} = g_{охлCA_i} \cdot G_{st(i,1)}$

$G_{охлPK_i} = g_{охлPK_i} \cdot G_{st(i,2)}$

$\mathrm{stack}\left(G_{охлCA}^{T}, G_{охлPK}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2 | 0.0 | 0.0 | 0.0 | 0.0 |

$$\sum_{i=1}^{Z} G_{охлCA_i} + \sum_{i=1}^{Z} G_{охлCA_i} \le G_{cooling} = 1$$

$G_{cooling} = 0.8$

$\mathrm{stack}\left(\mathrm{iteration}_{CA}{}^{T}, \mathrm{iteration}_{PK}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 2 |
| 2 | 2 | 2 | 2 | 2 |

$\mathrm{submatrix}\left(k^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.3 | 1.3 | 1.3 | 1.3 | 1.3 | 1.3 | 1.3 | 1.3 | 1.3 |

$\mathrm{submatrix}\left(P^{*T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 927.5 | 914.5 | 676.5 | 665.4 | 483.0 | 475.7 | 337.5 | 332.0 | 228.6 |

$\cdot 10^{3}$

$\mathrm{submatrix}\left(P^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 890.0 | 666.6 | 633.1 | 485.0 | 456.1 | 342.7 | 316.7 | 228.6 | 208.2 |

$\cdot 10^{3}$

$\mathrm{submatrix}\left(T^{*T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1368.9 | 1369.9 | 1281.1 | 1282.0 | 1192.4 | 1193.4 | 1102.5 | ... |

$\mathrm{submatrix}\left(T^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1356.0 | 1273.0 | 1261.5 | 1190.4 | 1176.4 | 1103.9 | 1085.9 | ... |

$\mathrm{submatrix}\left(T^{*}{}_{w}{}^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 1316.3 | 1317.8 | 1225.8 | 1228.5 | 1134.2 | 1136.8 | ... |

$\mathrm{submatrix}\left(T_{ад}{}^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 1268.9 | 1257.9 | 1185.9 | 1173.4 | 1100.0 | 1083.4 | ... |

$\mathrm{submatrix}\left(v^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

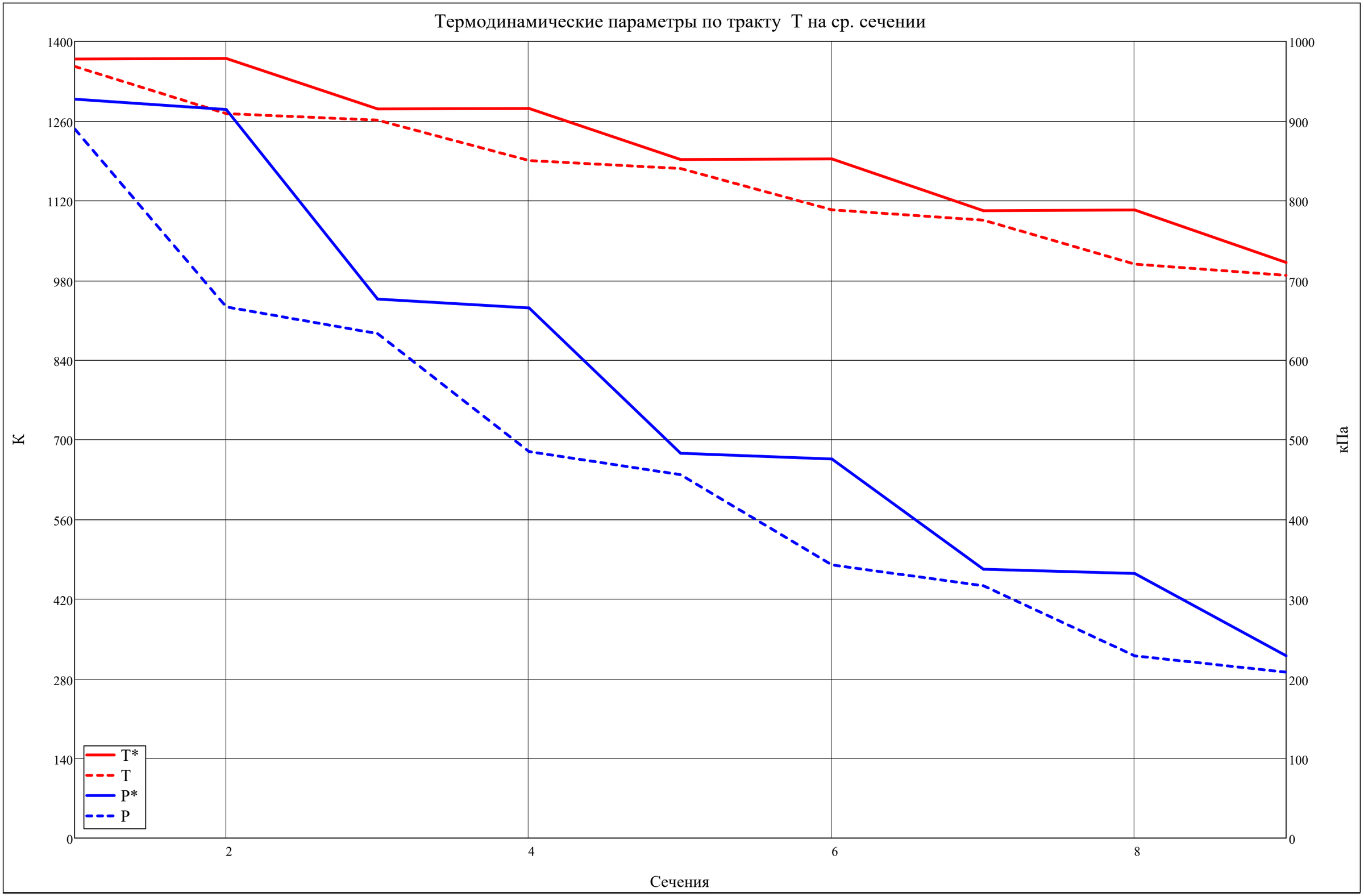
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.440 | 0.551 | 0.575 | 0.708 | 0.744 | 0.929 | 0.989 | 1.273 | 1.370 |

$\mathrm{submatrix}\left(\rho^{*T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.349 | 2.314 | 1.830 | 1.799 | 1.404 | 1.382 | 1.061 | 1.043 | 0.784 |

$\mathrm{submatrix}\left(\rho^{T}, \mathrm{av}\left(N_r\right), \mathrm{av}\left(N_r\right), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.275 | 1.815 | 1.740 | 1.412 | 1.344 | 1.076 | 1.011 | 0.786 | 0.730 |


Термодинамические параметры по тракту Т на ср. сечении
1400
1260
1120
980
840
700
560
420
280
140
0
К
1000
900
800
700
600
500
400
300
200
100
0
кПа
2
4
6
8
Сечения
T*
T
P*
P

$\mathrm{submatrix}\left(a_{зв}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 712.6 | 691.5 | 688.5 | 669.8 | 666.1 | 646.3 | 641.3 | 619.4 | 613.6 |

$\mathrm{submatrix}\left(a^{*}{}_{c}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 667.9 | 668.6 | 646.6 | 647.3 | 624.3 | 625.1 | 600.9 | 601.8 | 576.2 |

$\mathrm{submatrix}\left(a^{*}{}_{w}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 655.4 | 655.8 | 632.9 | 633.7 | 609.3 | 610.2 | 584.6 | 584.9 |

$\mathrm{submatrix}\left(c^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 180.0 | 490.9 | 220.6 | 474.7 | 197.9 | 465.9 | 200.4 | 476.7 | 232.6 |

$\mathrm{submatrix}\left(c_{u}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 451.9 | 135.7 | 435.3 | 85.6 | 419.9 | 39.1 | 415.9 | 10.3 |

$\mathrm{submatrix}\left(c_{a}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 180.0 | 191.7 | 173.9 | 189.4 | 178.5 | 202.0 | 196.6 | 232.9 | 232.4 |

$\mathrm{submatrix}\left(w^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 328.1 | 373.9 | 294.9 | 357.6 | 270.8 | 350.9 | 281.0 | 356.5 |

$\mathrm{submatrix}\left(w_{u}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 266.2 | 331.0 | 226.0 | 309.9 | 180.5 | 290.6 | 157.3 | 270.3 |

$\mathrm{submatrix}\left(w_{a}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 191.7 | 173.9 | 189.4 | 178.5 | 202.0 | 196.6 | 232.9 | 232.4 |

$\mathrm{submatrix}\left(c_{ад}^{T}, av(N_r), av(N_r), 1, 2Z\right) =$

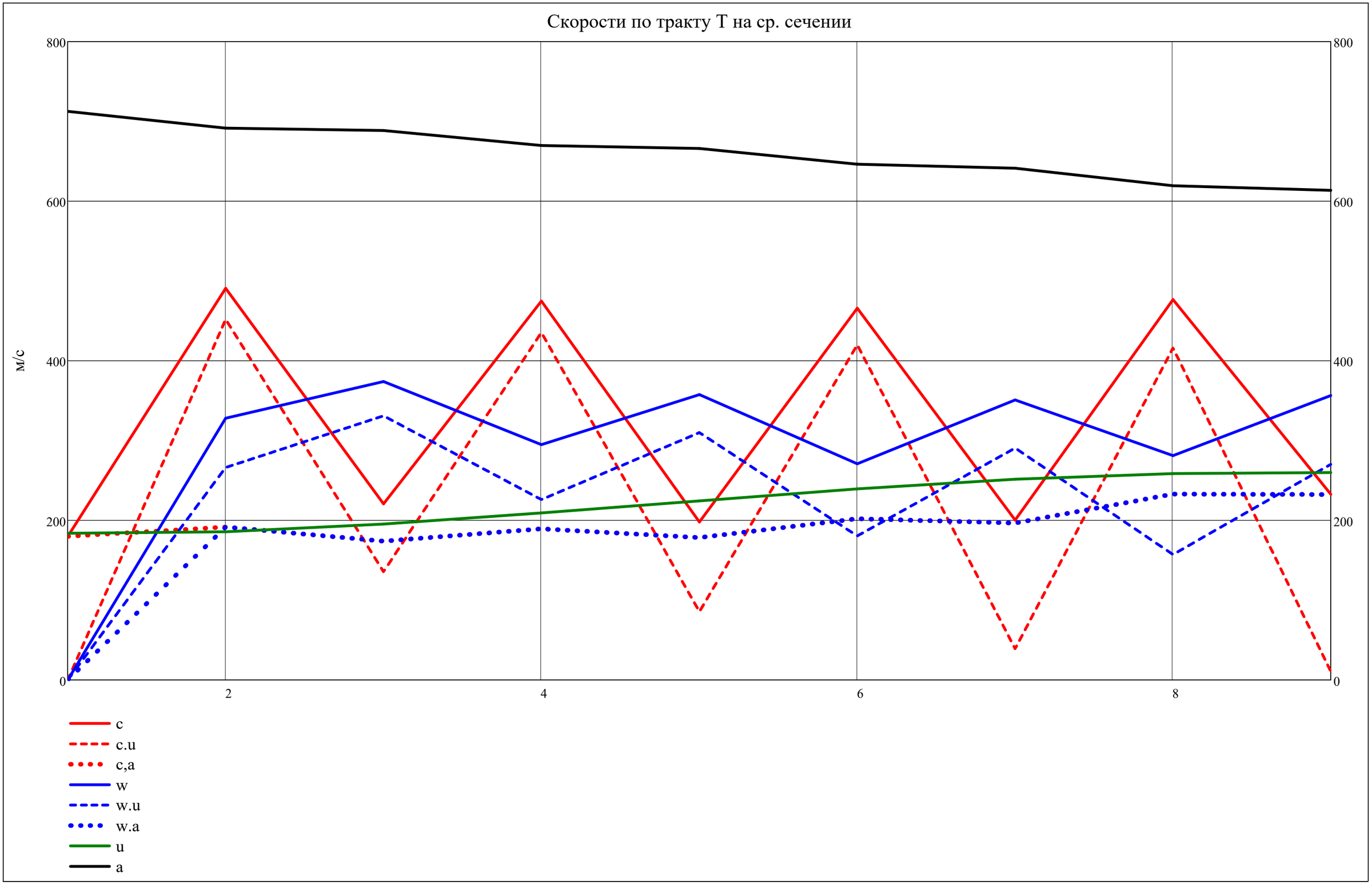
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 537.4 | 501.2 | 527.6 | 486.5 | 525.9 | 476.2 | 538.7 | 486.4 |

$\mathrm{submatrix}\left(w_{ад}^{T}, av(N_r), av(N_r), 1, 2Z+1\right) =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 0.0 | 386.0 | 0.0 | 368.0 | 0.0 | 359.5 | 0.0 | 365.5 |

$u^{T} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 172.2 | 172.2 | 180.7 | 193.8 | 208.2 | 222.8 | 234.2 | 240.3 | 240.3 |
| 2 | 183.8 | 185.7 | 195.4 | 209.3 | 224.4 | 239.4 | 251.6 | 258.7 | 260.0 |
| 3 | 195.3 | 199.1 | 210.0 | 224.7 | 240.5 | 256.1 | 268.9 | 277.0 | 279.7 |


Скорости по тракту Т на ср. сечении
м/с
800
600
400
200
0
2
4
6
8
c
c.u
c,a
w
w.u
w.a
u
a

$\text{submatrix}\left(\alpha, 1, 2\cdot Z + 1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 90.00 | 22.99 | 52.05 | 23.52 | 64.39 | 25.69 | 78.75 | 29.25 | 87.45 |

$.^{\circ}$

$\text{submatrix}\left(\alpha, 1, 2\cdot Z + 1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T \geq 11\cdot^{\circ} =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$\text{submatrix}\left(\beta, 1, 2\cdot Z + 1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.00 | 35.76 | 27.72 | 39.97 | 29.94 | 48.22 | 34.07 | 55.97 | 40.68 | | | | |

$.^{\circ}$

$\text{submatrix}\left(\varepsilon_{stator}, 1, Z, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

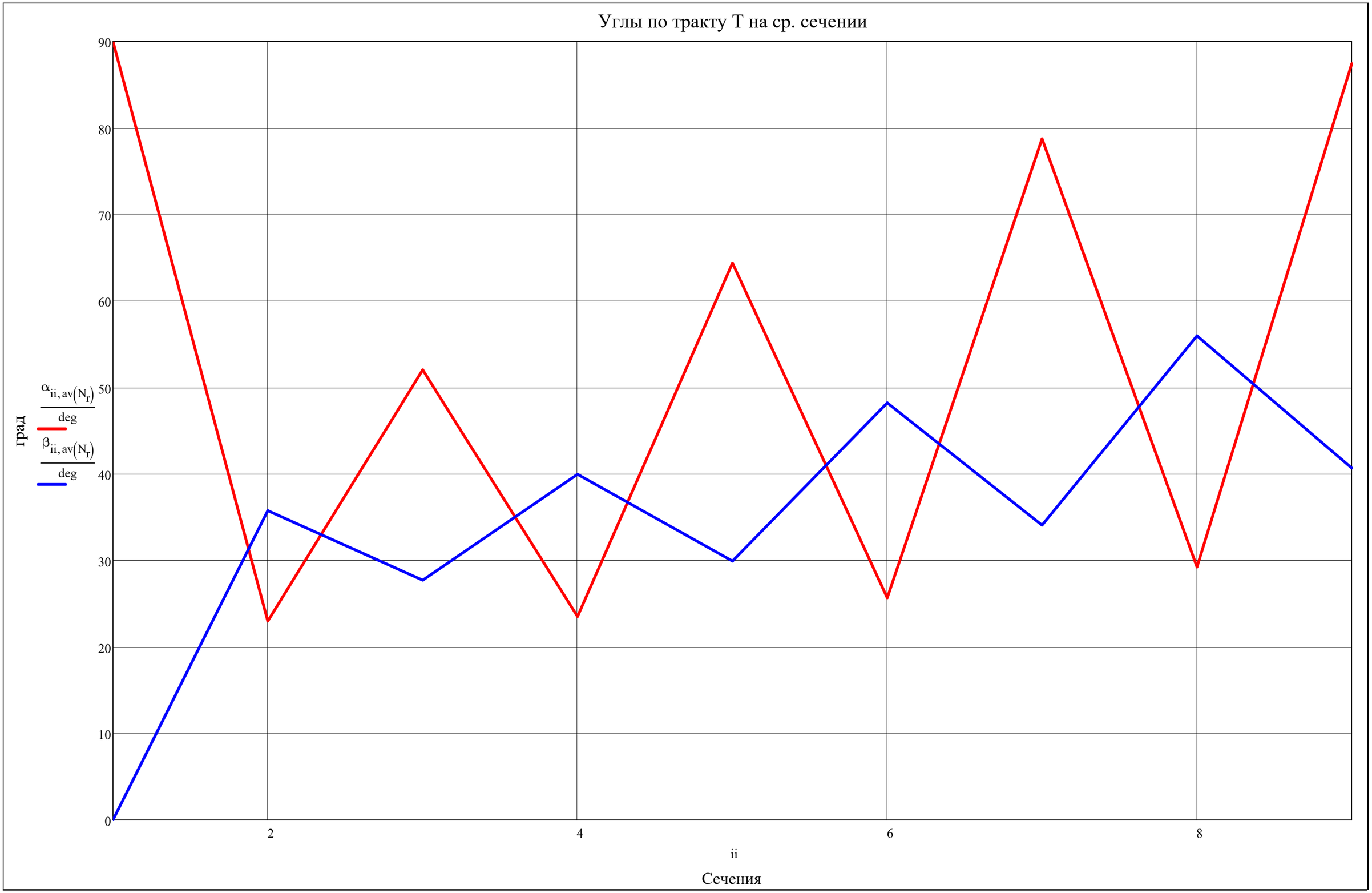
| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | -67.01 | -28.53 | -38.7 | -49.51 | | |

$.^{\circ}$

$\text{submatrix}\left(\varepsilon_{rotor}, 1, Z, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | -8.04 | -10.03 | -14.14 | -15.29 | | |

$.^{\circ}$


Углы по тракту Т на ср. сечении
90
80
70
60
50
40
30
20
10
0
град
$\frac{\alpha_{ii,av(N_r)}}{deg}$
$\frac{\beta_{ii,av(N_r)}}{deg}$
2
4
6
8
ii
Сечения

$\text{submatrix}\left(\lambda_c, 1, 2Z+1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.2695 | 0.7342 | 0.3411 | 0.7334 | 0.3170 | 0.7454 | 0.3335 | ... |

$\text{submatrix}\left(\lambda_w, 1, 2Z+1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.0000 | 0.5006 | 0.5702 | 0.4660 | 0.5643 | 0.4445 | 0.5750 | ... |

$\text{submatrix}\left(M_c, 1, 2Z+1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.2526 | 0.7099 | 0.3204 | 0.7088 | 0.2972 | 0.7209 | 0.3125 | ... |

$\text{submatrix}\left(M_c, 1, 2Z+1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T \le 1 =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$\text{submatrix}\left(M_w, 1, 2Z+1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.0000 | 0.4744 | 0.5431 | 0.4403 | 0.5369 | 0.4190 | 0.5471 | ... |

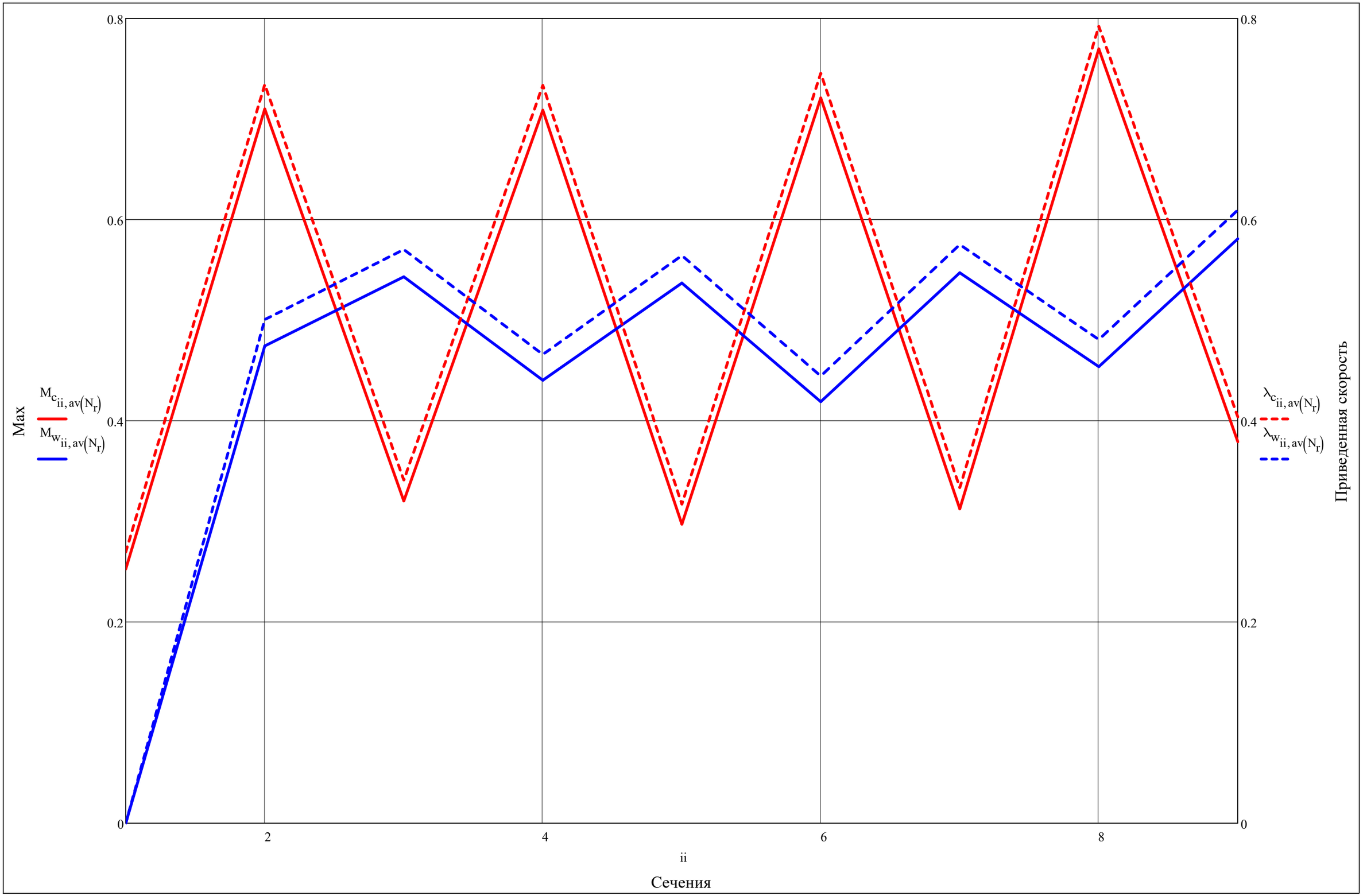
$\text{submatrix}\left(M_w, 1, 2Z+1, \text{av}\left(N_r\right), \text{av}\left(N_r\right)\right)^T \le 1 =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$\text{stack}\left(\upsilon_{stator}{}^T, \upsilon_{rotor}{}^T\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 43.08 | 63.04 | 58.94 | 53.66 |
| 2 | 68.71 | 68.31 | 67.38 | 67.10 |

$\cdot^{\circ}$


0.8
0.6
0.4
0.2
0
2
4
6
8
ii
Сечения
Max
$M_{c_{ii,av}(N_r)}$
$M_{w_{ii,av}(N_r)}$
$\lambda_{c_{ii,av}(N_r)}$
$\lambda_{w_{ii,av}(N_r)}$
Приведенная скорость

$t_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 40.6 | 27.9 | 35.9 | 43.3 |
| 2 | 43.6 | 30.1 | 38.6 | 46.6 |
| 3 | 46.5 | 32.4 | 41.3 | 49.8 |

$t_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 18.7 | 22.1 | 26.7 | 28.6 |
| 2 | 20.2 | 23.8 | 28.7 | 30.9 |
| 3 | 21.6 | 25.6 | 30.6 | 33.2 |

$\cdot 10^{-3}$

$\mathrm{submatrix}\left(\mathrm{chord}_{stator}{}^{T}, \mathrm{av}(N_r), \mathrm{av}(N_r), 1, Z\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 58.6 | 50.6 | 60.3 | 69.3 |

$\cdot 10^{-3}$

$\mathrm{submatrix}\left(\mathrm{chord}_{rotor}{}^{T}, \mathrm{av}(N_r), \mathrm{av}(N_r), 1, Z\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 31.5 | 36.3 | 40.9 | 42.4 |

$\cdot 10^{-3}$

$\mathrm{stack}\left(Z_{stator}{}^{T}, Z_{rotor}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 48 | 76 | 68 | 62 |
| 2 | 107 | 103 | 97 | 95 |

$\mathrm{stack}\left(\overline{t}_{ОПТСА}{}^{T}, \overline{t}_{ОПТРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.762 | 0.603 | 0.644 | 0.686 |
| 2 | 0.651 | 0.669 | 0.704 | 0.731 |

$\dfrac{t_{stator_{i,\mathrm{av}(N_r)}}}{\mathrm{chord}_{stator_{i,\mathrm{av}(N_r)}}} = 0.7 \le \dfrac{t_{stator_{i,\mathrm{av}(N_r)}}}{\mathrm{chord}_{stator_{i,\mathrm{av}(N_r)}}} \le 1 = \dfrac{t_{rotor_{i,\mathrm{av}(N_r)}}}{\mathrm{chord}_{rotor_{i,\mathrm{av}(N_r)}}} = 0.7 \le \dfrac{t_{rotor_{i,\mathrm{av}(N_r)}}}{\mathrm{chord}_{rotor_{i,\mathrm{av}(N_r)}}} \le 1 =$

| |
|---|
| 0.743 |
| 0.595 |
| 0.640 |
| 0.672 |

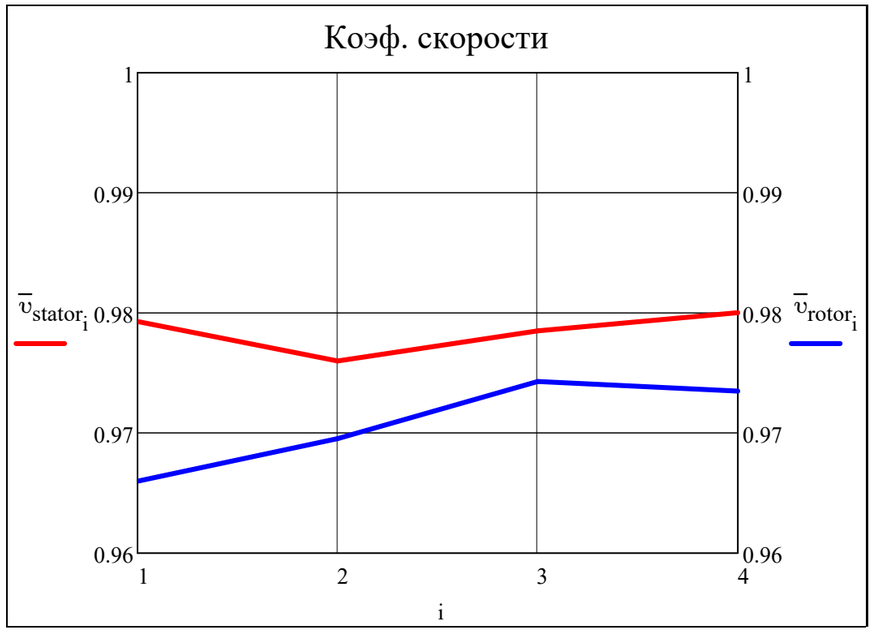
| |
|---|
| 1 |
| 0 |
| 0 |
| 0 |

| |
|---|
| 0.640 |
| 0.657 |
| 0.700 |
| 0.729 |

| |
|---|
| 0 |
| 0 |
| 1 |
| 1 |

$\mathrm{stack}\left(\overline{\upsilon}_{stator}^{\,T}, \overline{\upsilon}_{rotor}^{\,T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.9793 | 0.9760 | 0.9785 | 0.9800 |
| 2 | 0.9660 | 0.9695 | 0.9743 | 0.9735 |



$\mathrm{stack}\left(\xi_{трСА}{}^{T}, \xi_{трРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.396 | 1.811 | 1.520 | 1.394 |
| 2 | 3.905 | 3.265 | 2.464 | 2.488 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{крСА}{}^{T}, \xi_{крРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.517 | 1.852 | 1.587 | 1.340 |
| 2 | 0.739 | 0.671 | 0.561 | |

$\cdot\%$

$\mathrm{stack}\left(\xi_{ReСА}{}^{T}, \xi_{ReРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | -0.014 | 0.079 | 0.097 | 0.127 |
| 2 | 0.287 | 0.348 | 0.424 | 0.572 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{\lambda СА}{}^{T}, \xi_{\lambda РК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.218 | 0.219 | 0.201 | 0.134 |
| 2 | 0.370 | 0.381 | 0.360 | 0.295 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{втСА}{}^{T}, \xi_{втРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.982 | 0.780 | 0.847 | 0.961 |
| 2 | 1.384 | 1.337 | 1.265 | 1.414 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{тдСА}{}^{T}, \xi_{тдРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.000 | 0.000 | 0.000 | 0.000 |
| 2 | 0.000 | 0.000 | 0.000 | 0.000 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{смСА}{}^{T}, \xi_{смРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.000 | 0.000 | 0.000 | 0.000 |
| 2 | 0.000 | 0.000 | 0.000 | 0.000 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{СА}{}^{T}, \xi_{РК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.566 | 4.029 | 3.486 | 3.224 |
| 2 | 3.469 | 2.933 | 2.379 | 2.417 |

$\cdot\%$

$\mathrm{stack}\left(\xi_{прСА}{}^{T}, \xi_{прРК}{}^{T}\right) =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.117 | 3.961 | 3.405 | 2.995 |
| 2 | 5.301 | 4.665 | 3.809 | 3.818 |

$\cdot\%$

$\xi_{вых}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 16.848 | 14.071 | 14.525 | 18.643 |

$\cdot\%$

$\xi_{\Delta r}{}^{T} =$

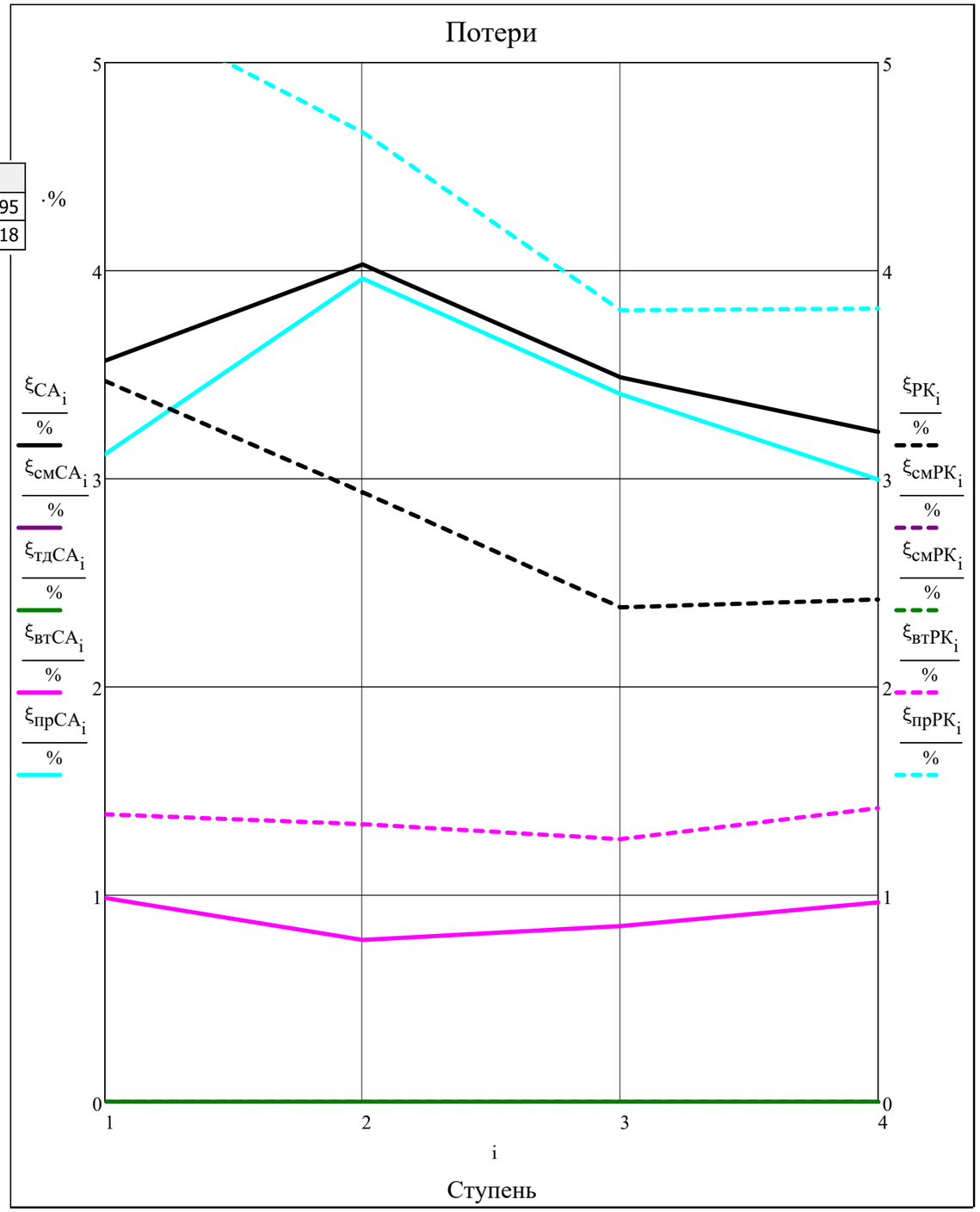
| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.810 | 1.782 | 1.822 | 1.658 |

$\cdot\%$

$\xi_{тр.в}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.138 | 0.221 | 0.298 | 0.244 |

$\cdot\%$




Параметры турбины

Вывод результатов поступенчатого расчета по ср. сечению ОТ в EXCEL:

Выбор закона профилирования Л по высоте

$$m = \begin{pmatrix} \begin{cases} \overline{\upsilon}_{stator_1} \cdot \cos\left(\alpha_{st(1,2),av}(N_r)\right)^2 & \text{if } Z = 1 \\ -0.5 & \text{otherwise} \end{cases} \\ -0.25 \\ 0 \\ 0.25 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \text{"}\alpha.2\text{=const"} \\ \text{"}\Gamma\text{=const"} \\ \text{"m=const"} \\ \text{"R=const"} \end{pmatrix} = \begin{pmatrix} \cos\left(\alpha_{st(i,2),av}(N_r)\right)^2 \cdot \overline{\upsilon}_{stator_i} \\ 1 \cdot \overline{\upsilon}_{stator_i} \\ 0.2 \\ -1 \cdot \overline{\upsilon}_{stator_i} \end{pmatrix}$$

$m^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | -0.5000 | -0.2500 | 0.0000 | 0.2500 | 1.0000 | 1.0000 |

Выбор закона профилирования Л по высоте

Расчет Л по высоте

Результаты расчета Л по высоте

$P*^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 927.5 | 914.5 | 676.5 | 665.4 | 483.0 | 475.7 | 337.5 | 332.0 | 228.6 |
| 2 | 927.5 | 914.5 | 676.5 | 665.4 | 483.0 | 475.7 | 337.5 | 332.0 | 228.6 |
| 3 | 927.5 | 914.5 | 676.5 | 665.4 | 483.0 | 475.7 | 337.5 | 332.0 | 228.6 |

$\cdot 10^3$

$P^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 890.0 | 631.4 | 635.8 | 459.5 | 458.6 | 325.0 | 317.3 | 215.2 | 208.2 |
| 2 | 890.0 | 660.5 | 638.1 | 481.1 | 459.0 | 339.4 | 317.3 | 225.8 | 208.2 |
| 3 | 890.0 | 686.7 | 639.3 | 500.3 | 459.2 | 352.3 | 317.3 | 235.2 | 208.2 |

$\cdot 10^3$

$T*^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1369 | 1370 | 1281 | 1282 | 1192 | 1193 | 1102 | 1104 | 1011 |
| 2 | 1369 | 1370 | 1281 | 1282 | 1192 | 1193 | 1102 | 1104 | 1011 |
| 3 | 1369 | 1370 | 1281 | 1282 | 1192 | 1193 | 1102 | 1104 | 1011 |

$T^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1355 | 1253 | 1262 | 1172 | 1177 | 1087 | 1086 | 991 | 988 |
| 2 | 1355 | 1267 | 1263 | 1185 | 1178 | 1098 | 1086 | 1003 | 988 |
| 3 | 1355 | 1278 | 1264 | 1196 | 1178 | 1108 | 1086 | 1013 | 988 |

$T*_w{}^T =$

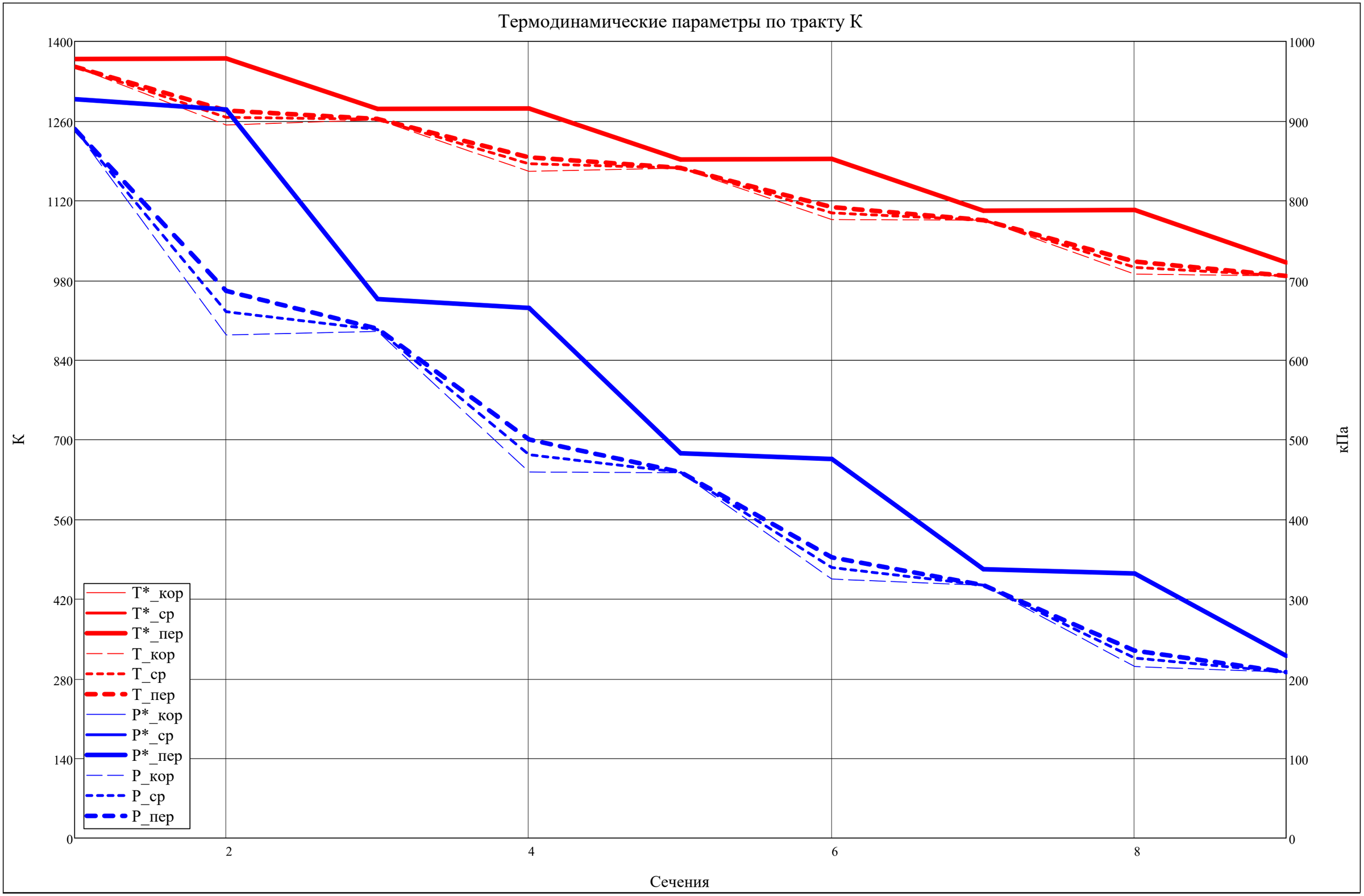
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1381.3 | 1313.8 | 1273.3 | 1222.9 | 1197.1 | 1130.2 | 1120.2 | 1036.6 |
| 2 | 1383.0 | 1313.1 | 1278.7 | 1222.6 | 1203.3 | 1130.7 | 1126.9 | 1038.0 |
| 3 | 1384.8 | 1312.5 | 1284.4 | 1222.5 | 1209.8 | 1131.5 | 1133.9 | ... |

$\rho*^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.349 | 2.314 | 1.830 | 1.799 | 1.404 | 1.382 | 1.061 | 1.043 | 0.784 |
| 2 | 2.349 | 2.314 | 1.830 | 1.799 | 1.404 | 1.382 | 1.061 | 1.043 | 0.784 |
| 3 | 2.349 | 2.314 | 1.830 | 1.799 | 1.404 | 1.382 | 1.061 | 1.043 | 0.784 |

$\rho^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.276 | 1.747 | 1.747 | 1.360 | 1.350 | 1.037 | 1.013 | 0.753 | 0.731 |
| 2 | 2.276 | 1.808 | 1.751 | 1.408 | 1.351 | 1.071 | 1.013 | 0.781 | 0.731 |
| 3 | 2.276 | 1.862 | 1.754 | 1.450 | 1.352 | 1.102 | 1.013 | 0.805 | 0.731 |


Термодинамические параметры по тракту К
К
кПа
Сечения
1400
1260
1120
980
840
700
560
420
280
140
0
1000
900
800
700
600
500
400
300
200
100
0
2
4
6
8
Т*_кор
Т*_ср
Т*_пер
Т_кор
Т_ср
Т_пер
Р*_кор
Р*_ср
Р*_пер
Р_кор
Р_ср
Р_пер

$k^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.317 | 1.317 | 1.321 | 1.321 | 1.326 | 1.326 | 1.331 | 1.331 | 1.338 |
| 2 | 1.317 | 1.317 | 1.321 | 1.321 | 1.326 | 1.326 | 1.331 | 1.331 | 1.338 |
| 3 | 1.317 | 1.317 | 1.321 | 1.321 | 1.326 | 1.326 | 1.331 | 1.331 | 1.338 |

$R_L^T =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.0536 | 0.0511 | 0.0925 | 0.1037 |
| 2 | 0.0913 | 0.1091 | 0.1574 | 0.1829 |
| 3 | 0.1251 | 0.1592 | 0.2136 | 0.2505 |

$R_L^T \geq 0.05 =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 2 | 1 | 1 | 1 | 1 |
| 3 | 1 | 1 | 1 | 1 |

$a^*_c{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 670.1 | 670.3 | 648.6 | 648.9 | 626.2 | 626.5 | 602.7 | 603.0 | 577.9 |
| 2 | 670.1 | 670.3 | 648.6 | 648.9 | 626.2 | 626.5 | 602.7 | 603.0 | 577.9 |
| 3 | 670.1 | 670.3 | 648.6 | 648.9 | 626.2 | 626.5 | 602.7 | 603.0 | 577.9 |

$u^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 172.2 | 172.2 | 180.7 | 193.8 | 208.2 | 222.8 | 234.2 | 240.3 | 240.3 |
| 2 | 183.8 | 185.7 | 195.4 | 209.3 | 224.4 | 239.4 | 251.6 | 258.7 | 260.0 |
| 3 | 195.3 | 199.1 | 210.0 | 224.7 | 240.5 | 256.1 | 268.9 | 277.0 | 279.7 |

$c^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 180.0 | 529.2 | 213.2 | 511.7 | 188.5 | 500.3 | 197.2 | 511.5 | 232.4 |
| 2 | 180.0 | 497.3 | 207.2 | 480.3 | 186.8 | 472.2 | 197.0 | 483.7 | 232.4 |
| 3 | 180.0 | 467.8 | 203.7 | 451.5 | 186.0 | 446.4 | 197.0 | 458.7 | 232.4 |

$c_u{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0 | 476.1 | 142.0 | 459.1 | 77.6 | 444.1 | 29.4 | 444.1 | 13.3 |
| 2 | 0.0 | 458.8 | 112.6 | 441.4 | 55.1 | 426.8 | 13.1 | 424.0 | 0.2 |
| 3 | 0.0 | 444.5 | 86.6 | 426.4 | 35.3 | 411.8 | -1.1 | 406.3 | -10.9 |

$c_a{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 180.0 | 231.2 | 159.0 | 226.1 | 171.8 | 230.5 | 195.0 | 253.8 | 232.0 |
| 2 | 180.0 | 191.7 | 173.9 | 189.4 | 178.5 | 202.0 | 196.6 | 232.9 | 232.4 |
| 3 | 180.0 | 145.6 | 184.4 | 148.3 | 182.6 | 172.4 | 197.0 | 212.9 | 232.1 |

$\Delta c_a{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 51.2 | -72.2 | 67.1 | -54.3 | 58.7 | -35.5 | 58.8 | -21.8 |
| 2 | 11.7 | -17.8 | 15.5 | -11.0 | 23.5 | -5.4 | 36.3 | -0.5 |
| 3 | -34.4 | 38.7 | -36.1 | 34.4 | -10.3 | 24.6 | 15.9 | 19.2 |

$a^*_w{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 673.1 | 656.4 | 646.6 | 633.7 | 627.5 | 609.7 | 607.5 | 584.4 | 584.3 |
| 2 | 673.5 | 656.3 | 648.0 | 633.7 | 629.1 | 609.8 | 609.4 | 584.8 | 586.3 |
| 3 | 674.0 | 656.1 | 649.5 | 633.6 | 630.8 | 610.0 | 611.2 | 585.3 | 588.3 |

$a_{зв}{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 717.7 | 690.0 | 693.6 | 668.3 | 671.1 | 644.7 | 645.8 | 616.9 | 617.4 |
| 2 | 717.7 | 693.8 | 693.9 | 672.0 | 671.1 | 648.2 | 645.8 | 620.6 | 617.4 |
| 3 | 717.7 | 697.0 | 694.0 | 675.2 | 671.2 | 651.1 | 645.8 | 623.7 | 617.5 |

$w^T =$

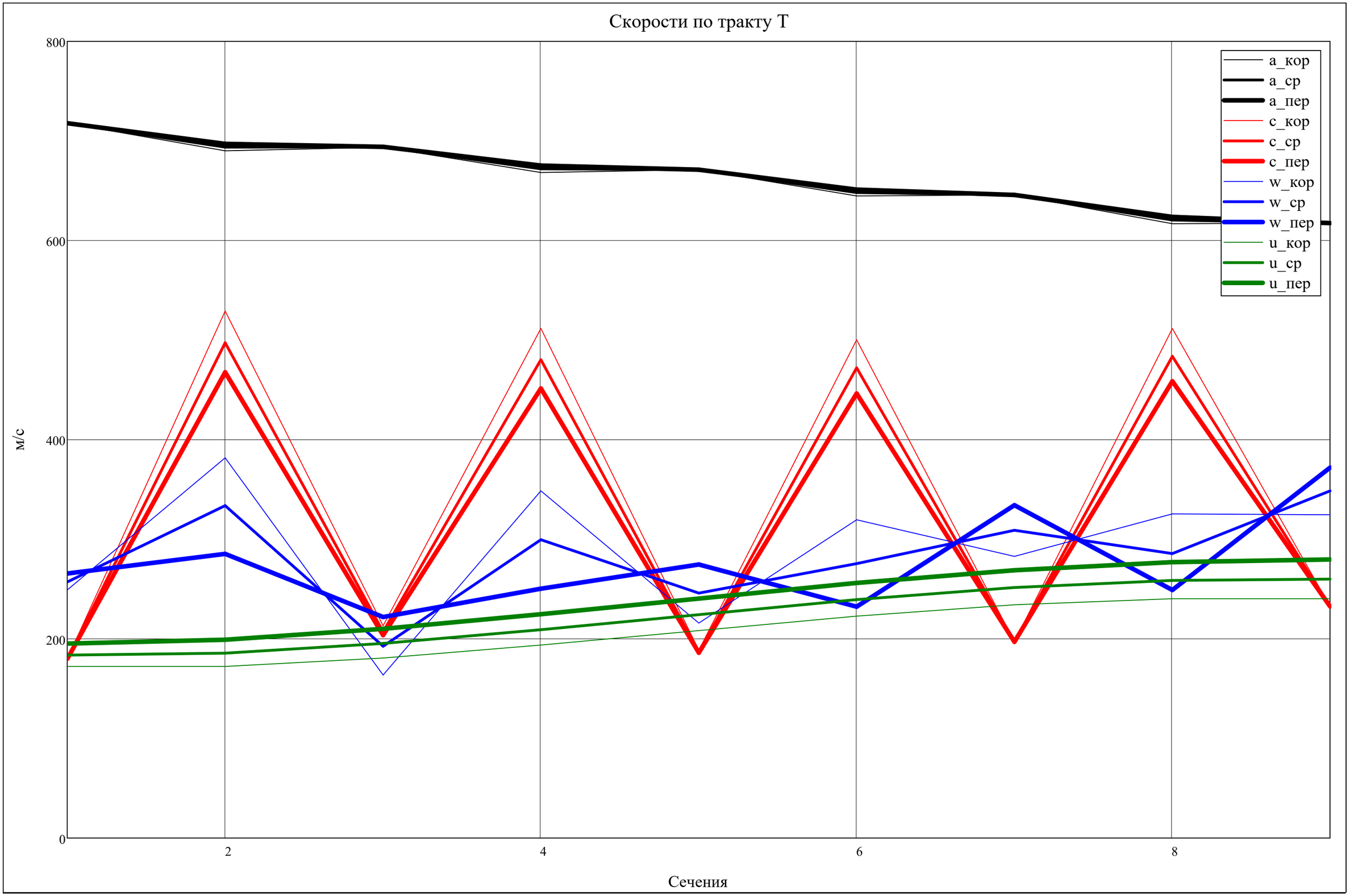
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 249.1 | 381.8 | 163.6 | 348.6 | 215.8 | 319.5 | 282.8 | 325.5 | 324.6 |
| 2 | 257.2 | 333.7 | 192.6 | 299.6 | 246.0 | 275.5 | 309.0 | 285.6 | 348.6 |
| 3 | 265.6 | 285.4 | 221.9 | 250.3 | 274.7 | 232.3 | 334.2 | 249.1 | 371.9 |

$w_u{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 172.2 | -303.8 | 38.6 | -265.3 | 130.6 | -221.3 | 204.8 | -203.8 |
| 2 | 183.8 | -273.2 | 82.8 | -232.1 | 169.3 | -187.4 | 238.5 | -165.3 |
| 3 | 195.3 | -245.4 | 123.5 | -201.7 | 205.2 | -155.7 | 270.0 | ... |

$w_a{}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 180.0 | 231.2 | 159.0 | 226.1 | 171.8 | 230.5 | 195.0 | 253.8 | 232.0 |
| 2 | 180.0 | 191.7 | 173.9 | 189.4 | 178.5 | 202.0 | 196.6 | 232.9 | 232.4 |
| 3 | 180.0 | 145.6 | 184.4 | 148.3 | 182.6 | 172.4 | 197.0 | 212.9 | 232.1 |


Скорости по тракту Т
800
600
400
200
0
м/с
2
4
6
8
Сечения
a_кор
a_ср
a_пер
c_кор
c_ср
c_пер
w_кор
w_ср
w_пер
u_кор
u_ср
u_пер

$\alpha^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 90.00 | 25.90 | 48.23 | 26.23 | 65.69 | 27.43 | 81.44 | 29.75 | 86.72 |
| 2 | 90.00 | 22.68 | 57.09 | 23.23 | 72.85 | 25.32 | 86.19 | 28.78 | 89.96 |
| 3 | 90.00 | 18.14 | 64.85 | 19.18 | 79.08 | 22.71 | 90.31 | 27.66 | 92.70 |

$\cdot^\circ$

$\beta^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 46.26 | 142.73 | 76.34 | 139.56 | 52.76 | 133.84 | 43.59 | 128.76 |
| 2 | 44.41 | 144.94 | 64.54 | 140.78 | 46.51 | 132.86 | 39.50 | 125.36 |
| 3 | 42.67 | 149.32 | 56.19 | 143.67 | 41.67 | 132.10 | 36.12 | ... |

$\cdot^\circ$

$80\cdot^\circ \le \alpha^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 2 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 3 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |

[1, с.78]

Угол поворота потока:

$\varepsilon_{stator}{}^T =$

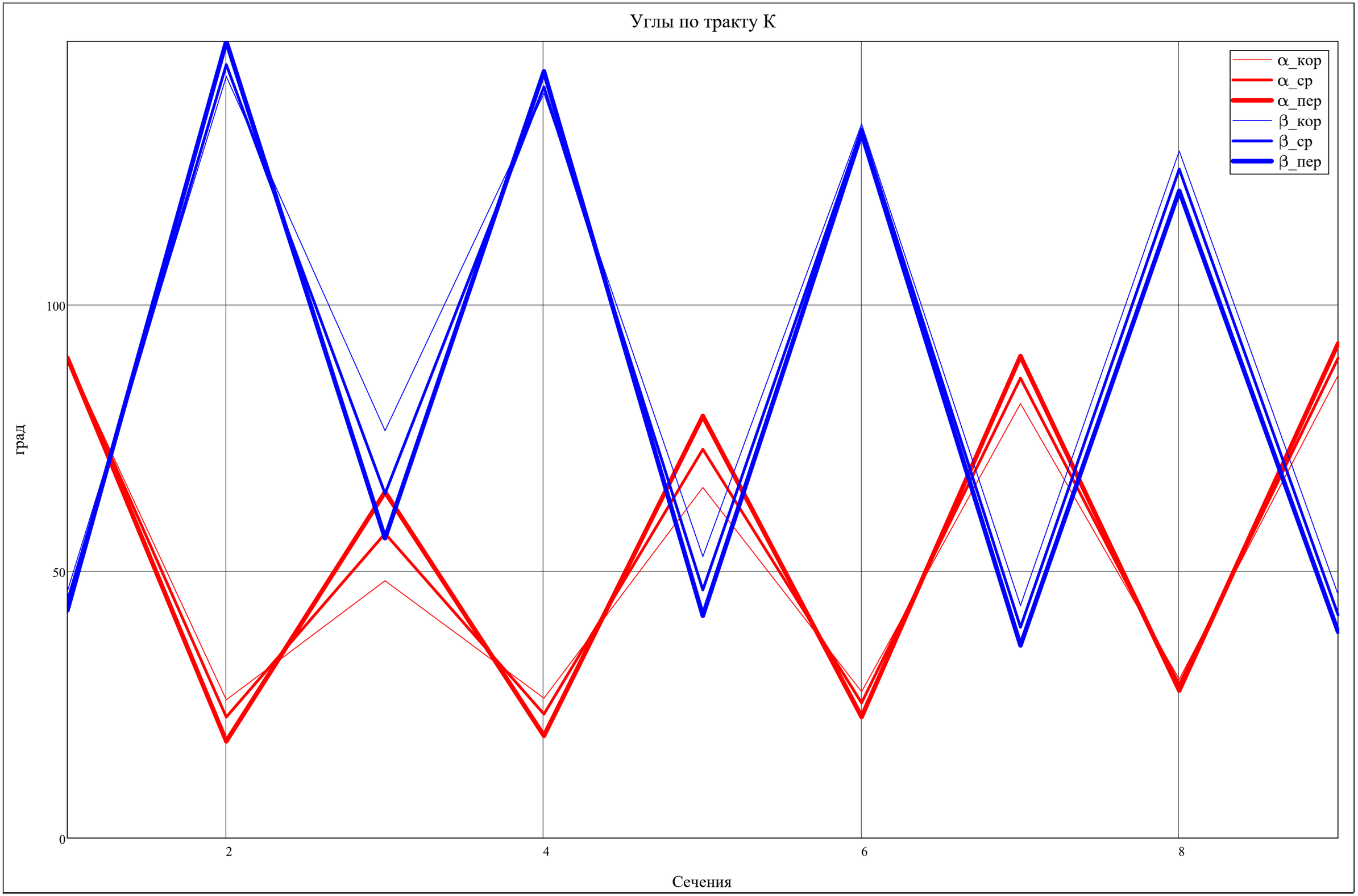
| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 64.10 | 22.00 | 38.26 | 51.69 |
| 2 | 67.32 | 33.86 | 47.53 | 57.40 |
| 3 | 71.86 | 45.67 | 56.37 | 62.65 |

$\cdot^\circ$

$\varepsilon_{rotor}{}^T =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 66.39 | 86.80 | 90.25 | 83.14 |
| 2 | 80.39 | 94.27 | 93.36 | 83.56 |
| 3 | 93.13 | 102.01 | 95.98 | 82.65 |

$\cdot^\circ$


Углы по тракту К
α_кор
α_ср
α_пер
β_кор
β_ср
β_пер
100
50
0
град
2
4
6
8
Сечения

$\lambda_c^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.269 | 0.790 | 0.329 | 0.789 | 0.301 | 0.799 | 0.327 | 0.848 | 0.402 |
| 2 | 0.269 | 0.742 | 0.319 | 0.740 | 0.298 | 0.754 | 0.327 | 0.802 | 0.402 |
| 3 | 0.269 | 0.698 | 0.314 | 0.696 | 0.297 | 0.713 | 0.327 | 0.761 | 0.402 |

$\lambda_w^T =$

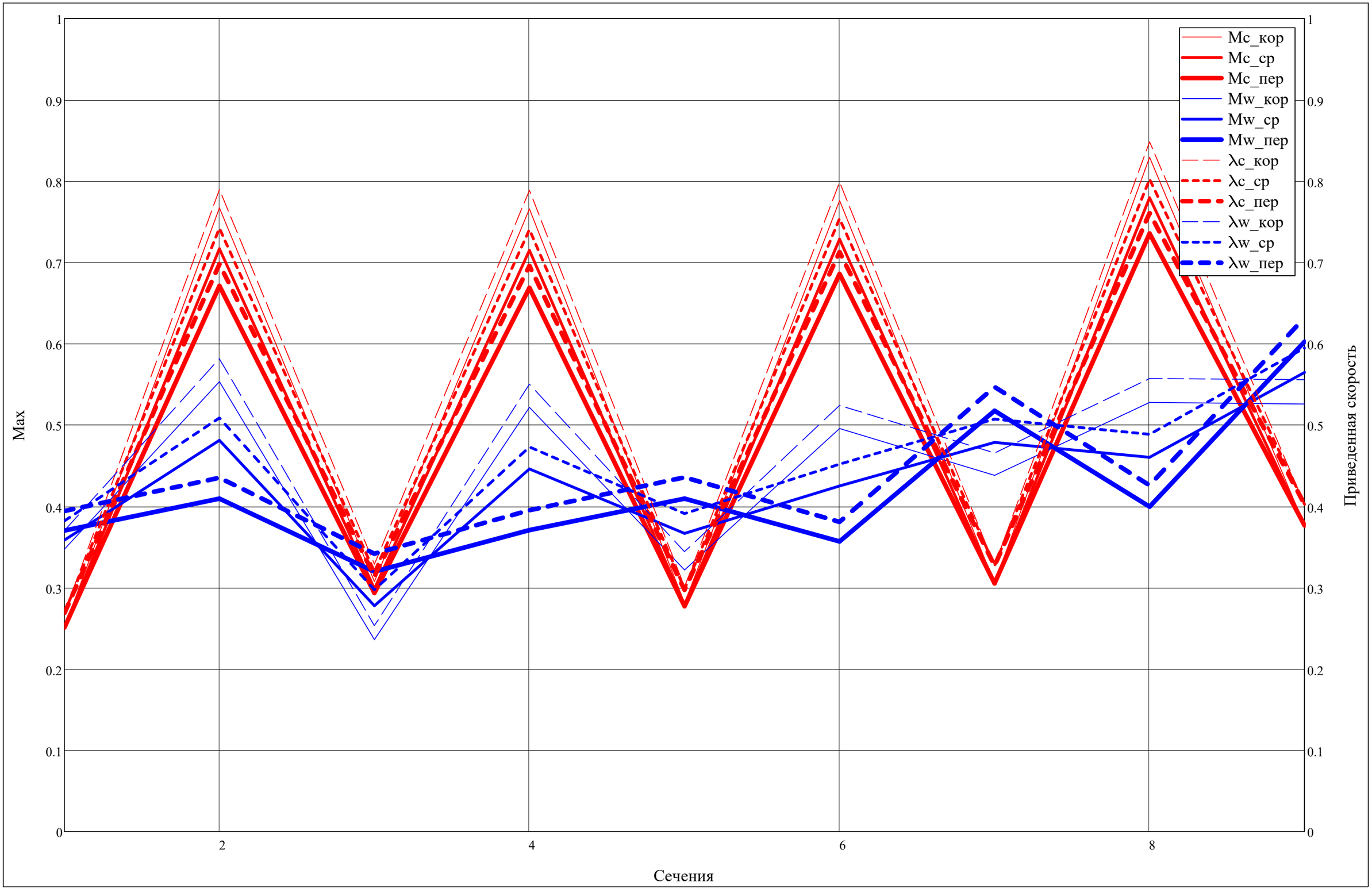
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.370 | 0.582 | 0.253 | 0.550 | 0.344 | 0.524 | 0.465 | 0.557 | 0.556 |
| 2 | 0.382 | 0.509 | 0.297 | 0.473 | 0.391 | 0.452 | 0.507 | 0.488 | 0.595 |
| 3 | 0.394 | 0.435 | 0.342 | 0.395 | 0.436 | 0.381 | 0.547 | 0.426 | 0.632 |

$M_c^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.251 | 0.767 | 0.307 | 0.766 | 0.281 | 0.776 | 0.305 | 0.829 | 0.376 |
| 2 | 0.251 | 0.717 | 0.299 | 0.715 | 0.278 | 0.728 | 0.305 | 0.779 | 0.376 |
| 3 | 0.251 | 0.671 | 0.293 | 0.669 | 0.277 | 0.686 | 0.305 | 0.735 | 0.376 |

$M_w^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.3 | 0.6 | 0.2 | 0.5 | 0.3 | 0.5 | 0.4 | 0.5 | 0.5 |
| 2 | 0.4 | 0.5 | 0.3 | 0.4 | 0.4 | 0.4 | 0.5 | 0.5 | 0.6 |
| 3 | 0.4 | 0.4 | 0.3 | 0.4 | 0.4 | 0.4 | 0.5 | 0.4 | 0.6 |


Mc_кор
Mc_ср
Mc_пер
Mw_кор
Mw_ср
Mw_пер
λc_кор
λc_ср
λc_пер
λw_кор
λw_ср
λw_пер
Mах
Приведенная скорость
Сечения
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
2
4
6
8

Вывод результатов расчета параметров потока по высоте Л

Рассматриваемая ступень: $j = \left| \begin{array}{l} j = Z \\ j = \left| \begin{array}{l} \text{"Такой ступени не существует!"} \quad \text{if } (j < 1) \vee (j > Z) \\ j \quad \text{otherwise} \end{array} \right. \end{array} \right. = 4$

Построение треугольников скоростей в 3х сечениях

$$\Delta_c(v,i,j,r) = \left| \begin{array}{l} \tan\left(\alpha_{st(i,j),r}\right)\cdot v \quad \text{if } \left(\tan\left(\alpha_{st(i,j),r}\right) \ge 0\right) \wedge \left(-\left|c_{st(i,j),r}\cdot\cos\left(\alpha_{st(i,j),r}\right)\right| \le v \le 0\right) \\ \tan\left(\alpha_{st(i,j),r}\right)\cdot v \quad \text{if } \left(\tan\left(\alpha_{st(i,j),r}\right) < 0\right) \wedge \left(0 \le v \le \left|c_{st(i,j),r}\cdot\cos\left(\alpha_{st(i,j),r}\right)\right|\right) \end{array} \right.$$

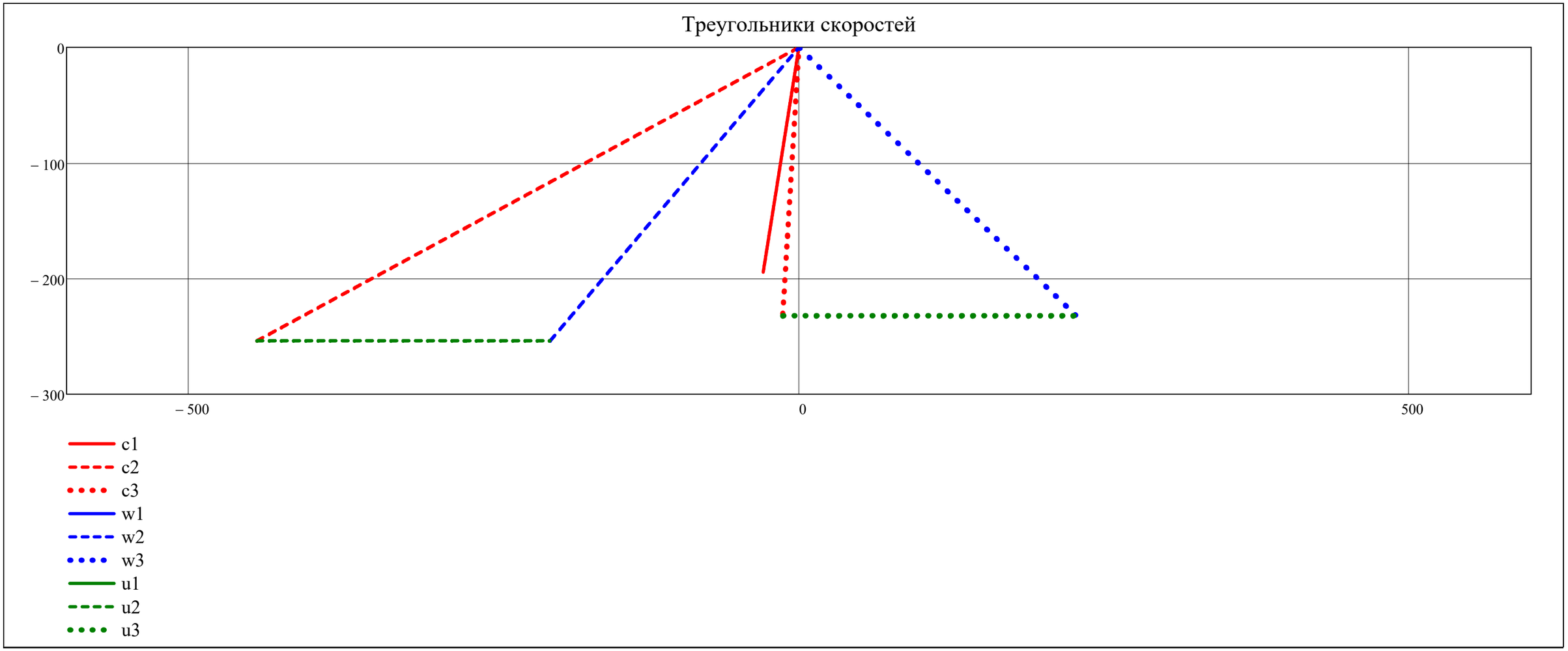
$$\Delta_w(v,i,j,r) = \left| \begin{array}{l} -\tan\left(\beta_{st(i,j),r}\right)\cdot v \quad \text{if } \left(-\tan\left(\beta_{st(i,j),r}\right) \ge 0\right) \wedge \left(-\left|w_{st(i,j),r}\cdot\cos\left(\beta_{st(i,j),r}\right)\right| \le v \le 0\right) \wedge (j \ne 1) \\ -\tan\left(\beta_{st(i,j),r}\right)\cdot v \quad \text{if } \left(-\tan\left(\beta_{st(i,j),r}\right) < 0\right) \wedge \left(0 \le v \le \left|w_{st(i,j),r}\cdot\cos\left(\beta_{st(i,j),r}\right)\right|\right) \wedge (j \ne 1) \end{array} \right.$$

$$\Delta_u(v,i,j,r) = \left| \begin{array}{l} -c_{a_{st(i,j),r}} \quad \text{if } \left(-c_{st(i,j),r}\cdot\cos\left(\alpha_{st(i,j),r}\right) \le v \le w_{st(i,j),r}\cdot\cos\left(\beta_{st(i,j),r}\right)\right) \wedge (j \ne 1) \\ \text{NaN} \quad \text{otherwise} \end{array} \right.$$

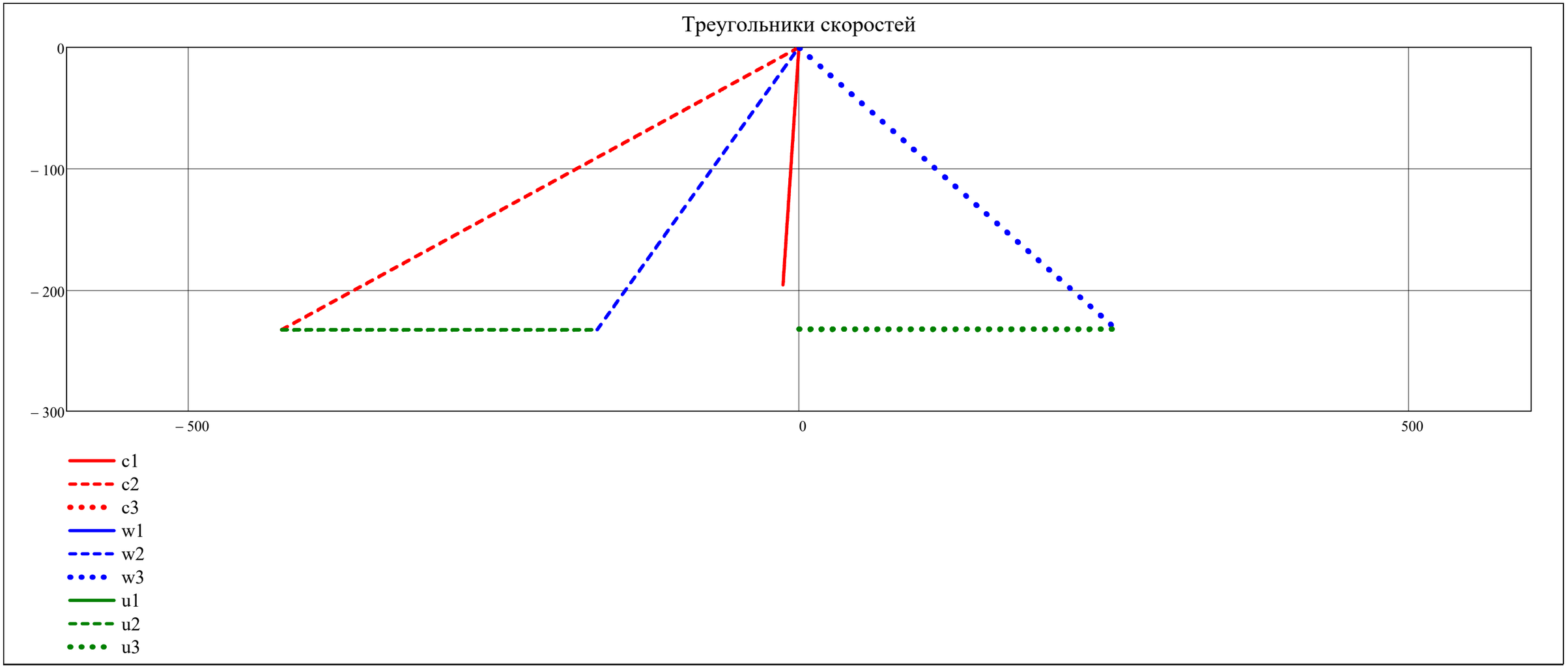
$$v_{lim} = \text{ceil}\left(\frac{\max(c,w,u)}{10^2}\right)\cdot 10^2 = 600.0$$

$$v = -\max(c,w,u), -\max(c,w,u) + \frac{\max(c,w,u)}{3000} .. \max(c,w,u)$$

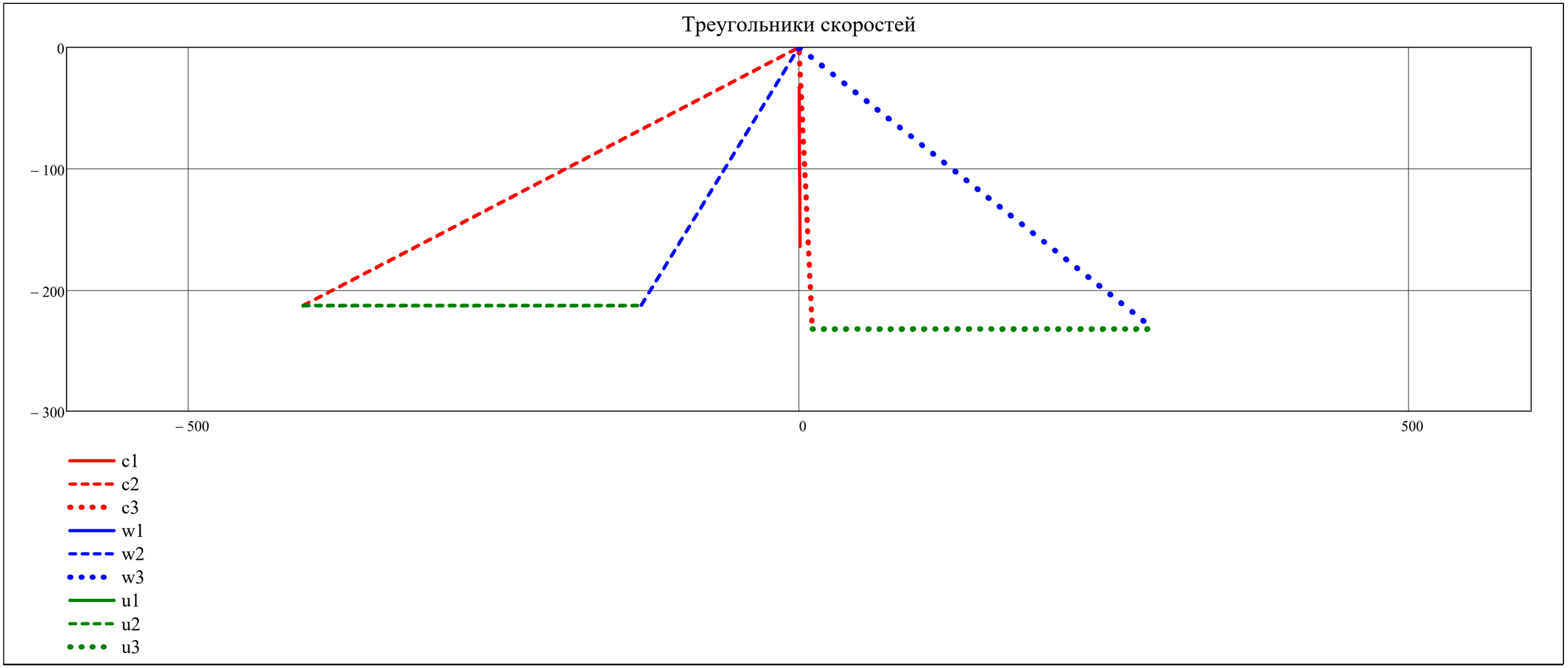
r = 1


Треугольники скоростей
0
– 100
– 200
– 300
– 500
0
500
c1
c2
c3
w1
w2
w3
u1
u2
u3

$r := av(N_r)$


Треугольники скоростей
0
– 100
– 200
– 300
– 500
0
500
c1
c2
c3
w1
w2
w3
u1
u2
u3

$r = N_r$




Построение треугольников скоростей в 3х сечениях

Парусность: $\begin{pmatrix} \overline{b}_{парусСА} \\ \overline{b}_{парусРК} \end{pmatrix} = \begin{pmatrix} 1 \\ 0.85 \end{pmatrix}$

Расчет хорд Л по парусности

Ср. линия профиля:
0.5 - дуга окружности
0.45 - парабола

$\overline{x}_f = 0.45$

Расчет параметров решетки

Результаты расчета параметров решеток

Относительные радиусы профилей ():

$\overline{r_inlet}_{stator}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 4.400 | 4.400 | 4.400 | 4.400 |
| 2 | 4.400 | 4.400 | 4.400 | 4.400 |
| 3 | 4.400 | 4.400 | 4.400 | 4.400 |

$\overline{r_inlet}_{rotor}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.900 | 3.900 | 3.900 | 3.900 |
| 2 | 3.300 | 3.300 | 3.300 | 3.300 |
| 3 | 2.100 | 2.100 | 2.100 | 2.100 |

$\overline{r_outlet}_{stator}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2.200 | 2.200 | 2.200 | 2.200 |
| 2 | 2.200 | 2.200 | 2.200 | 2.200 |
| 3 | 2.200 | 2.200 | 2.200 | 2.200 |

$\overline{r_outlet}_{rotor}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.300 | 1.300 | 1.300 | 1.300 |
| 2 | 1.100 | 1.100 | 1.100 | 1.100 |
| 3 | 0.700 | 0.700 | 0.700 | 0.700 |

Радиусы профилей:

$r_{вхCA}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2.58 | 2.23 | 2.66 | 3.05 |
| 2 | 2.58 | 2.23 | 2.66 | 3.05 |
| 3 | 2.58 | 2.23 | 2.66 | 3.05 |

$\cdot 10^{-3}$

$r_{вхPK}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.33 | 1.53 | 1.73 | 1.79 |
| 2 | 1.04 | 1.20 | 1.35 | 1.40 |
| 3 | 0.61 | 0.70 | 0.79 | 0.82 |

$\cdot 10^{-3}$

$r_{выхCA}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.29 | 1.11 | 1.33 | 1.52 |
| 2 | 1.29 | 1.11 | 1.33 | 1.52 |
| 3 | 1.29 | 1.11 | 1.33 | 1.52 |

$\cdot 10^{-3}$

$r_{выхPK}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.44 | 0.51 | 0.58 | 0.60 |
| 2 | 0.35 | 0.40 | 0.45 | 0.47 |
| 3 | 0.20 | 0.23 | 0.26 | 0.27 |

$\cdot 10^{-3}$

Относительная толщина профиля ():

$\overline{c}_{stator}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 11.000 | 11.000 | 11.000 | 11.000 |
| 2 | 11.000 | 11.000 | 11.000 | 11.000 |
| 3 | 11.000 | 11.000 | 11.000 | 11.000 |

$\overline{c}_{rotor}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 13.0 | 13.0 | 13.0 | 13.0 |
| 2 | 11.0 | 11.0 | 11.0 | 11.0 |
| 3 | 7.0 | 7.0 | 7.0 | 7.0 |

$\cdot\%$

Толщина профиля [м]:

$c_{CA}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6.5 | 5.6 | 6.6 | 7.6 |
| 2 | 6.5 | 5.6 | 6.6 | 7.6 |
| 3 | 6.5 | 5.6 | 6.6 | 7.6 |

$\cdot 10^{-3}$

$c_{PK}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 4.4 | 5.1 | 5.8 | 6.0 |
| 2 | 3.5 | 4.0 | 4.5 | 4.7 |
| 3 | 2.0 | 2.3 | 2.6 | 2.7 |

$\cdot 10^{-3}$

Относительный шаг решетки ():

$\overline{t}_{CA}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.6926 | 0.5508 | 0.5945 | 0.6251 |
| 2 | 0.7428 | 0.5952 | 0.6398 | 0.6721 |
| 3 | 0.7930 | 0.6396 | 0.6850 | 0.7192 |

$\overline{t}_{PK}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.5486 | 0.5637 | 0.6028 | 0.6251 |
| 2 | 0.6403 | 0.6573 | 0.7002 | 0.7292 |
| 3 | 0.7482 | 0.7675 | 0.8148 | 0.8517 |

Шаг решетки [м]:

$t_{stator}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 40.6 | 27.9 | 35.9 | 43.3 |
| 2 | 43.6 | 30.1 | 38.6 | 46.6 |
| 3 | 46.5 | 32.4 | 41.3 | 49.8 |

$\cdot 10^{-3}$

$t_{rotor}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 18.7 | 22.1 | 26.7 | 28.6 |
| 2 | 20.2 | 23.8 | 28.7 | 30.9 |
| 3 | 21.6 | 25.6 | 30.6 | 33.2 |

$\cdot 10^{-3}$

Относительная густота решетки ():

$b/t_{CA}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.444 | 1.815 | 1.682 | 1.600 |
| 2 | 1.346 | 1.680 | 1.563 | 1.488 |
| 3 | 1.261 | 1.563 | 1.460 | 1.391 |

$b/t_{PK}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1.823 | 1.774 | 1.659 | 1.600 |
| 2 | 1.562 | 1.521 | 1.428 | 1.371 |
| 3 | 1.337 | 1.303 | 1.227 | 1.174 |

Длина хорды профиля [м]:

$chord_{stator}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 58.6 | 50.6 | 60.3 | 69.3 |
| 2 | 58.6 | 50.6 | 60.3 | 69.3 |
| 3 | 58.6 | 50.6 | 60.3 | 69.3 |

$\cdot 10^{-3}$

$chord_{rotor}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 34.0 | 39.2 | 44.2 | 45.8 |
| 2 | 31.5 | 36.3 | 40.9 | 42.4 |
| 3 | 28.9 | 33.3 | 37.6 | 38.9 |

$\cdot 10^{-3}$

Угол поворота потока:

$\varepsilon_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 64.10 | 22.00 | 38.26 | 51.69 |
| 2 | 67.32 | 33.86 | 47.53 | 57.40 |
| 3 | 71.86 | 45.67 | 56.37 | 62.65 |

$.^{\circ}$

$\varepsilon_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 66.39 | 86.80 | 90.25 | 83.14 |
| 2 | 80.39 | 94.27 | 93.36 | 83.56 |
| 3 | 93.13 | 102.01 | 95.98 | 82.65 |

$.^{\circ}$

Угол установки профиля:

$\upsilon_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 134.2 | 124.4 | 130.1 | 136.5 |
| 2 | 129.9 | 123.9 | 129.5 | 136.7 |
| 3 | 123.2 | 120.9 | 127.8 | 136.3 |

$.^{\circ}$

$\upsilon_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 142.1 | 136.0 | 134.8 | 138.3 |
| 2 | 136.5 | 132.6 | 133.1 | 137.6 |
| 3 | 130.3 | 128.8 | 131.3 | 137.1 |

$.^{\circ}$

Угол изгиба профиля:

$\pi - \varepsilon_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 115.9 | 158.0 | 141.7 | 128.3 |
| 2 | 112.7 | 146.1 | 132.5 | 122.6 |
| 3 | 108.1 | 134.3 | 123.6 | 117.4 |

$.^{\circ}$

$\pi - \varepsilon_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 113.6 | 93.2 | 89.8 | 96.9 |
| 2 | 99.6 | 85.7 | 86.6 | 96.4 |
| 3 | 86.9 | 78.0 | 84.0 | 97.3 |

$.^{\circ}$

Результаты расчета параметров решеток

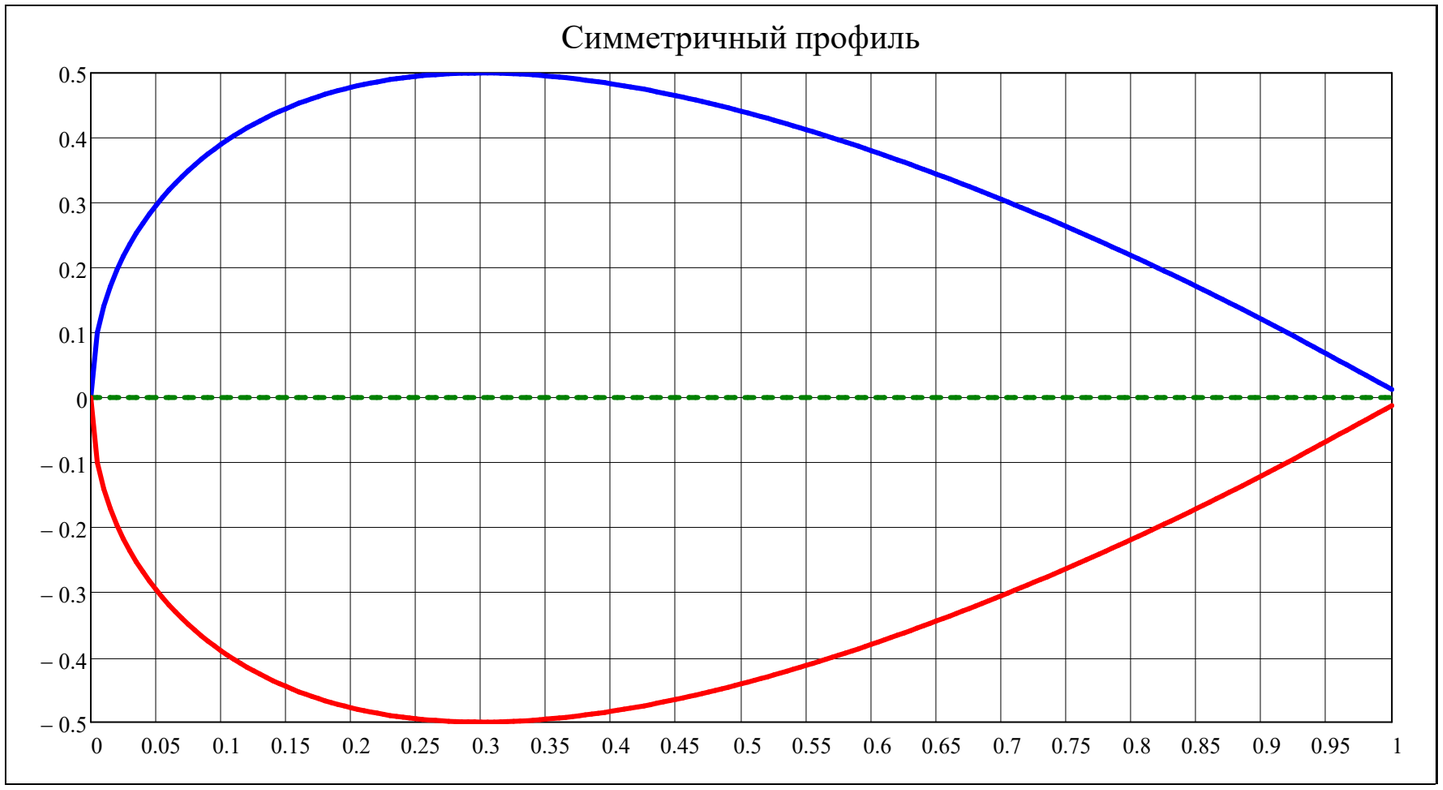
Подключение симметричного профиля

$$\begin{pmatrix} X_U & Y_U \\ X_L & Y_L \end{pmatrix} = \mathrm{NACA}(0, 0, 100\%, 1)$$

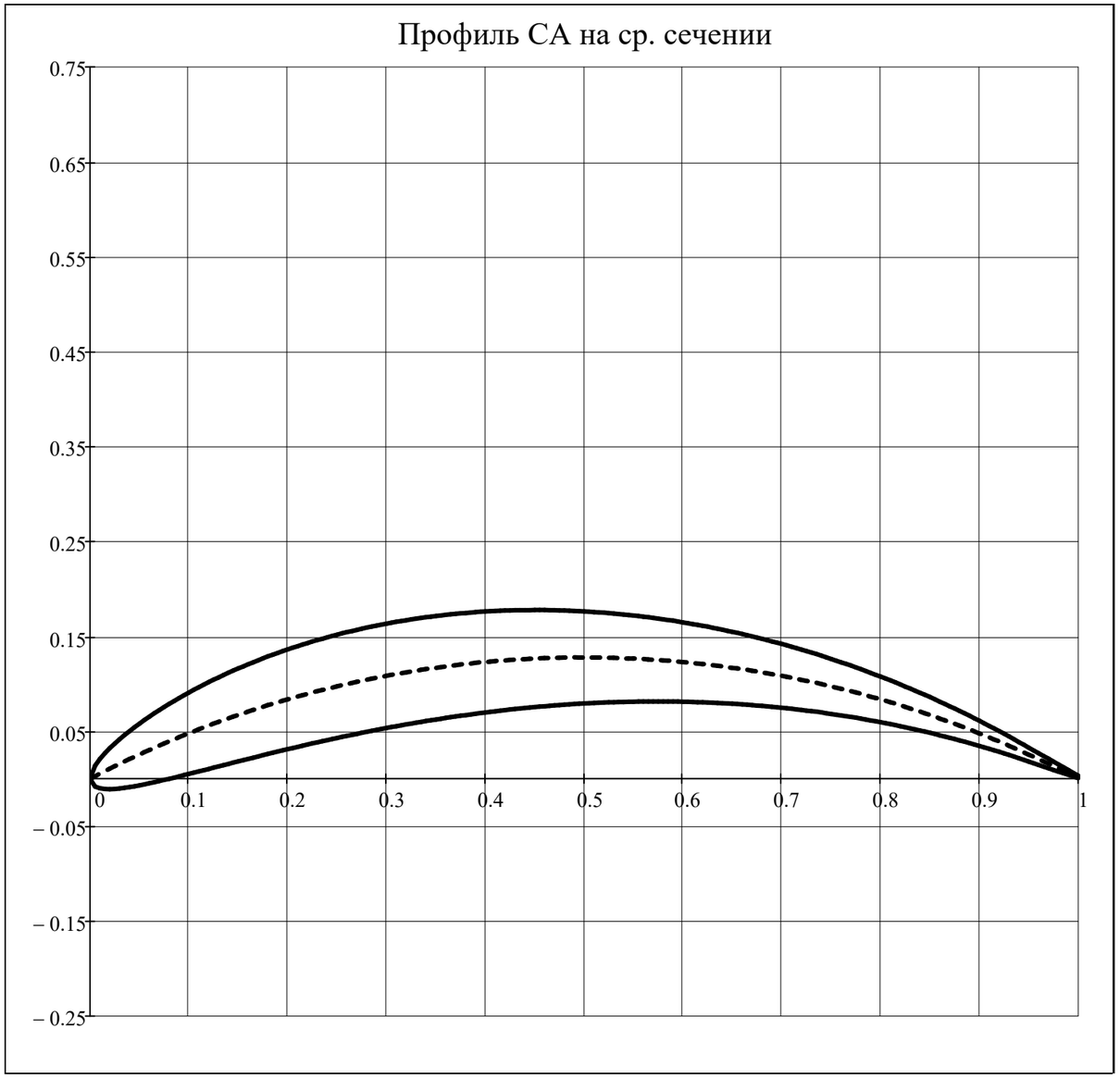
Относ. координаты профиля РК и СА:

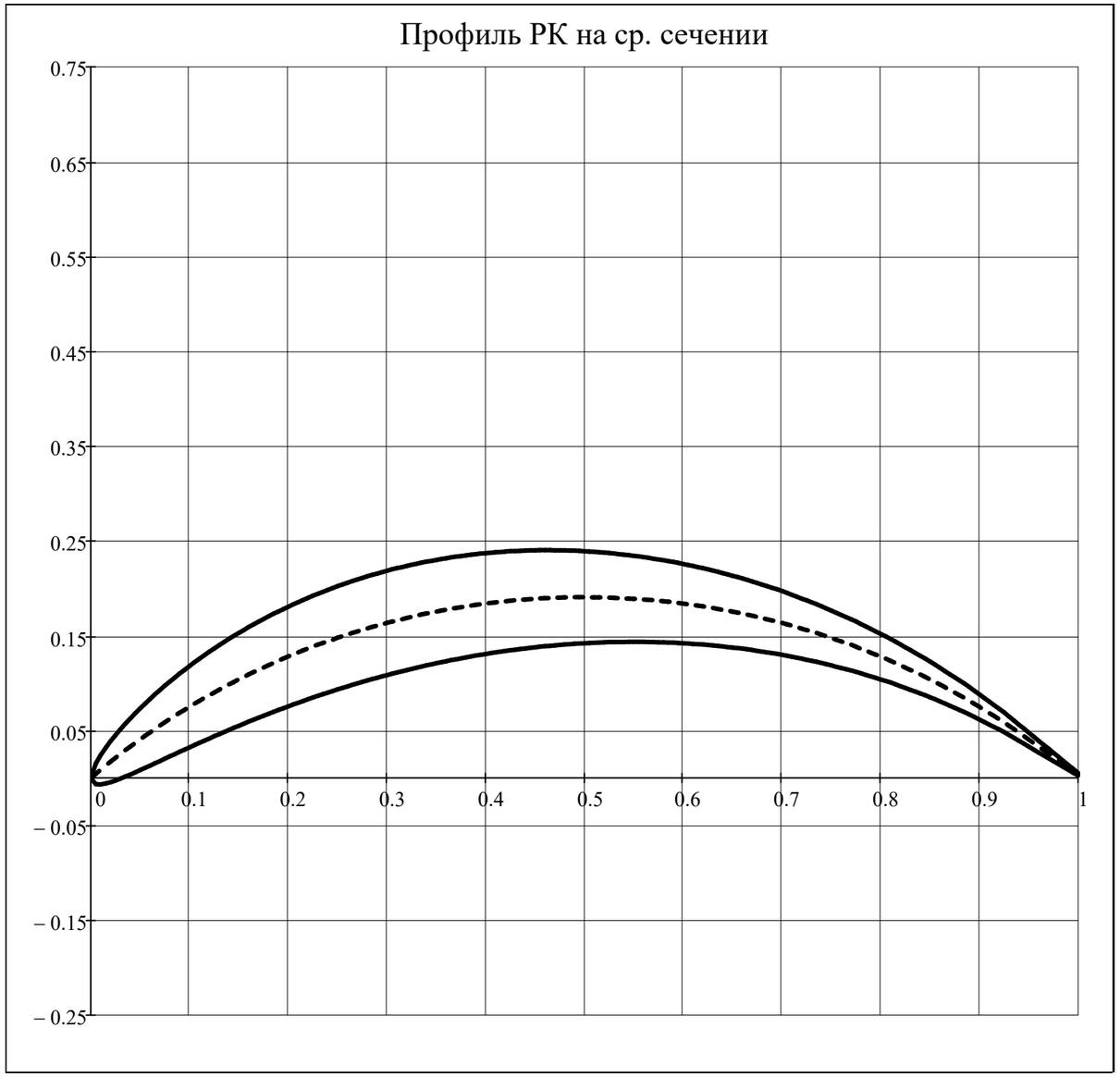
$$\mathrm{AIRFOIL}_0\left(x, \mathrm{line}, \overline{f}, \overline{x}_f, \overline{c}\right) = \begin{cases} \begin{cases} \mathrm{linterp}\left(X_U, Y_U, x\right) & \text{if } \mathrm{line} = \text{"+"} \\ \dfrac{\mathrm{linterp}\left(X_U, Y_U, x\right) + \mathrm{linterp}\left(X_L, Y_L, x\right)}{2} & \text{if } \mathrm{line} = \text{"0"} \\ \mathrm{linterp}\left(X_L, Y_L, x\right) & \text{if } \mathrm{line} = \text{"-"} \end{cases} & \text{if } 0 \le x \le 1 \\ \mathrm{NaN} & \text{otherwise} \end{cases}$$

$$x = 0, 0.005 .. 1$$



$$\mathrm{AIRFOIL}(x, line, \bar{c}, \theta) = \begin{cases} \mathrm{linterp}\left(X_U, y/b_{ср.л}\left(X_U, \theta\right) + Y_U \cdot \bar{c}, x\right) & \text{if } line = \text{"+"} \\ \dfrac{\mathrm{linterp}\left(X_U, y/b_{ср.л}\left(X_U, \theta\right) + Y_U \cdot \bar{c}, x\right) + \mathrm{linterp}\left(X_L, y/b_{ср.л}\left(X_L, \theta\right) + Y_L \cdot \bar{c}, x\right)}{2} & \text{if } line = \text{"0"} \\ \mathrm{linterp}\left(X_L, y/b_{ср.л}\left(X_L, \theta\right) + Y_L \cdot \bar{c}, x\right) & \text{if } line = \text{"-"} \\ \mathrm{NaN} & \text{otherwise} \end{cases}$$




Подключение симметричного профиля

Результат расчета абсолютных геометрических характеристик сечений Л

Результат расчета абсолютных геометрических характеристик сечений Л

$l_upper_{CA}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 64.56 | 52.34 | 63.63 | 74.60 |
| 2 | 64.97 | 53.07 | 64.52 | 75.34 |
| 3 | 65.58 | 53.97 | 65.48 | 76.07 |

$\cdot 10^{-3}$

$l_upper_{PK}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 37.98 | 45.76 | 52.09 | 53.03 |
| 2 | 35.86 | 42.70 | 48.09 | 48.63 |
| 3 | 33.32 | 39.27 | 43.63 | 43.76 |

$\cdot 10^{-3}$

$l_lower_{CA}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 60.35 | 51.07 | 61.02 | 70.56 |
| 2 | 60.56 | 51.13 | 61.28 | 70.87 |
| 3 | 60.88 | 51.36 | 61.66 | 71.26 |

$\cdot 10^{-3}$

$l_lower_{PK}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 35.06 | 41.44 | 47.03 | 48.17 |
| 2 | 33.06 | 38.98 | 43.92 | 44.73 |
| 3 | 31.40 | 36.88 | 41.06 | 41.42 |

$\cdot 10^{-3}$

$area_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 258.72 | 192.80 | 273.90 | 361.23 |
| 2 | 258.72 | 192.80 | 273.90 | 361.23 |
| 3 | 258.72 | 192.80 | 273.90 | 361.23 |

$\cdot 10^{-6}$

$area_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 102.96 | 136.54 | 173.93 | 186.58 |
| 2 | 74.54 | 98.85 | 125.93 | 135.08 |
| 3 | 40.06 | 53.12 | 67.67 | 72.59 |

$\cdot 10^{-6}$

$Sx_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1665.7 | 355.1 | 1055.4 | 2185.0 |
| 2 | 1756.8 | 549.9 | 1321.3 | 2441.5 |
| 3 | 1887.1 | 748.6 | 1581.0 | 2681.1 |

$\cdot 10^{-9}$

$Sx_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 399.6 | 823.4 | 1238.7 | 1251.8 |
| 2 | 330.8 | 607.3 | 863.2 | 843.2 |
| 3 | 193.6 | 329.7 | 440.5 | 411.2 |

$\cdot 10^{-9}$

$Sy_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6389.6 | 4110.4 | 6960.2 | 10541.9 |
| 2 | 6389.6 | 4110.4 | 6960.2 | 10541.9 |
| 3 | 6389.6 | 4110.4 | 6960.2 | 10541.9 |

$\cdot 10^{-9}$

$Sy_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1475.6 | 2253.4 | 3240.0 | 3599.6 |
| 2 | 988.2 | 1509.1 | 2169.8 | 2410.6 |
| 3 | 488.0 | 745.2 | 1071.4 | 1190.3 |

$\cdot 10^{-9}$

$x0_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 24.7 | 21.3 | 25.4 | 29.2 |
| 2 | 24.7 | 21.3 | 25.4 | 29.2 |
| 3 | 24.7 | 21.3 | 25.4 | 29.2 |

$\cdot 10^{-3}$

$x0_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 14.3 | 16.5 | 18.6 | 19.3 |
| 2 | 13.3 | 15.3 | 17.2 | 17.8 |
| 3 | 12.2 | 14.0 | 15.8 | 16.4 |

$\cdot 10^{-3}$

$y0_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6.4 | 1.8 | 3.9 | 6.0 |
| 2 | 6.8 | 2.9 | 4.8 | 6.8 |
| 3 | 7.3 | 3.9 | 5.8 | 7.4 |

$\cdot 10^{-3}$

$y0_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.9 | 6.0 | 7.1 | 6.7 |
| 2 | 4.4 | 6.1 | 6.9 | 6.2 |
| 3 | 4.8 | 6.2 | 6.5 | 5.7 |

$\cdot 10^{-3}$

$Jx_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 12406 | 1067 | 5183 | 15768 |
| 2 | 13721 | 2076 | 7721 | 19367 |
| 3 | 15720 | 3549 | 10739 | 23084 |

$\cdot 10^{-12}$

$Jx_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1820 | 5629 | 9958 | 9567 |
| 2 | 1658 | 4157 | 6598 | 5925 |
| 3 | 1029 | 2241 | 3150 | 2579 |

$\cdot 10^{-12}$

$Jy_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 206780 | 114833 | 231764 | 403124 |
| 2 | 206780 | 114833 | 231764 | 403124 |
| 3 | 206780 | 114833 | 231764 | 403124 |

$\cdot 10^{-12}$

$Jy_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 27712 | 48732 | 79084 | 90998 |
| 2 | 17166 | 30187 | 48990 | 56370 |
| 3 | 7789 | 13697 | 22229 | 25578 |

$\cdot 10^{-12}$

$Jxy_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 43693 | 8055 | 28524 | 67776 |
| 2 | 46072 | 12470 | 35696 | 75706 |
| 3 | 49471 | 16968 | 42691 | 83109 |

$\cdot 10^{-12}$

$Jxy_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6082 | 14405 | 24450 | 25610 |
| 2 | 4652 | 9819 | 15755 | 15956 |
| 3 | 2498 | 4894 | 7385 | 7151 |

$\cdot 10^{-12}$

$Jx0_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1681 | 413 | 1116 | 2551 |
| 2 | 1791 | 508 | 1347 | 2866 |
| 3 | 1955 | 642 | 1613 | 3185 |

$\cdot 10^{-12}$

$Jx0_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 269 | 664 | 1137 | 1168 |
| 2 | 190 | 426 | 681 | 662 |
| 3 | 94 | 195 | 283 | 250 |

$\cdot 10^{-12}$

$Jy0_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 48976 | 27198 | 54893 | 95480 |
| 2 | 48976 | 27198 | 54893 | 95480 |
| 3 | 48976 | 27198 | 54893 | 95480 |

$\cdot 10^{-12}$

$Jy0_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6564 | 11542 | 18731 | 21553 |
| 2 | 4066 | 7150 | 11603 | 13351 |
| 3 | 1845 | 3244 | 5265 | 6058 |

$\cdot 10^{-12}$

$Jxy0_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2554 | 485 | 1704 | 4010 |
| 2 | 2683 | 747 | 2119 | 4456 |
| 3 | 2865 | 1009 | 2515 | 4866 |

$\cdot 10^{-12}$

$Jxy0_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 355 | 816 | 1377 | 1460 |
| 2 | 266 | 549 | 882 | 909 |
| 3 | 140 | 269 | 411 | 408 |

$\cdot 10^{-12}$

$\alpha_major_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.1 | 1.0 | 1.8 | 2.5 |
| 2 | 3.2 | 1.6 | 2.3 | 2.7 |
| 3 | 3.5 | 2.2 | 2.7 | 3.0 |

$\cdot^{\circ}$

$\alpha_major_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 3.2 | 4.3 | 4.4 | 4.1 |
| 2 | 3.9 | 4.6 | 4.6 | 4.1 |
| 3 | 4.5 | 5.0 | 4.7 | 4.0 |

$\cdot^{\circ}$

$Ju_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1544 | 404 | 1063 | 2378 |
| 2 | 1639 | 487 | 1263 | 2652 |
| 3 | 1781 | 604 | 1495 | 2929 |

$\cdot 10^{-12}$

$Ju_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 249 | 603 | 1030 | 1064 |
| 2 | 172 | 382 | 610 | 597 |
| 3 | 83 | 171 | 249 | 221 |

$\cdot 10^{-12}$

$Jv_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 49113.3 | 27206.9 | 54947.2 | 95652.3 |
| 2 | 49127.8 | 27219.0 | 54977.0 | 95693.5 |
| 3 | 49149.7 | 27236.4 | 55011.7 | 95735.4 |

$\cdot 10^{-12}$

$Jv_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6583 | 11603 | 18838 | 21657 |
| 2 | 4084 | 7194 | 11674 | 13416 |
| 3 | 1856 | 3268 | 5299 | 6087 |

$\cdot 10^{-12}$

$Juv_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | -0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 0 | 0 |
| 3 | -0 | 0 | -0 | 0 |

$\cdot 10^{-12}$

$Juv_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | -0 | 0 | 0 | -0 |
| 2 | -0 | 0 | 0 | -0 |
| 3 | 0 | 0 | 0 | 0 |

$\cdot 10^{-12}$

$Jp_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 50657 | 27611 | 56010 | 98030 |
| 2 | 50767 | 27706 | 56240 | 98346 |
| 3 | 50931 | 27841 | 56507 | 98664 |

$\cdot 10^{-12}$

$Jp_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 6832 | 12206 | 19868 | 22721 |
| 2 | 4256 | 7576 | 12284 | 14013 |
| 3 | 1939 | 3439 | 5548 | 6308 |

$\cdot 10^{-12}$

$Wp_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1465.9 | 940.2 | 1593.7 | 2416.2 |
| 2 | 1466.3 | 940.9 | 1594.8 | 2417.3 |
| 3 | 1466.7 | 941.7 | 1595.9 | 2418.3 |

$\cdot 10^{-9}$

$Wp_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 340.2 | 519.9 | 747.5 | 830.5 |
| 2 | 226.9 | 346.5 | 498.2 | 553.6 |
| 3 | 111.2 | 169.7 | 244.2 | 271.4 |

$\cdot 10^{-9}$

$stiffness_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 2479.0 | 1376.7 | 2778.6 | 4832.9 |
| 2 | 2479.0 | 1376.7 | 2778.6 | 4832.9 |
| 3 | 2479.0 | 1376.7 | 2778.6 | 4832.9 |

$\cdot 10^{-12}$

$stiffness_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 464.0 | 816.0 | 1324.2 | 1523.7 |
| 2 | 205.8 | 361.9 | 587.3 | 675.8 |
| 3 | 37.8 | 66.5 | 107.9 | 124.2 |

$\cdot 10^{-12}$

$CPx_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 20.527 | 17.720 | 21.121 | 24.255 |
| 2 | 20.527 | 17.720 | 21.121 | 24.255 |
| 3 | 20.527 | 17.720 | 21.121 | 24.255 |

$\cdot 10^{-3}$

$CPx_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 11.912 | 13.717 | 15.482 | 16.035 |
| 2 | 11.018 | 12.688 | 14.321 | 14.832 |
| 3 | 10.125 | 11.660 | 13.160 | 13.630 |

$\cdot 10^{-3}$

$CPy_{stator}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 2 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 3 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

$\cdot 10^{-3}$

$CPy_{rotor}{}^{T} =$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 2 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 3 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

$\cdot 10^{-3}$

Результат расчета абсолютных геометрических характеристик сечений Л

Вывод результатов расчета геометрических хар-к сечений Л

Рассматриваемая ступень: $j := \begin{cases} j = 1 & \\ j = \begin{cases} \text{"Такой ступени не существует!"} & \text{if } (j < 1) \vee (j > Z) \\ j & \text{otherwise} \end{cases} \end{cases} = 1$

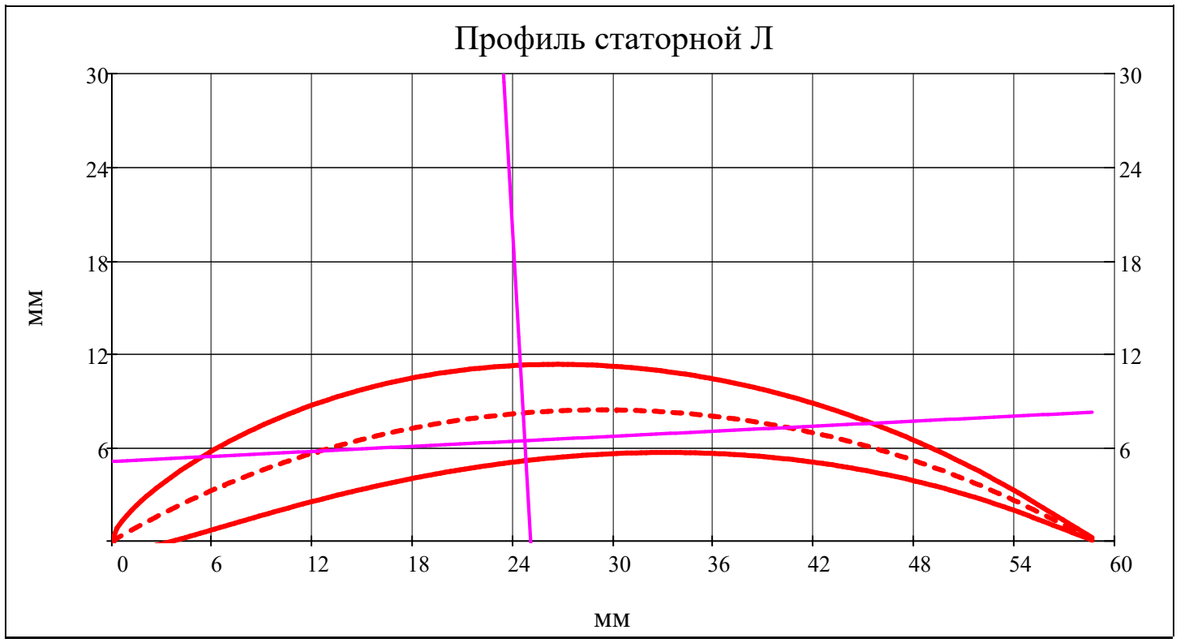
Построение профилей Л РК и НА

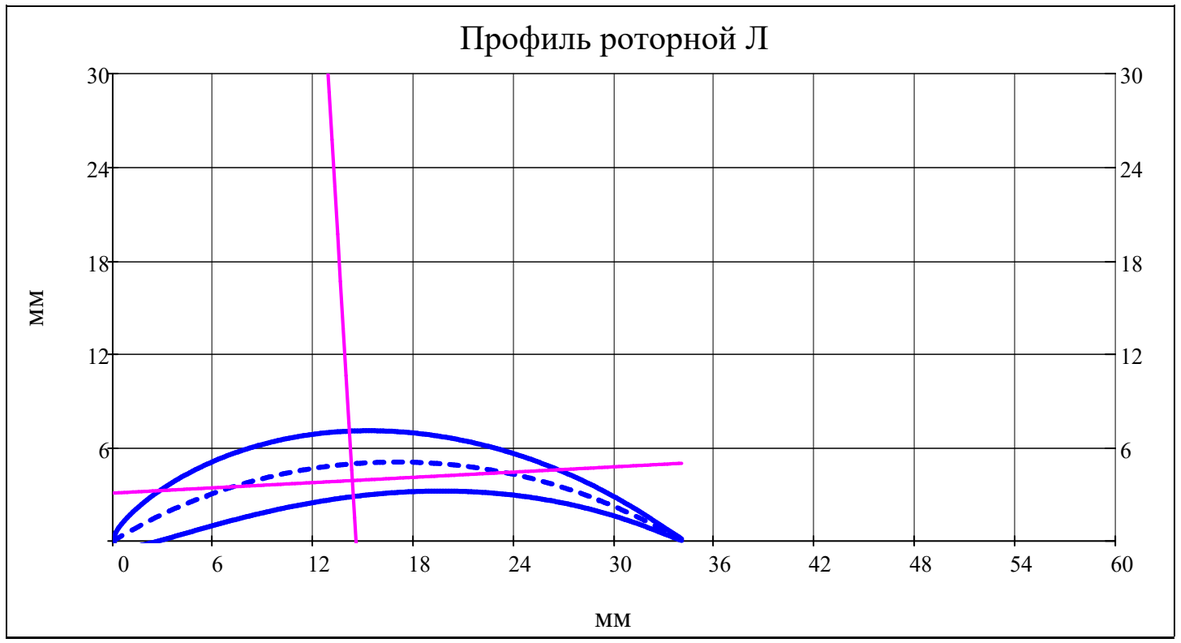
$$AXLE0(type, x, i, r) = \begin{cases} \dfrac{y0_{rotor_{i,r}}}{chord_{rotor_{i,r}}} + \tan\left(\alpha_major_{rotor_{i,r}}\right)\cdot\left(x - \dfrac{x0_{rotor_{i,r}}}{chord_{rotor_{i,r}}}\right) & \text{if } type = \text{"rotor"} \\ \dfrac{y0_{stator_{i,r}}}{chord_{stator_{i,r}}} + \tan\left(\alpha_major_{stator_{i,r}}\right)\cdot\left(x - \dfrac{x0_{stator_{i,r}}}{chord_{stator_{i,r}}}\right) & \text{if } type = \text{"stator"} \\ NaN & \text{otherwise} \end{cases}$$

$$b_{lim} = \frac{ceil\left(\max\left(chord_{rotor_{j,N_r}}, chord_{stator_{j,N_r}}\right)\cdot 10^2\right)}{10^2} = 60.0\cdot 10^{-3}$$

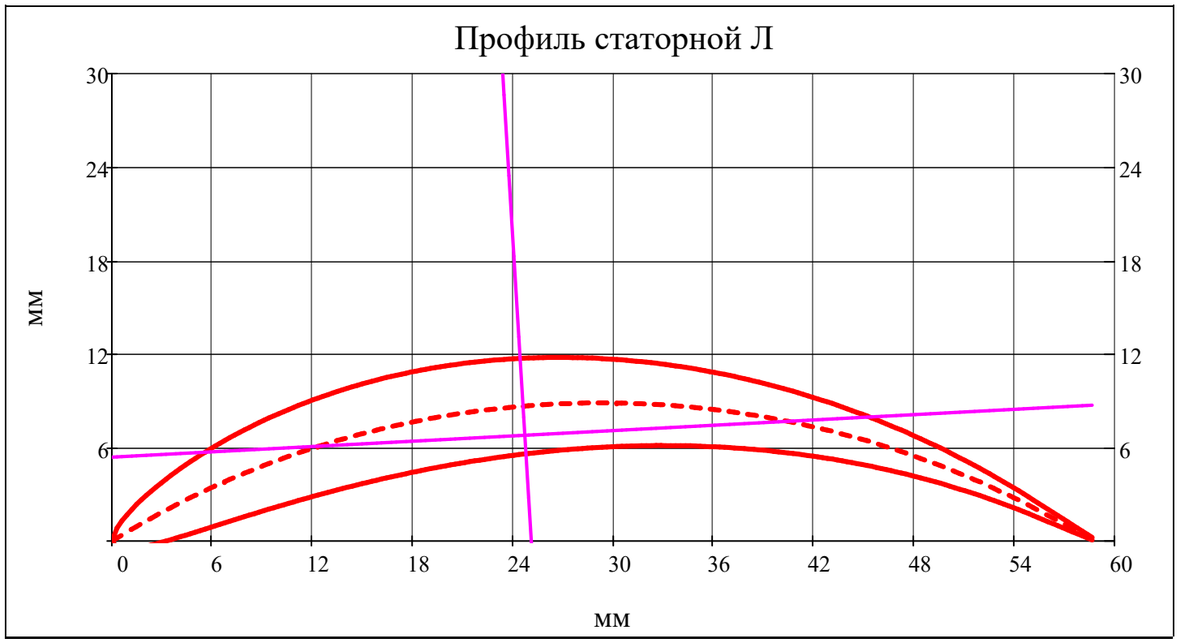
$$AXLE90(type, x, i, r) = \begin{cases} \dfrac{y0_{rotor_{i,r}}}{chord_{rotor_{i,r}}} + \tan\left(\alpha_major_{rotor_{i,r}} + \dfrac{\pi}{2}\right)\cdot\left(x - \dfrac{x0_{rotor_{i,r}}}{chord_{rotor_{i,r}}}\right) & \text{if } (type = \text{"rotor"}) \wedge \left|\alpha_major_{rotor_{i,r}}\right| \geq 1\cdot° \\ \dfrac{y0_{stator_{i,r}}}{chord_{stator_{i,r}}} + \tan\left(\alpha_major_{stator_{i,r}} + \dfrac{\pi}{2}\right)\cdot\left(x - \dfrac{x0_{stator_{i,r}}}{chord_{stator_{i,r}}}\right) & \text{if } (type = \text{"stator"}) \wedge \left|\alpha_major_{stator_{i,r}}\right| \geq 1\cdot° \\ NaN & \text{otherwise} \end{cases}$$

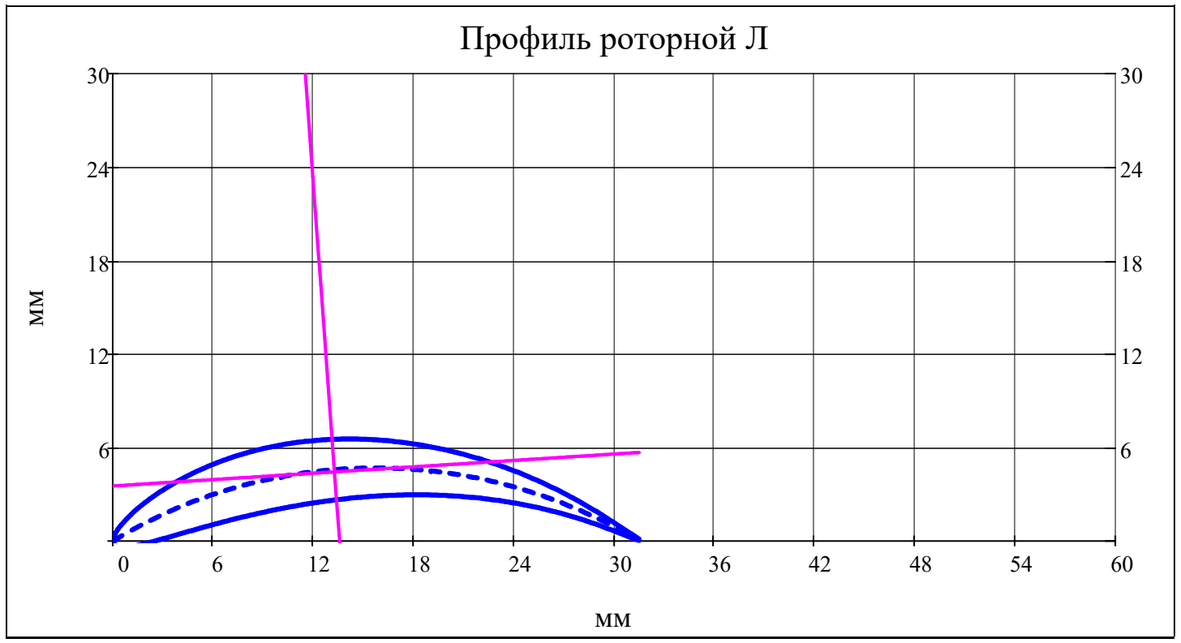
$r = 1$


Профиль статорной Л
мм
мм
0
6
12
18
24
30
36
42
48
54
60

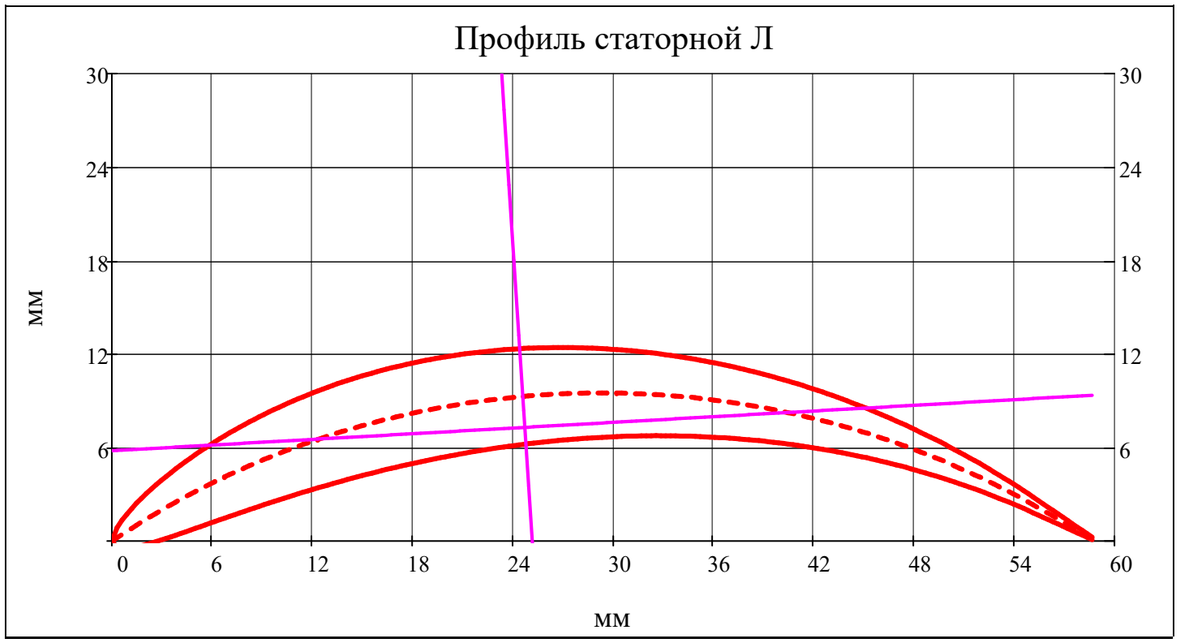

Профиль роторной Л
мм
мм
0
6
12
18
24
30
36
42
48
54
60

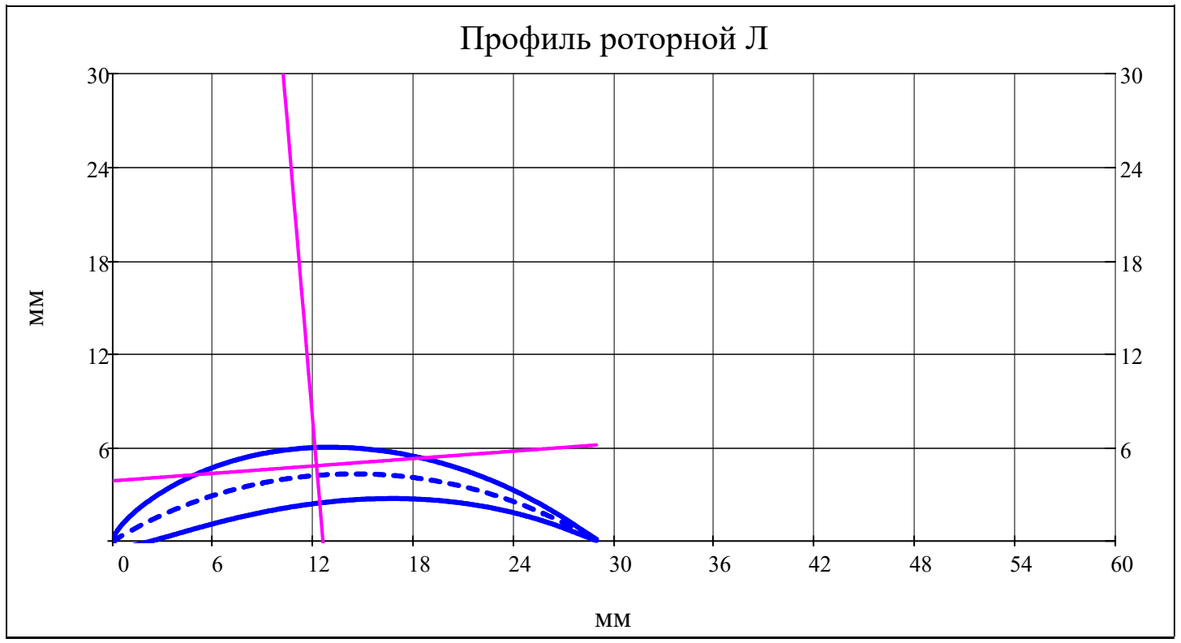
$r_{\sim} = av(N_r)$


Профиль статорной Л
мм
мм
30
24
18
12
6
0
6
12
18
24
30
36
42
48
54
60


Профиль роторной Л
мм
мм
30
24
18
12
6
0
6
12
18
24
30
36
42
48
54
60

$r = N_r$




Построение профилей Л РК и НА

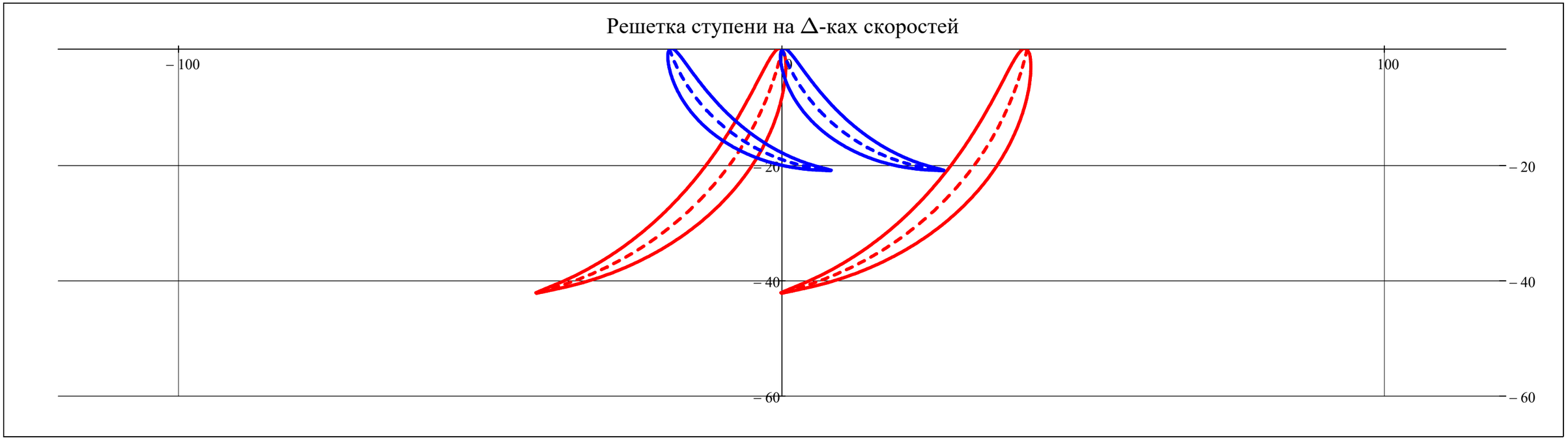
Вывод координат для построения профиля Л

Рассматриваемая ступень: $j = \begin{cases} j = 1 & \\ j = \begin{cases} \text{"Такой ступени не существует!"} & \text{if } (j < 1) \vee (j > Z) \\ j & \text{otherwise} \end{cases} \end{cases} = 1$

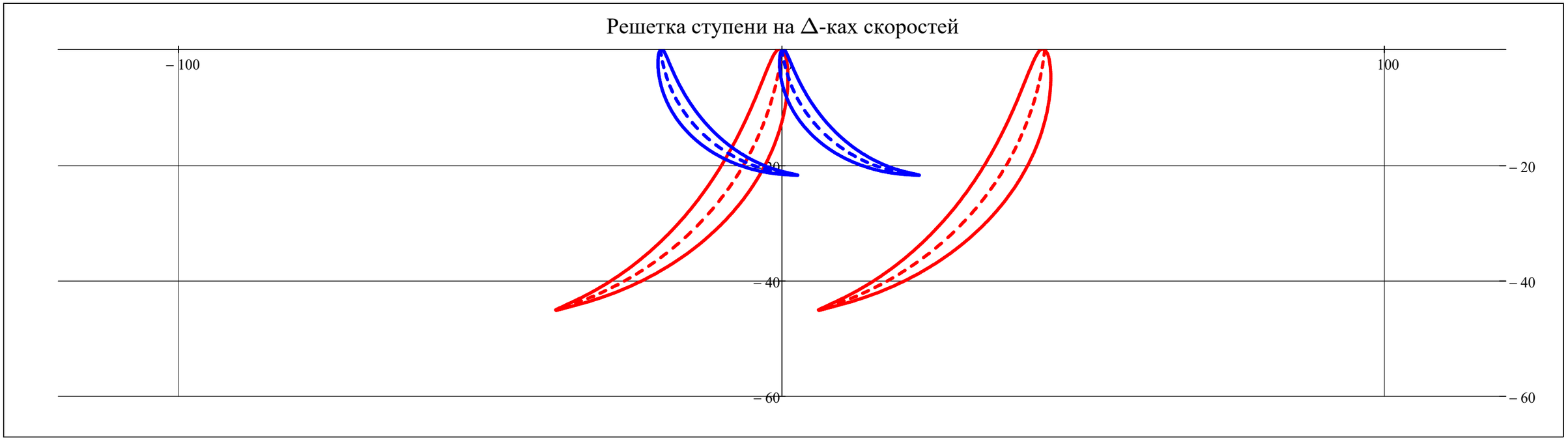
$$b_{lim} = \frac{\mathrm{ceil}\left(\max\left(\mathrm{chord}_{rotor_{j,N_r}}, \mathrm{chord}_{stator_{j,N_r}}\right)\cdot 10^2\right)}{10^2} = 60.0\cdot 10^{-3}$$

Построение плоских решеток профилей Л на треугольниках скоростей

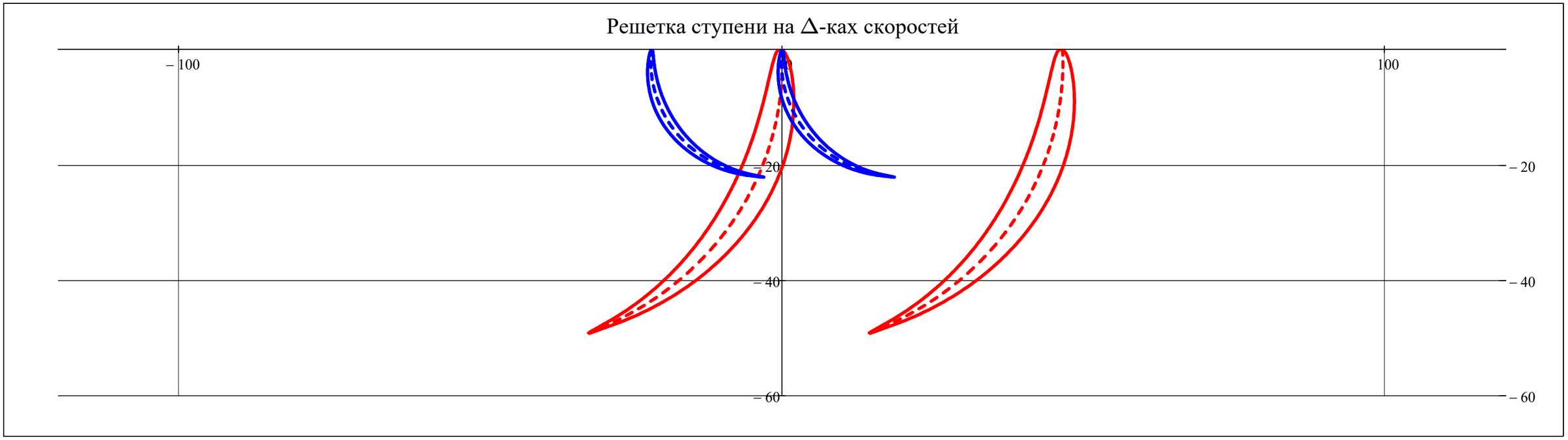
$r = 1$



$r = av(N_r)$



$r = N_r$




Построение плоских решеток профилей Л на треугольниках скоростей

Вывод данных по построению профиля Л РК и НА на треугольниках скоростей

Расчет собственных частот колебаний Л

$$\begin{pmatrix} \nu 0_{изгСА} & \nu 0_{изгРК} \\ \nu 0_{углСА} & \nu 0_{углРК} \\ \nu 0_{углСАб} & \nu 0_{углРКб} \end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } r \in av(N_r) \\ \quad\quad \text{for } mode \in 1..6 \\ \quad\quad\quad \nu 0_{изгСА_{i,mode}} = \nu 0_{изгиб}\left(mode, mean\left(h_{st(i,1)}, h_{st(i,2)}\right), E_Л, \rho_Л, area_{stator_{i,r}}, Ju_{stator_{i,r}}\right) \\ \quad\quad\quad \nu 0_{изгРК_{i,mode}} = \nu 0_{изгиб}\left(mode, mean\left(h_{st(i,2)}, h_{st(i,3)}\right), E_Л, \rho_Л, area_{rotor_{i,r}}, Ju_{rotor_{i,r}}\right) \\ \quad\quad\quad \nu 0_{углСА_{i,mode}} = \nu 0_{угл}\left(mode, 0, mean\left(h_{st(i,1)}, h_{st(i,2)}\right), Jung\left(2, \mu_Л, E_Л\right), \rho_Л, stiffness_{stator_{i,r}}, Jp_{stator_{i,r}}\right) \\ \quad\quad\quad \nu 0_{углРК_{i,mode}} = \nu 0_{угл}\left(mode, 0, mean\left(h_{st(i,2)}, h_{st(i,3)}\right), Jung\left(2, \mu_Л, E_Л\right), \rho_Л, stiffness_{rotor_{i,r}}, Jp_{rotor_{i,r}}\right) \\ \quad\quad\quad \nu 0_{углСАб_{i,mode}} = \nu 0_{угл}\left(mode, 1, mean\left(h_{st(i,1)}, h_{st(i,2)}\right), Jung\left(2, \mu_Л, E_Л\right), \rho_Л, stiffness_{stator_{i,r}}, Jp_{stator_{i,r}}\right) \\ \quad\quad\quad \nu 0_{углРКб_{i,mode}} = \nu 0_{угл}\left(mode, 1, mean\left(h_{st(i,2)}, h_{st(i,3)}\right), Jung\left(2, \mu_Л, E_Л\right), \rho_Л, stiffness_{rotor_{i,r}}, Jp_{rotor_{i,r}}\right) \\ \begin{pmatrix} \nu 0_{изгСА} & \nu 0_{изгРК} \\ \nu 0_{углСА} & \nu 0_{углРК} \\ \nu 0_{углСАб} & \nu 0_{углРКб} \end{pmatrix} \end{array} \right.$$

Частота собственных изгибных колебаний (Гц) [9, с.240]:

$$stack\left(\nu 0_{изгСА}, \nu 0_{изгРК}\right) = \blacksquare$$

Частота собственных угловых колебаний (Гц) [9, с.243] без и с бандажом:

$$stack\left(\nu 0_{углСА}, \nu 0_{углРК}\right) = \blacksquare$$

$$stack\left(\nu 0_{углСАб}, \nu 0_{углРКб}\right) = \blacksquare$$

Расчет собственных частот колебаний Л

Вывод результатов расчета собственных частот колебаний Л

Расчетный узел: $type = "turbine"$

Объем бандажной полки ($м^3$): $V_{бп} = 0$

Радиус положения ЦМ бандажной полки (м): $R_{бп} = 0$

Расчет Л на прочность

$$\text{neutral_line}(\text{type},x,i,r) = \begin{cases} \dfrac{y0_{rotor_{i,r}}}{chord_{rotor_{i,r}}} + \tan\left(\left(\alpha_major_{rotor_{i,r}} + \varphi_neutral_{rotor}\left(i, R_{st(i,2),r}\right)\right)\right)\cdot\left(x - \dfrac{x0_{rotor_{i,r}}}{chord_{rotor_{i,r}}}\right) & \text{if type = "rotor"} \\ \dfrac{y0_{stator_{i,r}}}{chord_{stator_{i,r}}} + \tan\left(\left(\alpha_major_{stator_{i,r}} + \varphi_neutral_{stator}\left(i, R_{st(i,2),r}\right)\right)\right)\cdot\left(x - \dfrac{x0_{stator_{i,r}}}{chord_{stator_{i,r}}}\right) & \text{if type = "stator"} \end{cases}$$

$$\text{epure}(\text{type},x,i,r) = \begin{cases} \dfrac{y0_{rotor_{i,r}}}{chord_{rotor_{i,r}}} + \dfrac{-1}{\tan\left(\alpha_major_{rotor_{i,r}} + \varphi_neutral_{rotor}\left(i, R_{st(i,2),r}\right) - \dfrac{\pi}{4}\right)}\cdot\left(x - \dfrac{x0_{rotor_{i,r}}}{chord_{rotor_{i,r}}}\right) & \text{if type = "rotor"} \\ \dfrac{y0_{stator_{i,r}}}{chord_{stator_{i,r}}} + \dfrac{-1}{\tan\left(\alpha_major_{stator_{i,r}} + \varphi_neutral_{stator}\left(i, R_{st(i,2),r}\right) - \dfrac{\pi}{4}\right)}\cdot\left(x - \dfrac{x0_{stator_{i,r}}}{chord_{stator_{i,r}}}\right) & \text{if type = "stator"} \end{cases}$$

Определение координат точек профиля Л, наиболее удаленных от НЛ

Наиболее удаленные точки от НЛ (мм):

$u_u_{rotor}^{T} = \blacksquare \cdot 10^{-3}$

$v_u_{rotor}^{T} = \blacksquare \cdot 10^{-3}$

$u_l_{rotor}^{T} = \blacksquare \cdot 10^{-3}$

$v_l_{rotor}^{T} = \blacksquare \cdot 10^{-3}$

$u_u_{stator}^{T} = \blacksquare \cdot 10^{-3}$

$v_u_{stator}^{T} = \blacksquare \cdot 10^{-3}$

$u_l_{stator}^{T} = \blacksquare \cdot 10^{-3}$

$v_l_{stator}^{T} = \blacksquare \cdot 10^{-3}$

$$\begin{pmatrix} \sigma_p_{rotor} & \sigma_n_{rotor} \\ \sigma_p_{stator} & \sigma_n_{stator} \end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } r \in 1..N_r \\ \quad \begin{pmatrix} \sigma_p_{rotor_{i,r}} & \sigma_n_{rotor_{i,r}} \\ \sigma_p_{stator_{i,r}} & \sigma_n_{stator_{i,r}} \end{pmatrix} = \begin{pmatrix} \dfrac{Mu_{rotor}(i, R_{st(i,2),r})}{Ju_{rotor_{i,r}}} \cdot v_u_{rotor_{i,r}} - \dfrac{Mv_{rotor}(i, R_{st(i,2),r})}{Jv_{rotor_{i,r}}} \cdot u_u_{rotor_{i,r}} & \dfrac{Mu_{rotor}(i, R_{st(i,2),r})}{Ju_{rotor_{i,r}}} \cdot v_l_{rotor_{i,r}} - \dfrac{Mv_{rotor}(i, R_{st(i,2),r})}{Jv_{rotor_{i,r}}} \cdot u_l_{rotor_{i,r}} \\ \dfrac{Mu_{stator}(i, R_{st(i,2),r})}{Ju_{stator_{i,r}}} \cdot v_u_{stator_{i,r}} - \dfrac{Mv_{stator}(i, R_{st(i,2),r})}{Jv_{stator_{i,r}}} \cdot u_u_{stator_{i,r}} & \dfrac{Mu_{stator}(i, R_{st(i,2),r})}{Ju_{stator_{i,r}}} \cdot v_l_{stator_{i,r}} - \dfrac{Mv_{stator}(i, R_{st(i,2),r})}{Jv_{stator_{i,r}}} \cdot u_l_{stator_{i,r}} \end{pmatrix} \\ \begin{pmatrix} \sigma_p_{rotor} & \sigma_n_{rotor} \\ \sigma_p_{stator} & \sigma_n_{stator} \end{pmatrix} \end{array} \right.$$

$$\begin{pmatrix} \sigma_p_{rotor.} & \sigma_p_{stator.} \\ \sigma_n_{rotor.} & \sigma_n_{stator.} \end{pmatrix} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \sigma_p_{rotor.}(i,z) = \text{interp}\left(\text{lspline}\left(\text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_p_{rotor}, i, i, 1, N_r)^T\right), \text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_p_{rotor}, i, i, 1, N_r)^T, z\right) \\ \quad \sigma_p_{stator.}(i,z) = \text{interp}\left(\text{lspline}\left(\text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_p_{stator}, i, i, 1, N_r)^T\right), \text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_p_{stator}, i, i, 1, N_r)^T, z\right) \\ \quad \sigma_n_{rotor.}(i,z) = \text{interp}\left(\text{lspline}\left(\text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_n_{rotor}, i, i, 1, N_r)^T\right), \text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_n_{rotor}, i, i, 1, N_r)^T, z\right) \\ \quad \sigma_n_{stator.}(i,z) = \text{interp}\left(\text{lspline}\left(\text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_n_{stator}, i, i, 1, N_r)^T\right), \text{submatrix}(R, st(i,1), st(i,1), 1, N_r)^T, \text{submatrix}(\sigma_n_{stator}, i, i, 1, N_r)^T, z\right) \\ \begin{pmatrix} \sigma_p_{rotor.} & \sigma_p_{stator.} \\ \sigma_n_{rotor.} & \sigma_n_{stator.} \end{pmatrix} \end{array} \right.$$

$\sigma_p_{rotor}^{T} = \blacksquare \cdot 10^6$ $\quad \sigma_p_{rotor}^{T} \le 70 \cdot 10^6 = \blacksquare$ $\quad \sigma_p_{stator}^{T} = \blacksquare \cdot 10^6$ $\quad \sigma_p_{stator}^{T} \le 70 \cdot 10^6 = \blacksquare$

$\sigma_n_{rotor}^{T} = \blacksquare \cdot 10^6$ $\quad \sigma_n_{rotor}^{T} \le 70 \cdot 10^6 = \blacksquare$ $\quad \sigma_n_{stator}^{T} = \blacksquare \cdot 10^6$ $\quad \sigma_n_{stator}^{T} \le 70 \cdot 10^6 = \blacksquare$

$$\begin{pmatrix}\sigma_{rotor}\\ \sigma_{stator}\end{pmatrix} = \left|\begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } r \in 1..N_r \\ \quad \left|\begin{array}{l} \sigma_{rotor_{i,r}} = \sqrt{\left(\sigma_z_{rotor}\left(i, R_{st(i,2),r}\right) + \max\left(\sigma_p_{rotor_{i,r}}, \sigma_n_{rotor_{i,r}}\right)\right)^2 + \tau_{rotor}\left(i, R_{st(i,2),r}\right)^2} \\ \sigma_{stator_{i,r}} = \sqrt{\left(0 + \max\left(\sigma_p_{stator_{i,r}}, \sigma_n_{stator_{i,r}}\right)\right)^2 + \tau_{stator}\left(i, R_{st(i,2),r}\right)^2} \end{array}\right. \\ \begin{pmatrix}\sigma_{rotor}\\ \sigma_{stator}\end{pmatrix} \end{array}\right.$$

$$\begin{pmatrix}\sigma_{rotor.}\\ \sigma_{stator.}\end{pmatrix} = \left|\begin{array}{l} \text{for } i \in 1..Z \\ \quad \left|\begin{array}{l} \sigma_{rotor.}(i,z) = \text{interp}\left(\text{lspline}\left(\text{submatrix}\left(R, st(i,2), st(i,2), 1, N_r\right)^T, \text{submatrix}\left(\sigma_{rotor}, i, i, 1, N_r\right)^T\right), \text{submatrix}\left(R, st(i,2), st(i,2), 1, N_r\right)^T, \text{submatrix}\left(\sigma_{rotor}, i, i, 1, N_r\right)^T, z\right) \\ \sigma_{stator.}(i,z) = \text{interp}\left(\text{lspline}\left(\text{submatrix}\left(R, st(i,2), st(i,2), 1, N_r\right)^T, \text{submatrix}\left(\sigma_{stator}, i, i, 1, N_r\right)^T\right), \text{submatrix}\left(R, st(i,2), st(i,2), 1, N_r\right)^T, \text{submatrix}\left(\sigma_{stator}, i, i, 1, N_r\right)^T, z\right) \end{array}\right. \\ \begin{pmatrix}\sigma_{rotor.}\\ \sigma_{stator.}\end{pmatrix} \end{array}\right.$$

$\sigma_{rotor}^{T} = \blacksquare \cdot 10^6$

$\sigma_{stator}^{T} = \blacksquare \cdot 10^6$

$$\begin{pmatrix}safety_{rotor}\\ safety_{stator}\end{pmatrix} = \left|\begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } r \in 1..N_r \\ \quad \left|\begin{array}{l} safety_{rotor_{i,r}} = \left|\begin{array}{ll} \dfrac{\sigma_{дл.Л}}{\sigma_{rotor_{i,r}}} & \text{if } \sigma_{rotor_{i,r}} \neq 0 \\ \infty & \text{otherwise} \end{array}\right. \\ safety_{stator_{i,r}} = \left|\begin{array}{ll} \dfrac{\sigma_{дл.Л}}{\sigma_{stator_{i,r}}} & \text{if } \sigma_{stator_{i,r}} \neq 0 \\ \infty & \text{otherwise} \end{array}\right. \end{array}\right. \\ \begin{pmatrix}safety_{rotor}\\ safety_{stator}\end{pmatrix} \end{array}\right.$$

$safety_{rotor}^{T} = \blacksquare$

$safety_{rotor}^{T} \geq safety = \blacksquare$

$safety_{stator}^{T} = \blacksquare$

$safety_{stator}^{T} \geq safety = \blacksquare$

Вывод результатов расчета Л на прочность

Рассматриваемая ступень: $j := \left| \begin{array}{l} j := \left| \begin{array}{ll} 1 & \text{if } type = \text{"compressor"} \\ Z & \text{if } type = \text{"turbine"} \end{array} \right. \\ j := \left| \begin{array}{ll} \text{"Такой ступени не существует!"} & \text{if } (j < 1) \vee (j > Z) \\ j & \text{otherwise} \end{array} \right. \end{array} \right. = 4$

$$b_{lim} = \frac{\mathrm{ceil}\left(\max\left(chord_{rotor_{j,N_r}}, chord_{stator_{j,N_r}}\right)\cdot 10^2\right)}{10^2} = 70.0\cdot 10^{-3}$$

Расстояния от оси ЛМ до рассматриваемой ступени (м): $Rj = \mathrm{submatrix}\left(R, 2\cdot Z - 1, 2\cdot Z + 1, 1, N_r\right) =$

| | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 422.0 | 453.2 | 484.5 |
| 2 | 433.0 | 466.0 | 499.1 |
| 3 | 433.0 | 468.5 | 503.9 |

$\cdot 10^{-3}$

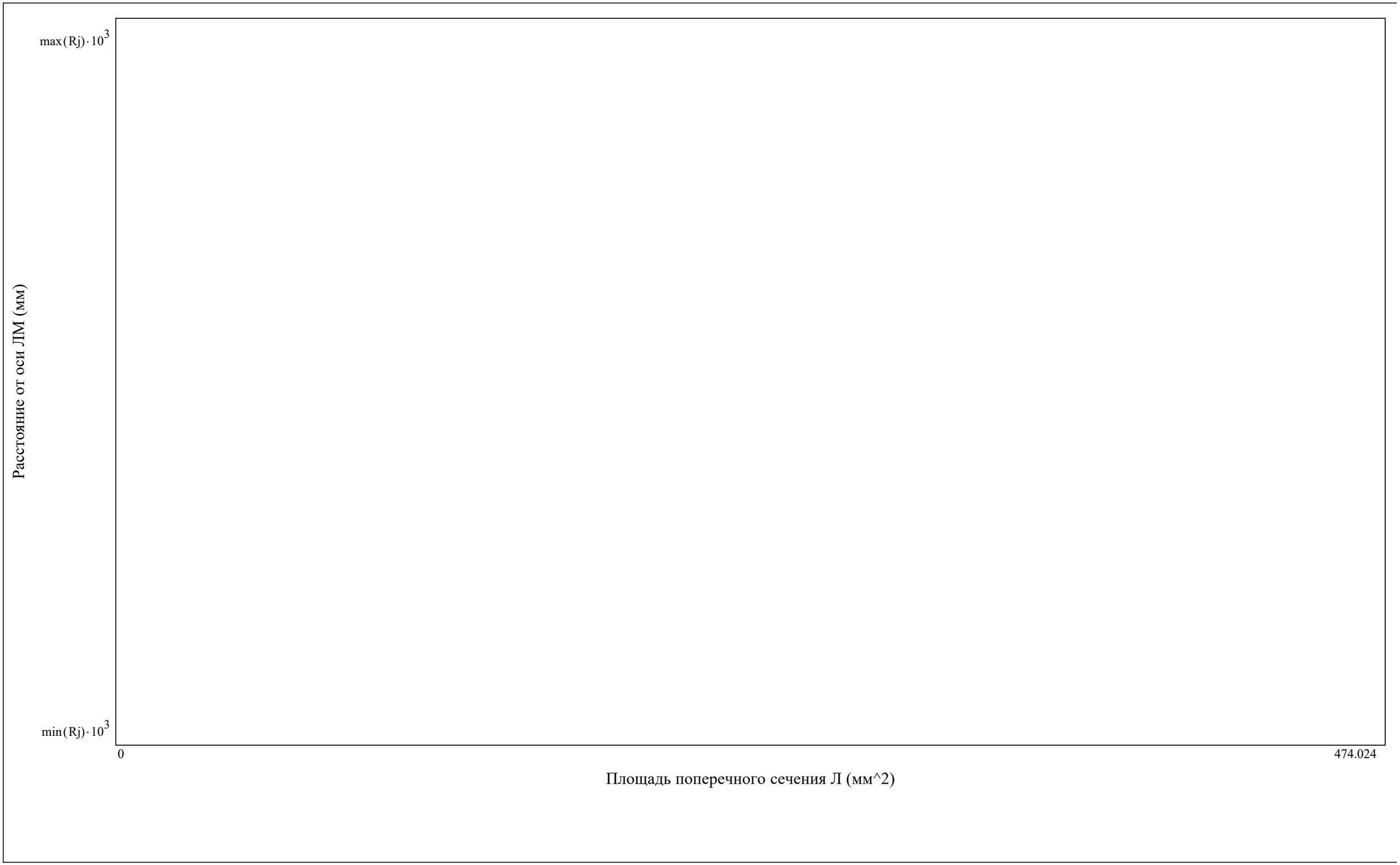
Дискретизация по высоте Л:

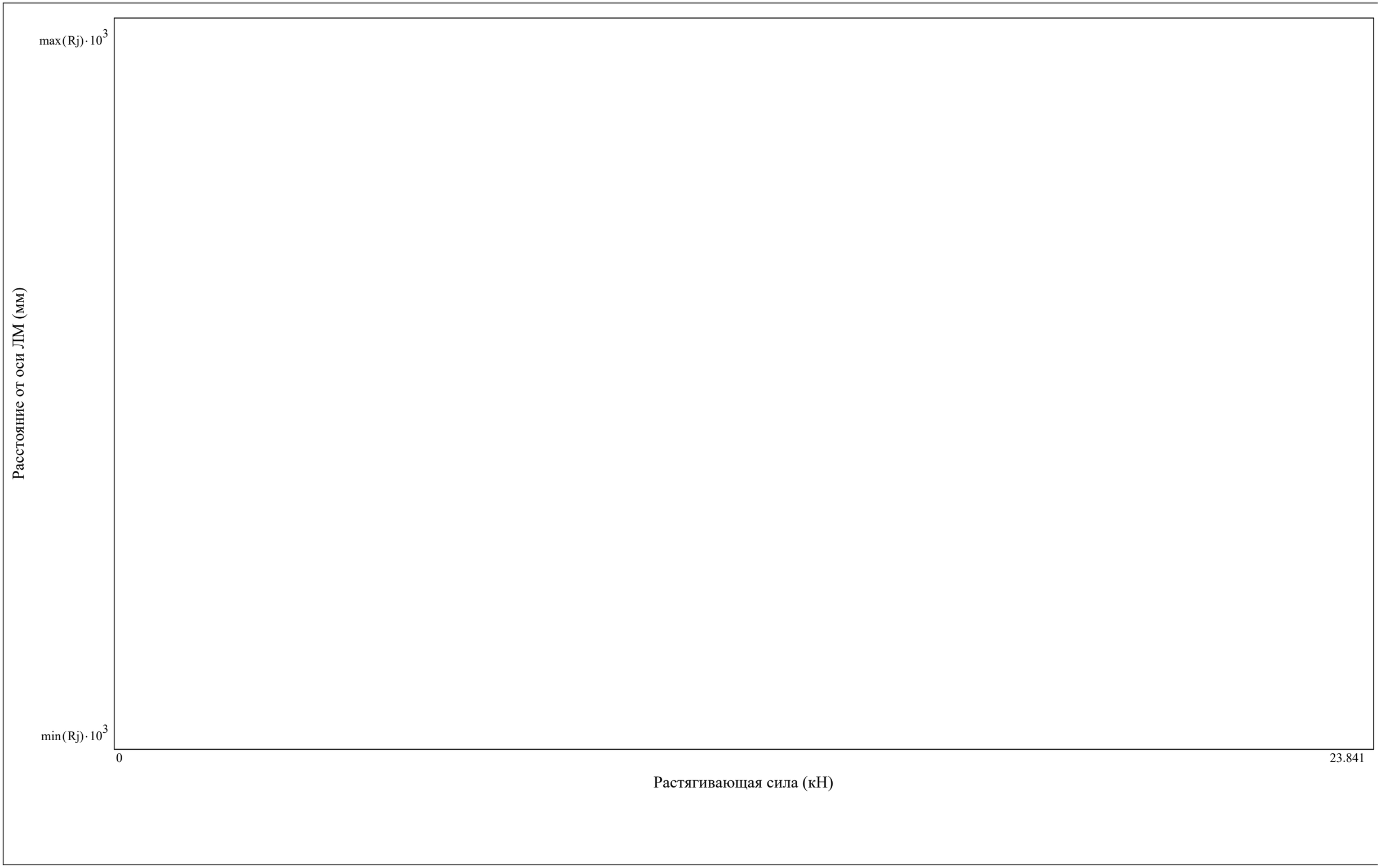
$$z = \min(Rj), \min(Rj) + \frac{\max(Rj) - \min(Rj)}{100} .. \max(Rj)$$

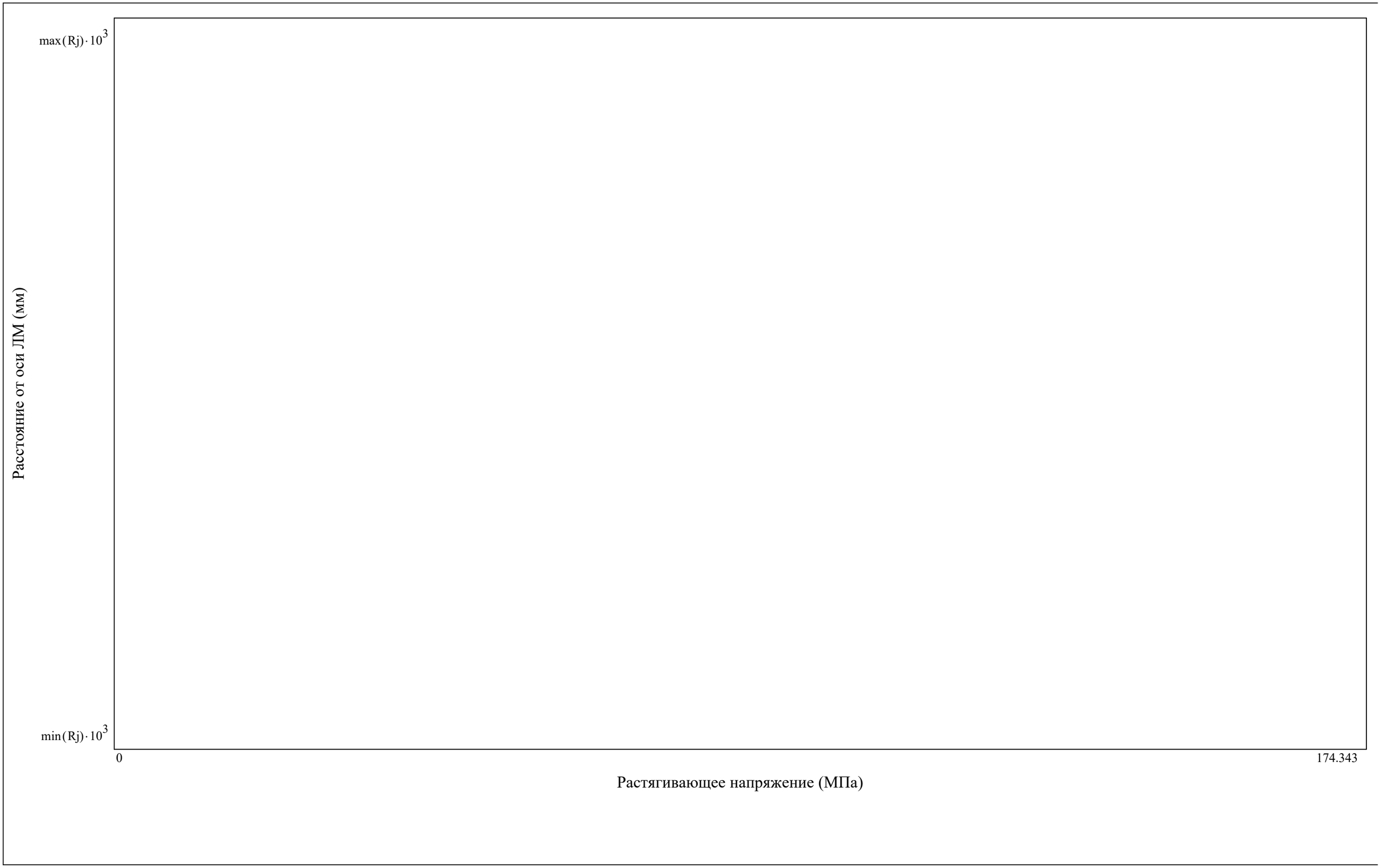
$$z_{rotor} = \left| \begin{array}{ll} \mathrm{mean}\left(Rj_{1,1}, Rj_{2,1}\right), \mathrm{mean}\left(Rj_{1,1}, Rj_{2,1}\right) + \dfrac{\mathrm{mean}\left(Rj_{1,N_r}, Rj_{2,N_r}\right) - \mathrm{mean}\left(Rj_{1,1}, Rj_{2,1}\right)}{100} .. \mathrm{mean}\left(Rj_{1,N_r}, Rj_{2,N_r}\right) & \text{if } type = \text{"compressor"} \\ \mathrm{mean}\left(Rj_{2,1}, Rj_{3,1}\right), \mathrm{mean}\left(Rj_{2,1}, Rj_{3,1}\right) + \dfrac{\mathrm{mean}\left(Rj_{2,N_r}, Rj_{3,N_r}\right) - \mathrm{mean}\left(Rj_{2,1}, Rj_{3,1}\right)}{100} .. \mathrm{mean}\left(Rj_{2,N_r}, Rj_{3,N_r}\right) & \text{if } type = \text{"turbine"} \end{array} \right.$$

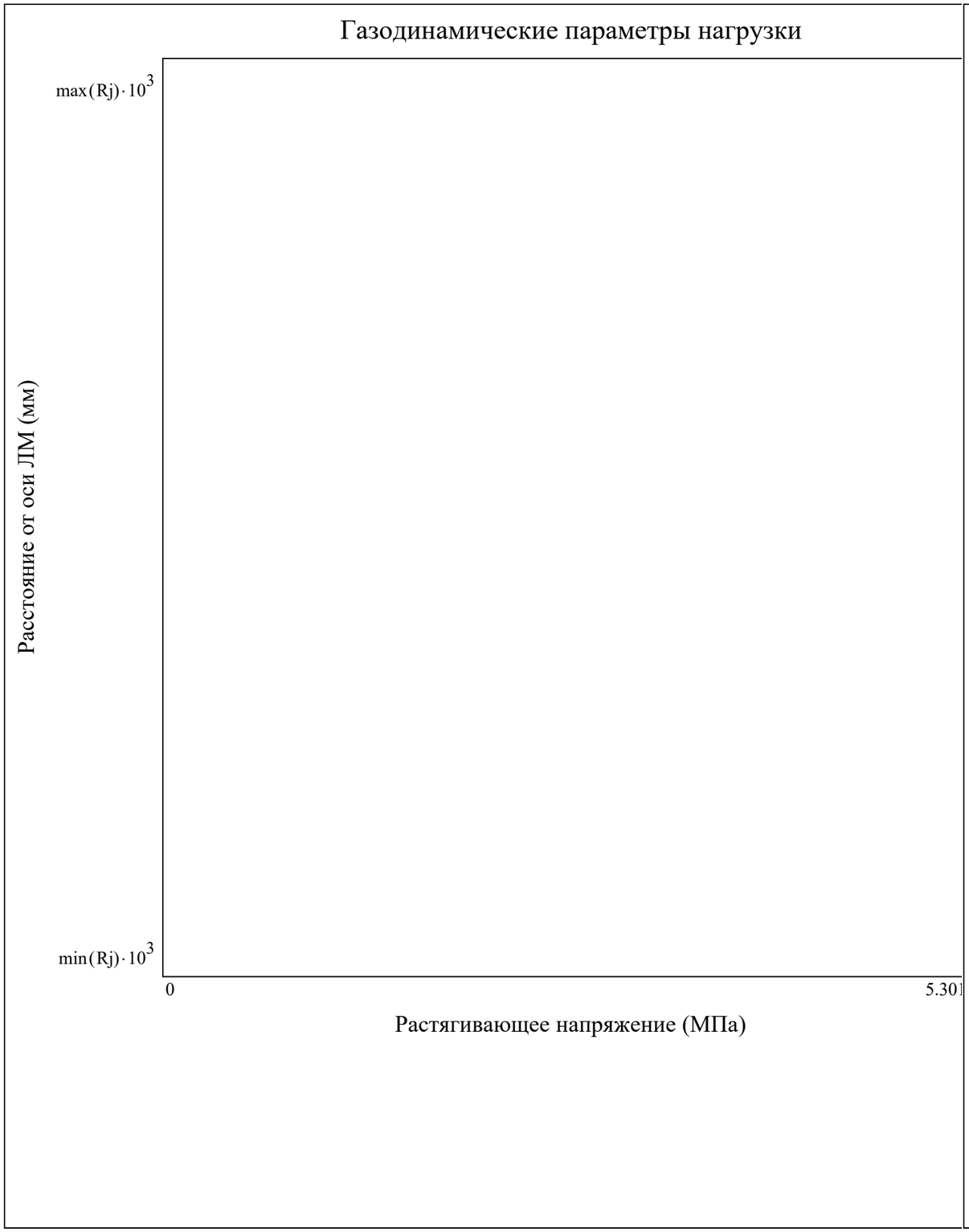
$$z_{stator} = \left| \begin{array}{ll} \mathrm{mean}\left(Rj_{2,1}, Rj_{3,1}\right), \mathrm{mean}\left(Rj_{2,1}, Rj_{3,1}\right) + \dfrac{\mathrm{mean}\left(Rj_{2,N_r}, Rj_{3,N_r}\right) - \mathrm{mean}\left(Rj_{2,1}, Rj_{3,1}\right)}{100} .. \mathrm{mean}\left(Rj_{2,N_r}, Rj_{3,N_r}\right) & \text{if } type = \text{"compressor"} \\ \mathrm{mean}\left(Rj_{1,1}, Rj_{2,1}\right), \mathrm{mean}\left(Rj_{1,1}, Rj_{2,1}\right) + \dfrac{\mathrm{mean}\left(Rj_{1,N_r}, Rj_{2,N_r}\right) - \mathrm{mean}\left(Rj_{1,1}, Rj_{2,1}\right)}{100} .. \mathrm{mean}\left(Rj_{1,N_r}, Rj_{2,N_r}\right) & \text{if } type = \text{"turbine"} \end{array} \right.$$

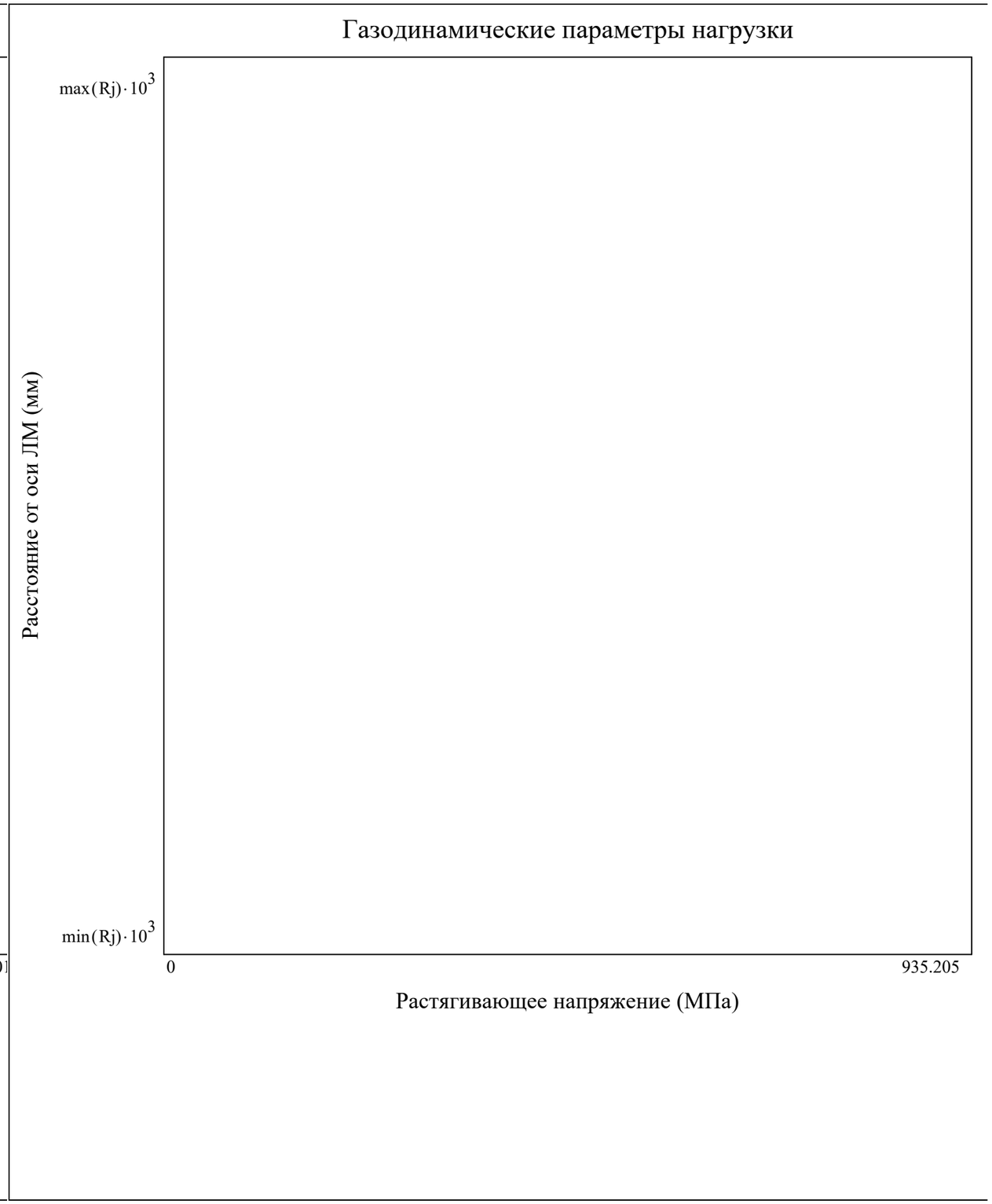
Результаты расчета на прочность Л

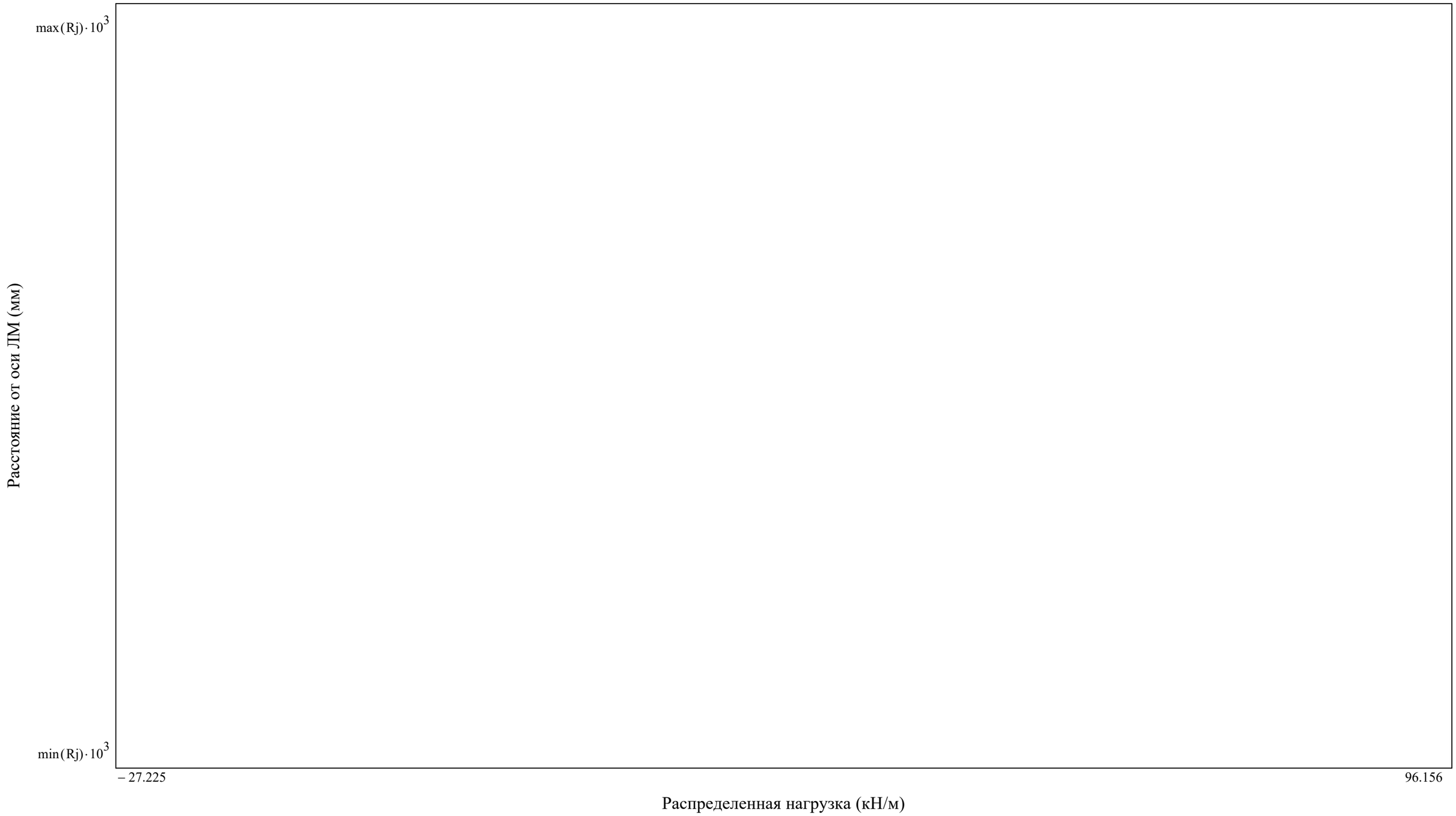

max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
474.024
Площадь поперечного сечения Л (мм^2)

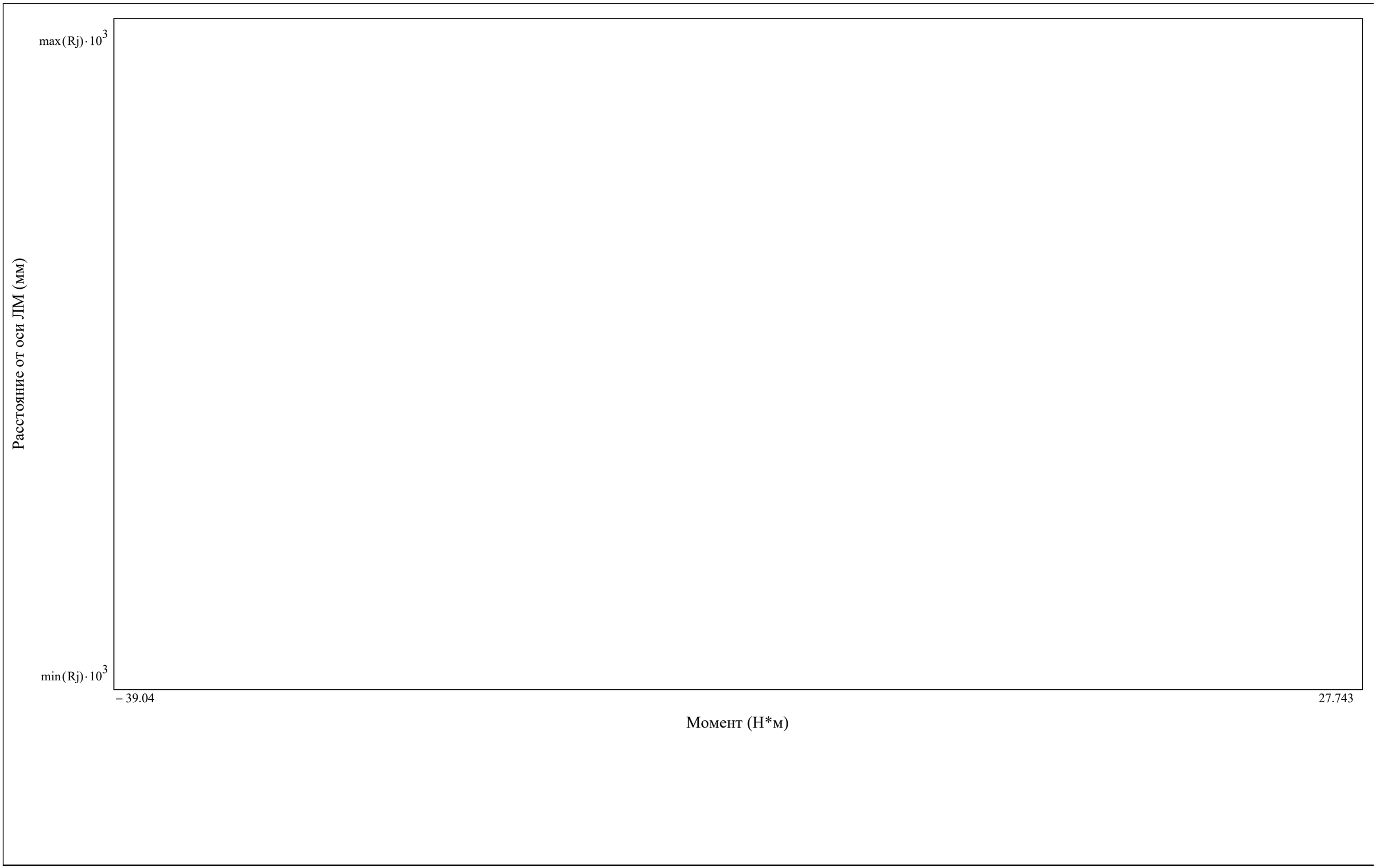

max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
23.841
Растягивающая сила (кН)

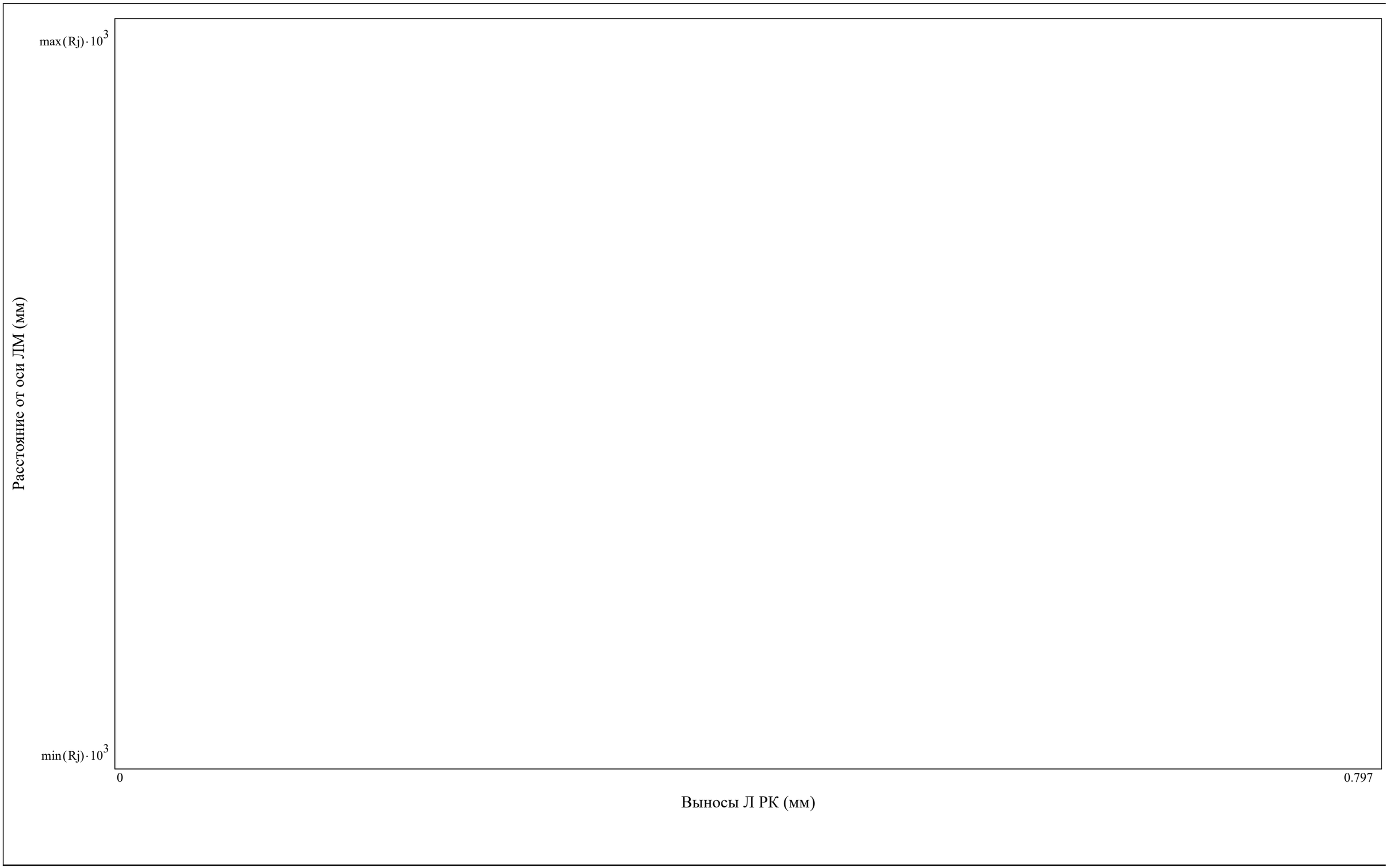

max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
174.343
Растягивающее напряжение (МПа)

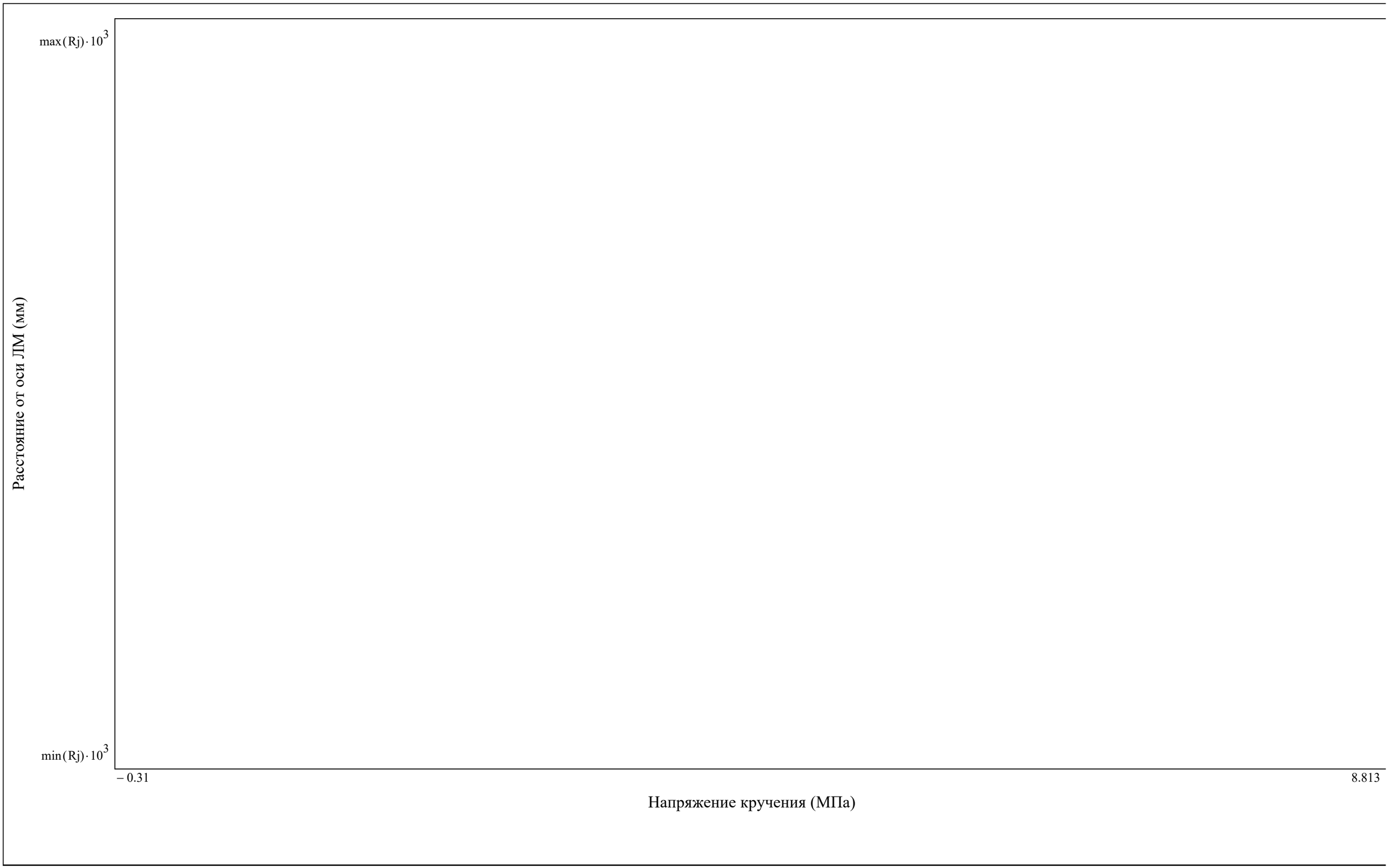

Газодинамические параметры нагрузки
max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
5.301
Растягивающее напряжение (МПа)

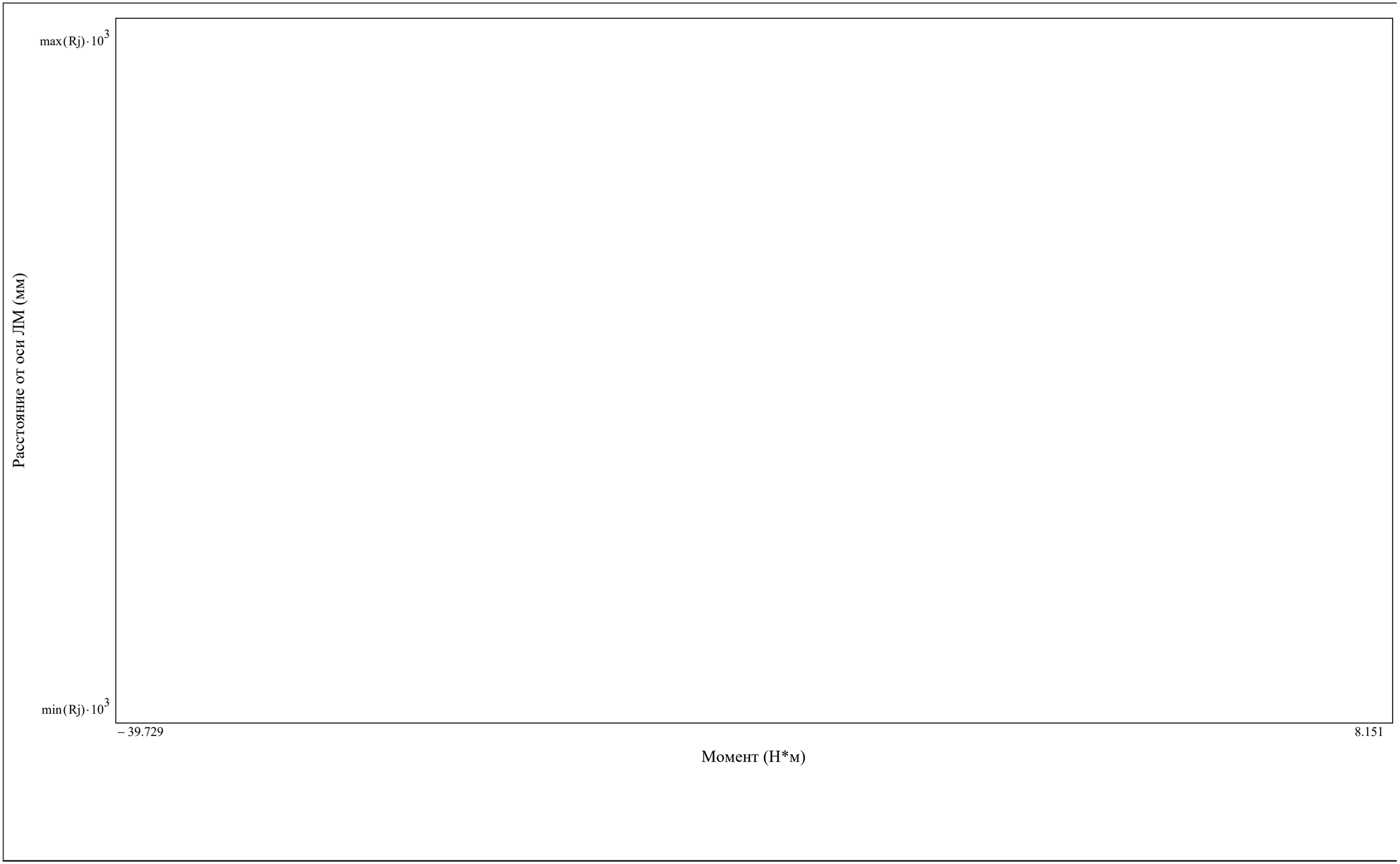

Газодинамические параметры нагрузки
max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
935.205
Растягивающее напряжение (МПа)

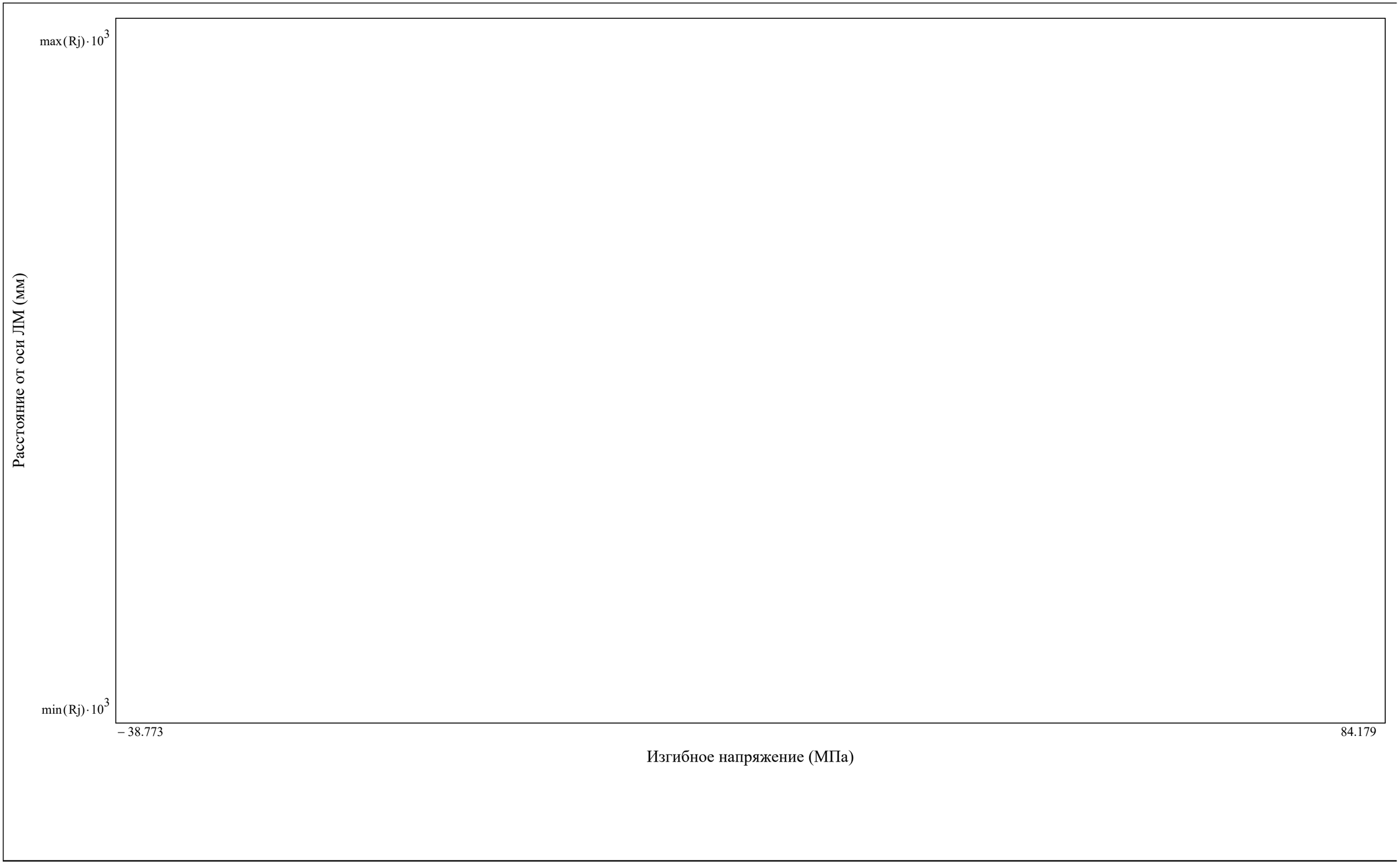

max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
– 27.225
96.156
Распределенная нагрузка (кН/м)

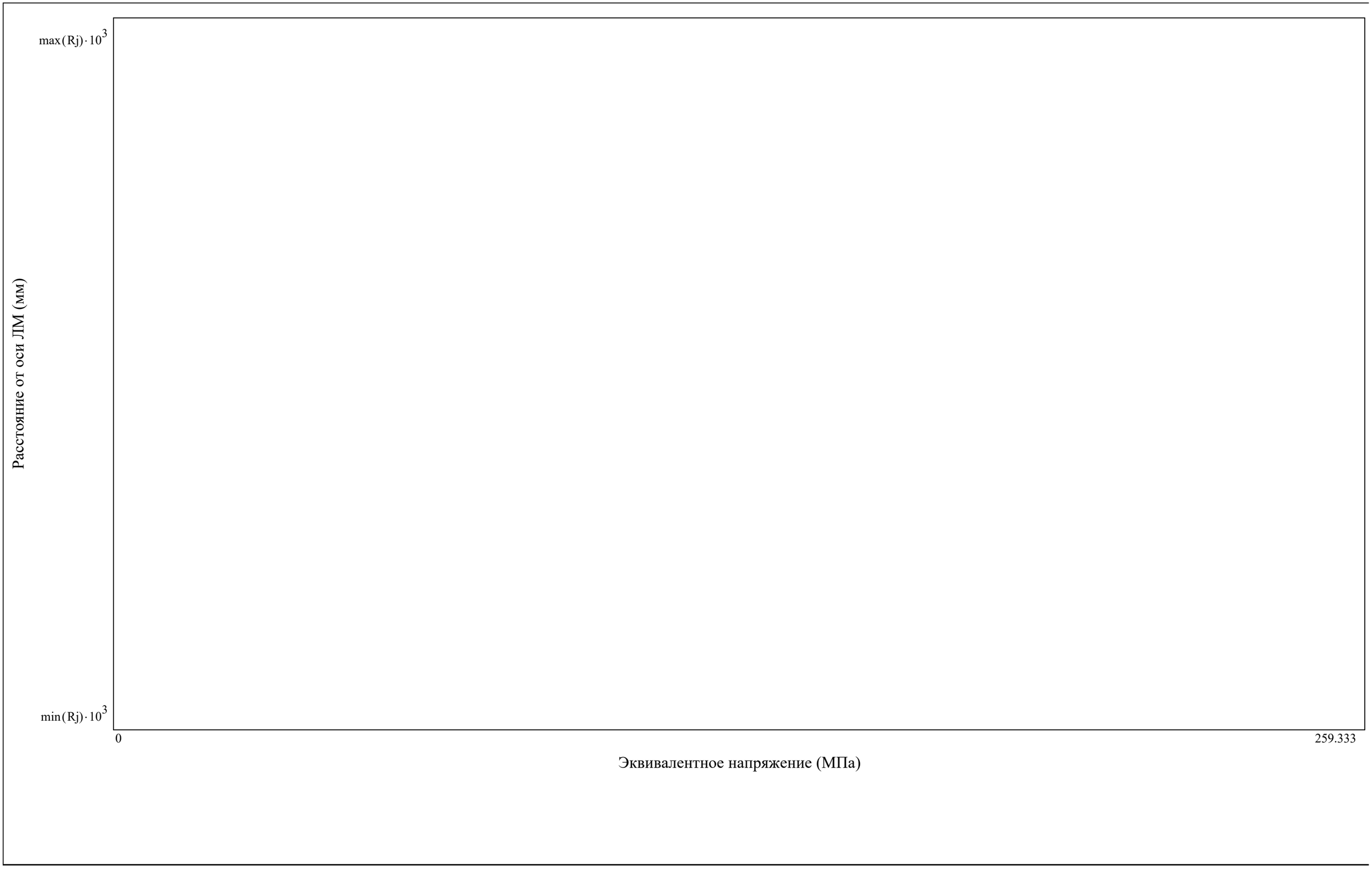

max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
– 39.04
27.743
Момент (Н*м)


max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
0.797
Выносы Л РК (мм)


max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
− 0.31
8.813
Напряжение кручения (МПа)


max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
– 39.729
8.151
Момент (Н*м)


max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
– 38.773
84.179
Изгибное напряжение (МПа)

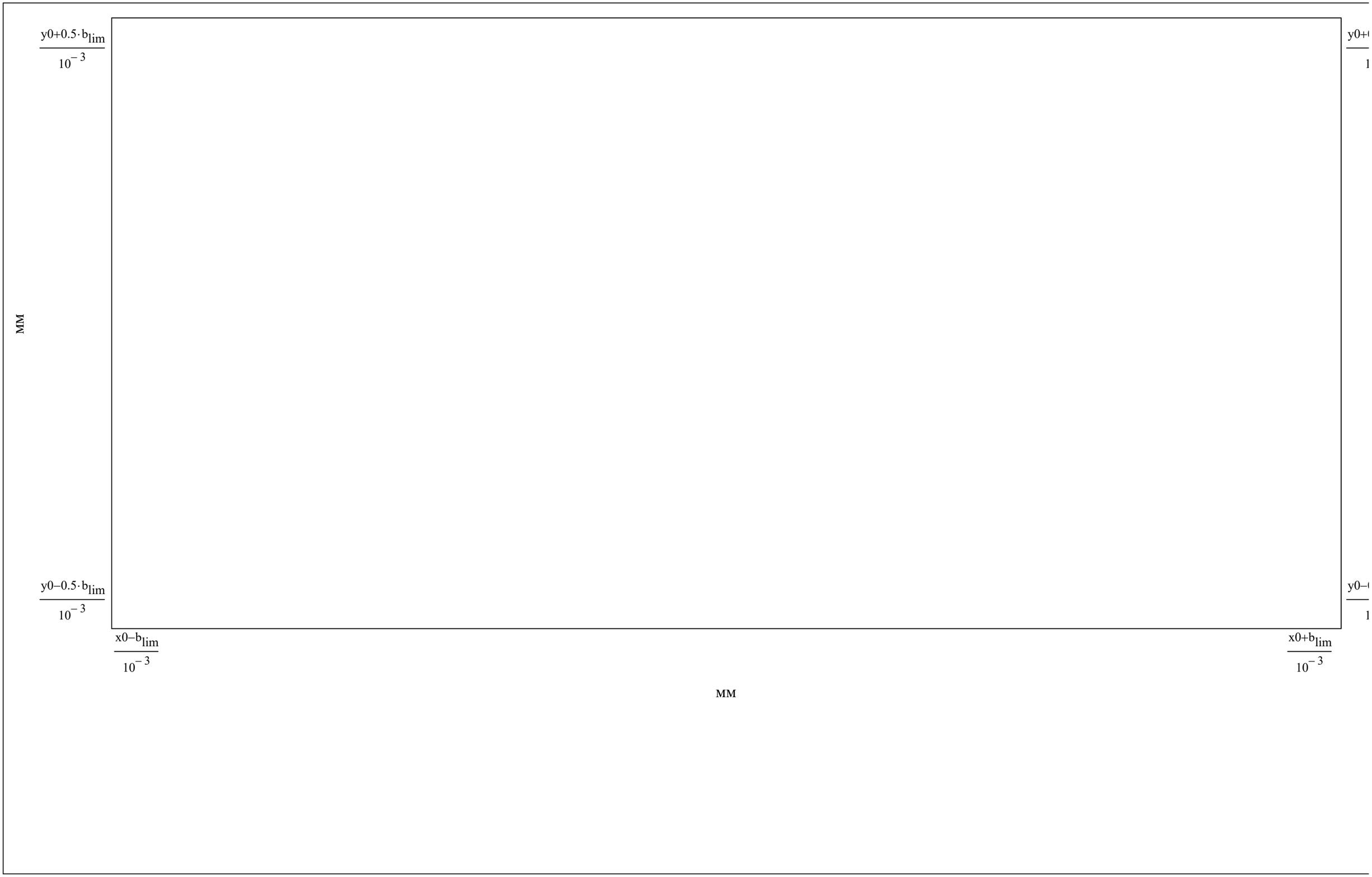

max(Rj)·10^3
min(Rj)·10^3
Расстояние от оси ЛМ (мм)
0
259.333
Эквивалентное напряжение (МПа)

$$\begin{pmatrix} blade \\ r \end{pmatrix} = \begin{pmatrix} \text{"rotor"} \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} v_p \\ v_n \end{pmatrix} = \begin{cases} \begin{pmatrix} v_u_{rotor_{j,r}} \\ v_l_{rotor_{j,r}} \end{pmatrix} & \text{if } blade = \text{"rotor"} \\ \begin{pmatrix} v_u_{stator_{j,r}} \\ v_l_{stator_{j,r}} \end{pmatrix} & \text{otherwise} \end{cases} = \blacksquare \cdot 10^{-3}$$

$$\begin{pmatrix} x0 \\ y0 \end{pmatrix} = \begin{cases} \begin{pmatrix} x0_{rotor_{j,r}} \\ y0_{rotor_{j,r}} \end{pmatrix} & \text{if } blade = \text{"rotor"} \\ \begin{pmatrix} x0_{stator_{j,r}} \\ y0_{stator_{j,r}} \end{pmatrix} & \text{otherwise} \end{cases} = \begin{array}{|c|c|} \hline & 1 \\ \hline 1 & 17.846 \\ \hline 2 & 6.242 \\ \hline \end{array} \cdot 10^{-3}$$

$$chord = \begin{cases} chord_{rotor_{j,r}} & \text{if } blade = \text{"rotor"} \\ chord_{stator_{j,r}} & \text{if } blade = \text{"stator"} \end{cases} = 42.4 \cdot 10^{-3}$$


$\frac{y0+0.5\cdot b_{lim}}{10^{-3}}$
$\frac{y0-0.5\cdot b_{lim}}{10^{-3}}$
$\frac{x0-b_{lim}}{10^{-3}}$
$\frac{x0+b_{lim}}{10^{-3}}$
MM
MM

Наиболее удаленные точки от НЛ (мм):

$$\begin{pmatrix} u_u_{rotor_{j,r}} & v_u_{rotor_{j,r}} \\ u_l_{rotor_{j,r}} & v_l_{rotor_{j,r}} \\ u_u_{stator_{j,r}} & v_u_{stator_{j,r}} \\ u_l_{stator_{j,r}} & v_l_{stator_{j,r}} \end{pmatrix} = \blacksquare \cdot 10^{-3}$$

$$\begin{pmatrix} \sigma_p_{rotor_{j,r}} & \sigma_p_{stator_{j,r}} \\ \sigma_n_{rotor_{j,r}} & \sigma_n_{stator_{j,r}} \end{pmatrix} = \blacksquare \cdot 10^{6}$$

$$\begin{pmatrix} \sigma_{stator_{j,r}} \\ \sigma_{rotor_{j,r}} \end{pmatrix} = \blacksquare \cdot 10^{6}$$

$$\begin{pmatrix} safety_{stator_{j,r}} \\ safety_{rotor_{j,r}} \end{pmatrix} = \blacksquare$$

Результаты расчета на прочность Л

$$type := \begin{cases} type = \text{"stator"} & \\ type = \begin{cases} \text{"Нет такого типа!"} & \text{if } type \neq \text{"stator"} \wedge type \neq \text{"rotor"} \\ type & \text{otherwise} \end{cases} \end{cases} = \text{"stator"}$$

Рассматриваемая ступень:

$$j := \begin{cases} j = 1 & \\ j = \begin{cases} \text{"Такой ступени не существует!"} & \text{if } (j < 1) \vee (j > Z) \\ j & \text{otherwise} \end{cases} \end{cases} = 1$$

Приведенная скорость по профилю Л

Общие параметры охлаждения

Конвективное охлаждение

## Результаты конвективного охлаждения

## Результат конвективно-пленочного охлаждения

$D^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 650.0 | 650.0 | 617.2 | | | | | | |
| 2 | 678.5 | 678.5 | 662.1 | | | | | | |
| 3 | 707.0 | 707.0 | 707.0 | | | | | | |

$\cdot 10^{-3}$

3

$$R := \frac{D}{2}$$

$R^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 325.0 | 325.0 | 308.6 | | | | | | |
| 2 | 339.2 | 339.2 | 331.0 | | | | | | |
| 3 | 353.5 | 353.5 | 353.5 | | | | | | |

$\cdot 10^{-3}$

$$\bar{d} = \left| \begin{array}{l} \text{for } i \in 1..Z \\ \quad \text{for } a \in 1..3 \\ \quad\quad \bar{d}_{st(i,a)} = \dfrac{D_{st(i,a),1}}{D_{st(i,a),N_r}} \\ \bar{d} \end{array} \right.$$

$\bar{d}^T =$

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.9194 | 0.9194 | 0.8730 | | | | | | |

$\bar{d}^T \le 0.9 =$

| | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 0 | 0 | 1 |

$$h = \left| \begin{array}{l} \text{for } i \in 1..2Z+1 \\ \quad h_i = \dfrac{F_i}{\pi \cdot D_{i,av}(N_r)} \\ h \end{array} \right.$$

$h^T =$

| | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 28.50 | 28.50 | 44.88 |

$\cdot 10^{-3}$

$$\begin{pmatrix} c_{st(j,1),r} \\ c_{st(j,2),r} \\ c_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 197.2 \\ 511.5 \\ 232.4 \end{pmatrix}$$

$$\begin{pmatrix} c_{a_{st(j,1),r}} \\ c_{a_{st(j,2),r}} \\ c_{a_{st(j,3),r}} \end{pmatrix} = \begin{pmatrix} 195.0 \\ 253.8 \\ 232.0 \end{pmatrix}$$

$$\begin{pmatrix} w_{st(j,1),r} \\ w_{st(j,2),r} \\ w_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 282.8 \\ 325.5 \\ 324.6 \end{pmatrix}$$

$$\begin{pmatrix} \alpha_{st(j,1),r} \\ \alpha_{st(j,2),r} \\ \alpha_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 81.44 \\ 29.75 \\ 86.72 \end{pmatrix} \cdot °$$

$$\begin{pmatrix} u_{st(j,1),r} \\ u_{st(j,2),r} \\ u_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 234.2 \\ 240.3 \\ 240.3 \end{pmatrix}$$

$$\begin{pmatrix} \beta_{st(j,1),r} \\ \beta_{st(j,2),r} \\ \beta_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 43.59 \\ 128.76 \\ 45.62 \end{pmatrix} \cdot °$$

$$\varepsilon_{stator_{j,r}} = 51.69 \cdot °$$

$$\varepsilon_{rotor_{j,r}} = 83.14 \cdot °$$

$$\begin{pmatrix} c_{st(j,1),r} \\ c_{st(j,2),r} \\ c_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 197.0 \\ 483.7 \\ 232.4 \end{pmatrix}$$

$$\begin{pmatrix} c_{a_{st(j,1),r}} \\ c_{a_{st(j,2),r}} \\ c_{a_{st(j,3),r}} \end{pmatrix} = \begin{pmatrix} 196.6 \\ 232.9 \\ 232.4 \end{pmatrix}$$

$$\begin{pmatrix} w_{st(j,1),r} \\ w_{st(j,2),r} \\ w_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 309.0 \\ 285.6 \\ 348.6 \end{pmatrix}$$

$$\begin{pmatrix} \alpha_{st(j,1),r} \\ \alpha_{st(j,2),r} \\ \alpha_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 86.19 \\ 28.78 \\ 89.96 \end{pmatrix} \cdot °$$

$$\begin{pmatrix} u_{st(j,1),r} \\ u_{st(j,2),r} \\ u_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 251.6 \\ 258.7 \\ 260.0 \end{pmatrix}$$

$$\begin{pmatrix} \beta_{st(j,1),r} \\ \beta_{st(j,2),r} \\ \beta_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 39.50 \\ 125.36 \\ 41.81 \end{pmatrix} \cdot °$$

$$\varepsilon_{stator_{j,r}} = 57.4 \cdot °$$

$$\varepsilon_{rotor_{j,r}} = 83.56 \cdot °$$

$$\begin{pmatrix} c_{st(j,1),r} \\ c_{st(j,2),r} \\ c_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 197.0 \\ 458.7 \\ 232.4 \end{pmatrix}$$

$$\begin{pmatrix} c_{a_{st(j,1),r}} \\ c_{a_{st(j,2),r}} \\ c_{a_{st(j,3),r}} \end{pmatrix} = \begin{pmatrix} 197.0 \\ 212.9 \\ 232.1 \end{pmatrix}$$

$$\begin{pmatrix} w_{st(j,1),r} \\ w_{st(j,2),r} \\ w_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 334.2 \\ 249.1 \\ 371.9 \end{pmatrix}$$

$$\begin{pmatrix} \alpha_{st(j,1),r} \\ \alpha_{st(j,2),r} \\ \alpha_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 90.31 \\ 27.66 \\ 92.70 \end{pmatrix} \cdot °$$

$$\begin{pmatrix} u_{st(j,1),r} \\ u_{st(j,2),r} \\ u_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 268.9 \\ 277.0 \\ 279.7 \end{pmatrix}$$

$$\begin{pmatrix} \beta_{st(j,1),r} \\ \beta_{st(j,2),r} \\ \beta_{st(j,3),r} \end{pmatrix} = \begin{pmatrix} 36.12 \\ 121.27 \\ 38.62 \end{pmatrix} \cdot °$$

$$\varepsilon_{stator_{j,r}} = 62.65 \cdot °$$

$$\varepsilon_{rotor_{j,r}} = 82.65 \cdot °$$

$$\frac{0.5 \cdot b_{lim}}{10^{-3}}$$

$$\frac{0.5 \cdot b_{lim}}{10^{-3}}$$